

5 Conceptualising a future oceans management system



5.1 Introduction

So far, we have outlined a number of problems and challenges that a future system will need to address. We have also highlighted some important contextual matters that any reform effort will need to be sensitive to. The scene is now set for Part 2 where we ask: what could a new system look like?

It is not particularly useful to simply offer a laundry list of possible changes in random order. We need to conceptualise the system as a connected whole, and divide it into manageable pieces for analysis. In this chapter we briefly explain options for how the system could be conceptualised/constructed, and describe the approach we have taken. In the chapters that follow, we then use our adopted structure to explore reform options.

5.2 Ways to divide up the oceans management system

The oceans management system could be conceptualised in a variety of ways (see Figure 5.1). For example, it is possible to look at the system on a sector-by-sector basis (eg chapters outlining reform options for fishing, aquaculture, mining, urban development); a resource-by-resource basis

(eg chapters on fish, coastal space, minerals); or a space-by-space basis (eg chapters on the Hauraki Gulf, Fiordland, a particular fisheries management area). It is also possible to go through existing statutory frameworks (eg chapters looking at possible changes to the RMA, the Fisheries Act and conservation legislation), or to explore reforms on a problem-by-problem basis (eg options for addressing biodiversity loss, climate change, chemical pollution). All are valid ways of thinking about “parts” of the system. Indeed, a number of more targeted analyses have looked at the system through these lenses in the past, including with respect to fisheries and the management of the EEZ.¹

Focusing on specific lenses like domains, spaces, sectors and resources remains important. We need to consider what makes the management of some of them different from others. For example, extensive property rights on land (one space) mean aspects of the system (and therefore reform options) may look quite different to those for the sea (another space).² Some specific locations (eg Fiordland) have features that may warrant dedicated protection or focus. We may have objectives or visions for some sectors or resources (eg renewable energy or aquaculture) that do not apply to others (eg oil and gas extraction). Each requires targeted analysis within a broader reform effort.

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Salmon farm, Marlborough Sounds

Lens	Explanation	Examples
Domain	An interrelated set of resources, or a “part” of the environment that is valued. A single domain can exist across multiple spaces,³ be impacted by multiple sectors, and contain multiple resources. Multiple domains are relevant to the marine context because they can impact on it (eg biodiversity, land, freshwater, soil, climate), even though “marine” is often considered to be a domain in its own right.	Marine, freshwater, biodiversity (or flora and fauna), ⁴ soil, land, air, climate.
Space	A geographical area, or a category of area defined by its predominant use. A space can contain multiple domains, ⁵ support multiple sectors, and contain many resources.	Taranaki (a region); the EEZ (a jurisdictional area); rural/urban (a category of area); ⁶ Fiordland (an area defined by physical characteristics).
Sector	The different ways in which humans use resources. ⁷ A single sector can use multiple resources, ⁸ impact multiple domains, and operate across multiple spaces.	Fishing, mining, aquaculture, navigation, tourism, scientific study, conservation.
Resource ⁹	The category of thing being used/consumed. ¹⁰ A single resource can be used by multiple sectors, ¹¹ exist in multiple domains, ¹² and exist across different spaces. ¹³	Fish, oil and gas, water, sand, wind.
Legislation	Existing statutory frameworks.	The RMA, Fisheries Act, Conservation Act, Marine Mammals Protection Act.
Problem	Specific problems or issues that need to be solved or addressed.	Biodiversity loss, water quality, climate change.

Figure 5.1: Some possible ways to conceptualise the oceans management system for analysis

However, an approach that only looks through one of these lenses risks becoming siloed, where interactions between spaces, sectors, resources, problems and domains are either lost or underappreciated. For example, a focus on fishing (a single sector) may ignore the fact that the many land-based pressures (eg sediment) affect fish stocks directly or through habitat degradation. This has been the case in practice, where a sectoral silo for fisheries management has meant that there has been little cognisance of decisions taken on land that impact that sector.¹⁴ Similarly, a focus on marine mining (another sector) risks ignoring the tensions that might arise between potential competing uses of seabed space (for example, between oil and gas extraction and carbon geo-sequestration in partially depleted wells).¹⁵ Overall, as we explained in a previous report:¹⁶

the greatest risk is that domain-based, spatial and sectoral approaches all have a tendency to presuppose at least something. A domain-based approach assumes that each receiving environment can be considered,

at least partly, in isolation of the others. In reality, they are intimately connected across varying spatial scales [for example, the important link between freshwater catchments and estuaries].

Similarly, a spatial approach tends to focus on spaces artificially delineated by humans [for example, the arbitrary line between the territorial sea and EEZ, or between QMAs]... It also runs the risk of ignoring ecological and other units (eg landscape or cultural) that frequently exist across lines on a map.

Furthermore, considering sectors is not only hard to do (there are thousands of potential categories, from horticulture to fishing), but also risks neglecting the importance of cumulative effects on receiving environments (which may be impacted by a large number of different sectors at the same time).

As shall be seen in Chapter 11, one issue with the current oceans management system is that it is fragmented across legislation, institutions and tools that are focused on a narrow range of sectors, spaces or resources. Thus looking at each existing statute in turn (eg the RMA, the Fisheries Act, the Marine Reserves Act) may fail to capture some important bigger picture questions like the tools that can be deployed to link silos together (eg marine spatial planning), the objectives underpinning them (eg te oranga o te taiao¹⁷ versus MSY), or the potential for fundamental legislative redesign (eg combining separate statutes into a single Oceans Act). Thus while going through existing frameworks is a useful way to describe the current system, it is a less useful for imagining the future. Individual “problems” are also often too interlinked to be tackled one by one.¹⁸

Part 2 of the work is about exploring options for reform. This involves dividing the system into different parts for analysis. How this is done is important, as it reflects how one conceptualises the system as a whole.

There are risks in approaching the system as a series of discrete sectors, geographical spaces, individual resources, separate domains, individual problems or existing statutes. Doing so may not capture the important links between and across these things.

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Danish seining, Hauraki Gulf

5.3 Our preferred approach: thematic layers

We think it is most useful to conceptualise the oceans management system as a series of broad thematic layers that cover the whole system. This enables us to split the system up in a way that cuts across all the things listed above, rather than analysing separate pieces (sectors, spaces, legislation) that then need to be stitched together. These broad themes are explored below, and comprise norms (eg ethics, principles, objectives), tools (different kinds of intervention) and structures (legislative and institutional design). For all of these matters, “networks” (things that move across and between other elements of the system) are relevant. These include the flow of money, information and public involvement.

A thematic structure is arguably more sensitive than others (eg sectoral, statute-by-statute) to the holistic approach of te ao Māori. Māori matters are not simply things the system has to address or “do”. Te ao Māori needs to pervade all tiers of the system (including how it is conceptualised) so that it is fully integrated, not treated as an add-on, afterthought, or a

group of matters placed in opposition to (or as grudging concessions to) a dominant Western paradigm.

Of course, splitting analysis in any way, including into themes, creates risks. Themes like legislative design, institutional settings and public participation are closely connected to each other and cannot be treated in isolation. It is therefore important to draw all these threads together to consider how themes work together to form a new system (see Part 3 of the report). However, as a whole, we think these cross-theme connections are easier to make, and provide deeper system-wide insights, than those relying on other analytical building blocks like statutes or sectors.

When considering system-wide reform options we have structured our analysis according to themes, including norms, tools and structures. Across these things, networks (eg money, information and public involvement) are also important to consider.



Te Werahi Beach

Craig Potton

The order in which we tackle these themes is important. Is it better, for example, to adopt a “top-down” approach? That would see us begin with overarching questions around ethics and principles, through to legislative structures and institutional design, then broad planning tools and finally small-scale interventions like consents and economic incentives. Broadly speaking, we took a top-down approach in our resource management reform project, saying:¹⁹

It makes sense to see relationships as broadly linear by starting with norms (what outcomes we ought to pursue, such as principles), then considering components of the system needed to achieve them (what kinds of things it needs to do, and the structures we need to establish, such as legislation and institutions), and ending with how we implement them (mechanisms such as plans, consents, processes and incentives). The conclusions of the prior exercise inform the consideration of the latter.

Alternatively, we could build our way from the bottom-up, focusing on the easy wins first (eg improvements to individual tools) and moving to larger scale interventions like changing institutional settings (eg restructuring local government) and legislative boundaries (eg creating an integrated Oceans Act). Only at the end would we consider options for more fundamental shifts to our ethics and principles. Essentially, this would be about moving from “least change” options to “most change” options, where the reader would be invited to draw a line at the point at which his or her appetite for reform abated.

In our view, big picture normative questions need to be tackled first. Even smaller scale changes (eg the deployment of underutilised tools like sustainability measures or MPAs) requires us to consider first *what we are aiming for and why*. Any change needs a solid normative framing. For example, deploying tools/mechanisms like legal personhood for nature, or putting a price on the natural world, requires us first to think deeply about potentially conflicting worldviews (is it right to treat nature as a commodity? Is it naïve to think of the oceans as a non-human person?). Deploying marine reserves requires us to ask what their purpose is – scientific research? Overall biodiversity enhancement? Tourism and recreation?

The theme of “norms” can be broken down into a series of sub-themes to be considered in turn, starting with what the system itself should and should not cover (ie when there is “overreach”), through to its ethical foundations, its principles, and the specific objectives it pursues (see Chapter 7).

It is important to think about the toolkit before considering the structural features of the system (how to design legislation and institutions) (see Chapters 8, 9 and 10). The latter need to be designed around the former. “Tools” are, essentially, any form of public intervention within the scope of the oceans management system.

Ultimately, statutes and institutions are about providing the most effective frameworks through which interventions can be made. In other words, form (structures like statutes and public institutions) should really follow function (the tools they deploy). For example, we should not create a Fisheries Act and then ask what it should contain, nor should we create an environmental regulator and then ask what it should do and why.

The mix of tools we use is not normatively neutral. To choose which tools are “best” we first need to have a sense of what they are trying to achieve. The risks of getting the order wrong can be seen in tools like MPAs, where a single type of tool is currently provided for in multiple statutes with different purposes, processes and institutional guardians. It can also be seen in the Tiriti clauses that differ significantly across legislation with no clear rationale for such differences.

The structural features of the system – legislation (see Chapter 11) and institutions (see Chapter 12) – then need to be considered closely together. As mentioned above, statutory and institutional boundaries, purposes/ mandates and relationships need to provide the most effective “home” for tools to be deployed and supported. But the design of legislation and institutions can also provide additional complementary measures. For example, fragmented *statutory* arrangements can be overcome by giving a single *institution* powers or oversight over all of them (eg the Department of Conservation administers a wide range of conservation legislation under the umbrella framework of the Conservation Act).

We are seeking to construct reform options for a future system through a series of themes. Essentially, this looks at:

- norms – what the system should be seeking to achieve;
- tools – the mechanisms/interventions by which the system seeks to achieve them; and
- structures – the legislative and institutional arrangements framing the system’s tools.

Finally, it is worth reiterating that this report is about presenting *options* for reform. Many different options could be put together across themes to create potentially hundreds of specific models for the future. In Part 3, we offer several possible starting points for what a system could look like overall.

It is also worth noting that a new system (no matter what it looks like) will not simply snap into place instantly. There will be a journey to get there over time. A sound transition plan, and one that is just and equitable in managing the process of change, will be crucial, but can only be considered in detail once there is a sense of what a new system should look like.

5.4 Concluding comments

The above discussion has two purposes. First, it outlines the way in which we are thinking about, and splitting up, the oceans management system for analysis. It provides a structure for the report and way of thinking about options that breaks down an otherwise overwhelmingly large subject into manageable chunks. But the discussion is also intended to highlight the importance of having *some* form of overarching conceptual framework, and for that to be clearly articulated in any future reform process the government undertakes. A theme-based conceptualisation is the one we have adopted (norms, tools, structures), and is one that could be usefully used for a future reform process, although others would also be possible. We turn now to a thorny question: why do we have a system at all and what do we expect it to do?



Picton

Endnotes

1

For example, see Raewyn Peart, Kate Mulcahy and Kelsey Serjeant *Governing Our Oceans: Environmental Reform for the EEZ* (Environmental Defence Society, Auckland, April 2011) and Office of the Prime Minister’s Chief Science Advisor *The Future of Commercial Fishing in Aotearoa New Zealand* (February 2021).

2

Although there are still some property rights in the marine context (eg quota, privately held parts of the coastal marine area).

3

For example, land spans the rural-urban divide and exists under lakes, rivers and the sea; flora and fauna/biodiversity does too.

4

Strictly speaking, “biodiversity” is less a domain than it is an objective (a diversity of flora and fauna, usually referring to indigenous flora and fauna). However, it is a convenient shorthand.

5

For example, Fiordland spans land and sea, while Taranaki contains freshwater and biodiversity.

6

Despite some treating “urban” as a domain alongside water, air and marine, it is better characterised as a space.

7

This is not exclusively for commercial gain. A sector can include recreation and conservation.

8

This includes the “resource” of a receiving environment for waste/pollution.

9

We recognise that the term “resource” tends to suggest an instrumentalist approach to value, and that other words may be more appropriate (eg taonga, te taiao or environment).

10

Not all resources are consumed. For example, marine mammals are a valuable resource for tourism (whale watching).

11

“Fish” and “fishing” are quite different. Fish can be used by sectors other than fishing (eg for conservation), and fishing has impacts on resources other than fish (eg habitats, marine mammals, other forms of bycatch).

12

For example, fish exist in domains like “marine”, “freshwater”, and “biodiversity”.

13

For example, across urban-rural boundaries and jurisdictional boundaries like the EEZ-coastal marine area.

14

Fisheries Act 1996.

15

Carbon sequestration is a process by which captured and compressed carbon dioxide emissions are pumped into depleted or partially depleted oil and gas formations. See Barry Barton, Kimberley Jane Jordan and Greg Severinsen *Carbon capture and storage: Designing the legal and regulatory framework for New Zealand*, (Centre for Environmental, Energy and Resources Law, Te Piringa Faculty of Law, University of Waikato, September 2013).

16

Greg Severinsen and Raewyn Peart *Reform of the Resource Management System: The Next Generation* (Environmental Defence Society, Auckland, December 2018) at 34.

17

See *Natural and Built Environments Bill* (Exposure Draft, 2021), cl 5; Ministry for the Environment *Natural and Built Environments Bill Parliamentary Paper on the Exposure Draft* (June, 2021), [93]-[99] at <www.environment.govt.nz/publications/natural-and-built-environments-bill-parliamentary-paper-on-the-exposure-draft/>.

18

For example, multiple overlapping categories of problems identified in environmental reporting.

19

Greg Severinsen and Raewyn Peart *Reform of the Resource Management System: The Next Generation* (Environmental Defence Society, Auckland, December 2018) at 35. In the Phase 1 report we structured our thinking according to a cascade of themes: from norms (ethics and principles), to functions (what roles the system should and should not play), to structures (laws, institutions, public participation), and finally tools (concrete interventions, like regulations and taxes, which shape people’s actions).

6 What is the rationale for having a system?



6.1 Introduction

This project is about the formal system, not people's personal ethics. The normative basis of those two things can be quite different. The most fundamental question to ask is, therefore, why we need an oceans management system at all. This can be recast in an even simpler way: *do* we need an oceans management system? People pursue individual and collective aims every day without a complex array of laws, institutions and public interventions to help them.

6.2 The rationale for having a system

All bar the most libertarian are, in modern times,¹ likely to accept that *some* kind of formal system is required, even if it is just to manage inevitable conflict and to prevent a tragedy of the commons in the context of finite resources. Yet it is still important to think about the reasons *why* the system is there. This is not so much about justifying its existence – that is easy to do by looking at conflicts and environmental impacts in the past – but rather about identifying where its boundaries should be, and what should be included (or excluded) at the margins. In other words, the reasons we think a system is necessary shed light on the places it is not required: the point where public authorities should have little or no role,² and where freedom, personal choices and unfettered markets should reign supreme.

Although there is unlikely to be an actual provision in a statute that specifically outlines what the rationale for the system is, it is important to have this discussion up front to avoid arguments later on about whether a particular tool or institution is going “too far” and overreaching what it is meant to be doing.

Some kind of formal system is needed for marine management. However, considering the basic reasons why we have a system sheds light on where its boundaries should be and what it should and should not do.

Perhaps surprisingly, there is no clear explanation as to where the boundaries of the *current* system lie. There is no statute that tells us why, overall, a system is required at sea. Instead, more specific interventions tend to be justified on their own merits in response to changing values,³ new technologies,⁴ or emerging issues.⁵ The lack of an overall sense of the system's scope can be seen in the case of MPAs. New proposals for protection often meet accusations that the system is overreaching

its proper boundaries by infringing people's rights (eg to fish), whereas others see new spatial protections as a core part of what the system should be doing.

This kind of debate about the limits of public power has been particularly noticeable over the past few years in the wake of Covid-19 and evolving expectations around te Tiriti o Waitangi. What is the relationship between the state and its citizens' choices? And between the Crown and Māori? We can give this a constitutional or political flavour – what limits should be placed on government power? Equally, it can be cast as a moral question – when is it right for others to constrain my freedom of choice? We can even speak in economic terms – to what extent should free markets be influenced or “distorted” by restrictions?

The oceans are a particularly fraught place in which these questions play out. On the one hand, the sea has long been seen as a place where people can escape the constraints of society, as reflected in the concept of “running away to sea”. Yet to some it is also a common or shared space, where the government has a legitimately *stronger* role to play than on land. We tackled this subject in the context of the broader resource management system, where we said:⁶

Public authorities can intervene in a wide range of things. The real issue is whether they should. Are there occasions where we should categorically not use public interventions to influence outcomes, and instead let private persons get on with doing it themselves? If so, what test should we use to determine what those occasions are?

The oceans management system is about public intervention. Defining its boundaries requires us to think carefully about the proper role of public authorities vis a vis private freedoms. It is useful to consider such questions to establish a common foundation for what the system is there for.

6.3 Narrow rationales for intervention

Some may be of the view that the system should only step in where something has “gone wrong”. In economics language, intervention might be justified when there is a market failure, such as an “externality” that needs to be “internalised”. A negative externality is created when a person does not bear the full cost of his or her actions (eg when a polluter does not pay).⁷ The task of public authorities is said to be to correct or

“internalise” externalities by imposing (at least part of) their true cost on those who created them.⁸ This increases the cost of undertaking the activity which, in turn, acts as a disincentive to carrying it out (thereby potentially reducing its adverse effects).

There are different ways to do that, but all require a degree of public intervention. It might be possible to calculate the financial cost of harm and recover that amount from the person causing it (such as through a tax on pollution or a payment for the use of a receiving environment). Alternatively, one could create regulations to prevent the harm from occurring in the first place (such as through identifying prohibited activities under the RMA), or to mitigate it to an acceptable level (such as through conditions placed on resource consents). In some cases, the components of the environment to which harm is being done could be privatised, which would in theory provide an incentive for the owners to protect their property from the impacts of others (such as through the creation of fisheries quota). We explore the use of property rights as a tool further in Chapter 8.

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Stormwater outfall, Mangawhai

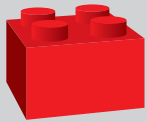
Multiple externalities exist in the marine environment, given that it is often the final recipient of pollution happening upstream in a catchment. Many impacts (such as the effects of contaminated stormwater on coastal habitats) are not priced or even recognised, let alone prevented. That is the case even where the effects of one commercial activity have a direct and measurable financial impact on others. For example, *E. coli* pollution of seawater requires shellfish harvesting to cease due to human health concerns. This means, for example, that mussel farmers in the Firth of Thames are unable to harvest after heavy rainfall due to potential pollution from dairy farms in the Hauraki Plains and oyster farms in the Bay of Islands were closed for eight years due to wastewater pollution.⁹ A system focused on preventing harm could address these kinds of anomalies. However, allocating proportional blame to particular individuals for their contribution to large systemic problems (eg runoff from roads or sediment from catchments), may prove very tricky in practice.

Despite potentially addressing such issues of concern, such a narrowly based system would not necessarily do all the things that some may expect of it. Government would not be in the business of driving particular outcomes or resource uses, or shaping people's behaviours, except to prevent the harm that they cause. Should that really be the role of the system here? Aside from practical difficulties (how would one measure or monetise the true value of harm to something like a marine reserve or the extinction of a species?) there are deeper ethical issues at play.

A narrow scope for intervention could also prevent actions that might be desirable from a long-term societal perspective. For instance, some may think that the current distribution of marine resources is unfair and they need to be reallocated (not least in the context of te Tiriti o Waitangi, but also between different commercial fishers and the commercial and recreational sectors).¹⁰

A narrow approach may also tend towards short-term and reactive management – intervening only when problems become apparent. If we are just putting out fires by assessing and mitigating the harm from individual activities, cumulative effects may be hard to spot until it is too late to prevent poor outcomes. It also seems unlikely that proactive measures, like supporting the growth of a new seaweed farming sector or even the creation of a network of MPAs, could be justified as the internalisation of externalities. They are about much more than preventing or compensating for damage.

A relatively narrow rationale for having an oceans management system could be to internalise negative externalities – essentially, to prevent or make people pay for any harm they cause. Beyond that, intervention from the system would not be justified. As well as being difficult to implement in practice, this approach may not reflect what people expect a system to do in the future. Indeed, it does not fully account for the reasons the current system intervenes.



A future system's ability to intervene could be based on internalising externalities and addressing market failures.

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Shoreline restoration at Mangere, Manukau harbour

6.4 Broader rationales for intervention

Broader rationales for a future oceans management system are possible, and arguably desirable. One important rationale may be to pre-empt and resolve disputes between users or uses of the marine space, which may not always be about preventing (or compensating for) harm caused by one person to another. It could instead be about encouraging a fair allocation of value from their use (such as through coastal occupation charging or royalties), or achieving synergies between mutually beneficial activities by coordinating them (such as aquaculture and shellfish bed restoration; or MPAs and tourism). This is more about “planning” than it is about remedying market failures.

Moreover, although it is by no means as complex as on land,¹¹ there may also be a need to proactively provide for public goods and services at sea (eg safety and navigation infrastructure and public transport). The very concept of a “public good” might even evolve in the future to include things like offshore renewable energy facilities (eg tidal or wind energy), marine parks (akin to green spaces provided by councils on land), or ecological infrastructure (eg to filter water or sequester carbon). It may involve proactive ecological restoration initiatives, such as the reinstatement of wetlands to reduce runoff into the sea, or restoration of coastal habitat (eg the removal of the Mangere wastewater treatment ponds in the Manukau Harbour).

Further rationales for a future system might be to resolve disputes and ensure the provision of public goods. This could encompass marine infrastructure that markets may not provide in an efficient or equitable manner as well as restoration initiatives. These would go well beyond the prevention of harm or the internalisation of externalities.

One might go even further and say that the system has a legitimate role to play whenever it would be in the *public interest*. This would reflect the idea that authorities are active stewards or trustees of a “shared” resource or space, not just dispute resolvers or providers of specific services where the market has failed.¹²

One benefit of contemplating intervention where there is a public interest is that what the “public interest” means can change over time, and sometimes quite quickly. The value of agility can be seen in the Covid-19 response, where it was seen to be in the public interest to place strong controls

over freedom of movement. It was also seen in the recent relatively fast shift between actively encouraging offshore petroleum exploration and phasing it out. It allows the system to create a wide variety of future-focused strategies to enhance wellbeing and to move our collective human endeavour forwards, and not just mitigate the “bad stuff” in a spiral towards environmental mediocrity. The ability for the system to expand or contract at the margins may be especially valuable given the extent of challenges (ie environmental and climate change) the oceans are facing.

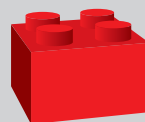
Of course, there are risks in contemplating a system that intervenes too much. Allowing interventions whenever the public interest is at stake creates less certainty and more room for argument. People can also have legitimate differences of opinion as to whether a public interest actually exists (especially when it clashes with private interests),¹³ and the “public” is not a uniform mass with a comprehensively defined set of interests. There are also questions as to whether policies designed to support (or at least

with the effect of supporting) private interests (such as the NES for marine aquaculture) fall within the category of the public interest. Some may see a risk in the government being able to “pick winners” when it comes to who (or what kinds of activities) can use the seabed or receive subsidies or other support (eg penalising offshore petroleum exploration while supporting offshore aquaculture or energy projects through fast track consenting or proactive spatial allocations). We need to be careful here:¹⁴

If we abandon a specific test based on externalities in favour of one that is based on a much more nebulous and evolutionary concept like the public interest, we recognise that it can be a slippery slope to unreasonably intrusive regulation.

Instead of the government, well-functioning markets could be relied on to determine what the public good is. After all, “it is not from the benevolence of the butcher, the brewer, or the baker that we expect our dinner, but from their regard to their own interest”.¹⁵ Yet there is no ethical magic in the market either. Market participants are not inherently “bad” or greedy, but nor is a democratically elected government inherently incompetent or draconian.¹⁶ Each brings a different set of strengths and weaknesses, and a rich conversation about public power.

One of the broadest rationales for having an oceans management system would be to pursue the “public interest”. This could encompass a whole range of interventions that could evolve over time and when circumstances change.



A future system could be designed to intervene whenever the public interest is at stake, providing more flexibility, but also less certainty, about scope creep and overreach.

It is also important to consider a tikanga Māori perspective when thinking about the scope of a future system. While that will need to be determined by Māori themselves, some thoughts may be ventured. On the one hand, tikanga does not support fixed or artificial boundaries in management between the public and private spheres (eg through tests focused on externalities). It is also oriented towards the wellbeing of the community (rather than economic efficiency or individualised rights), which is in turn linked to the wellbeing of the environment through whanaungatanga/ kinship connection. Tikanga also values agility over time – it is responsive and

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Beach replenishment, Point Chevalier, Auckland

constantly changing as the environment shifts – suggesting that a system may need a broad rationale that can expand and contract.

However, the rationale for intervention provided by tikanga is not exactly the same as a broad Western conception of the “public interest”, not least because tikanga has a lot to say about *who* gets to use or benefit from resources and how it is done. For example, a system that allocates coastal space on a first in time basis, or grants tourism concessions according to who can pay the most, may arguably fulfil the public interest but fall short of what tikanga and te Tiriti require.¹⁷ The most culturally significant use of the marine environment may not be the most “efficient” from an economic standpoint or the most protective from an environmental perspective. For example, it might be that an MPA managed according to local tikanga allows some customary and recreational harvest of fish, excludes commercial operators, and is subject to legally binding rāhui from time to time.

On the other hand, some might think tikanga supports a *narrow* scope for the system, because the system being talked about here is ultimately a “Western” one of formal laws, institutions and tools. From this perspective, the system has no business interfering with the Māori exercise of rangatiratanga and kaitiakitanga over resources or taonga, and its influence should be minimised.

In short, whether tikanga supports a broad or narrow rationale for the system may depend on whether the system itself is considered to be a constraint on tikanga (a negative intrusion), or rather a tool for Māori

to exercise tikanga (a positive enabler). This in turn may depend on the extent to which a future system can evolve to accept Māori as partners in decision-making and incorporate te ao Māori norms (see Chapter 7 on worldviews and Chapter 12 on institutional design) rather than something that just prevents Māori from doing things or paternalistically protects things that Māori value. What seems clear, however, is that a broad rationale for a future system cannot be based *only* on what the “public interest” demands. The interests of the public as a whole are not necessarily the interests of Māori safeguarded by te Tiriti, and a future system needs to be able to respond to both.

It is not immediately clear whether a tikanga perspective would support a broad or narrow scope for the system. Much may depend on whether that “system” integrates Māori perspectives and decision-making (which may legitimise a broad scope for intervention) or whether it is a Western construct operating in parallel to tikanga (which may suggest a narrower scope).



A future system could be able to intervene where necessary to meet te Tiriti o Waitangi obligations, even where those go beyond (or are different to) the public interest.

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Maketu Taiāhure

6.5 What is the right starting point?

Both broad and narrow rationales for a system have risks and benefits. But it is interesting to ponder in a first principles review why we tend to assume we need a justification for when the *system* should intervene. Instead, could we not flip that on its head by asking: when can we justify complete private freedom of action or the existence of private rights in *te moana*?

We could start with the premise that people are inherently part of a collective enterprise: a society or culture. From this view, the bestowal of many individual freedoms (including property rights) may well be good for society, but it is not assumed to be the natural way of things when it comes to how people interact with the marine environment. Unlike the strongly entrenched Western tradition on land, we still haven't privatised or even "tamed" the sea, and we may still be uncomfortable claiming that it is "ours". As Rousseau put it:¹⁸

The first person who, having enclosed a plot of land, took it into his head to say this is mine, and found people simple enough to believe him, was the true founder of civil society. What crimes, wars, murders, what miseries and horrors would the human race have been spared, had someone pulled up the stakes or filled in the ditch and cried out to his fellow men: 'Do not listen to this imposter. You are lost if you forget that the fruits of the earth belong to all and the earth to no one!'

On land, that ship has sailed (so to speak) for any but the most ardent socialist – it would be impossible to unpick private property rights even if one wanted to. At sea, however, such an approach might still be possible. There are many rights and interests, but there are few ditches or stakes. Indeed, *te moana*, and arguably the conceptual starting point for its management, is different to land in many ways. Some of these differences are legal, some are biophysical, and some are a matter of perception or tradition. But they suggest that a more expansive rationale for intervention and management in the marine space (eg the public interest) may be appropriate. For example:

- The arrangement of spatially fixed property rights in the marine area is substantially different than for those on land. There are few freehold land titles that extend under the territorial sea. Most of the area is designated "common marine and coastal area" which means it cannot be owned by the Crown or any person.¹⁹ Public rights of access to this area have been enshrined in law.²⁰ As most of the marine area is not "owned" by any party, greater public interest considerations come into

play than when managing activities undertaken on land. For example, the MACA Act recognises the importance of the marine and coastal area, for "its intrinsic worth" and "for the benefit, use and enjoyment of the public of New Zealand,"²¹ which is a statement that would be unthinkable with respect to private land.

- Due to this lack of ownership, the oceans management system must have a more active role in allocating rights to occupy and use space and resources than on land, since the Torrens system (governing private title) does not perform this role.
- The fundamentally interconnected nature of the marine environment means that it can be virtually impossible to confine the impacts of an activity to its immediate site. Flow-on effects often spread over wide areas and cumulate with multiple impacts from other places. This means that a project-by-project consenting approach (eg where impacts on neighbours are managed) is unlikely to be effective on its own without broader strategic planning, requiring a stronger role for public authorities.
- The sea is more of a "natural" environment than our land. Human activity has a less visible imprint at sea, which can be reflected in people's expectations about how it should look. So while farming on land is generally seen as a "picturesque" activity (rolling hills, fences and shelterbelts),²² there can be greater opposition to similar farming operations at sea (eg fish pens and buoys). Some might contend that this natural environment is more akin to the conservation estate, than to "built" or "developed" environments like cities or farms, therefore it requires an active public custodian.
- It is hard to see and understand what is going on below the waves, and marine management relies heavily on expert scientific measurement. While also important on land, it is particularly crucial that the oceans management system plays a proactive role in research and monitoring at sea, given that information is much harder and more expensive to obtain, and that different components (eg fish, habitats, climate change, water quality) are so interconnected.
- Aotearoa New Zealand does not have full sovereignty beyond the territorial sea, and "ownership" of the EEZ and extended continental shelf is limited. Although the country has broad jurisdiction over these areas, the ultimate basis for intervention rests on international law (quite different to on land and in the territorial sea). This suggests that public management is the starting point in these areas at least.

There are important contextual differences between marine and terrestrial environments. The oceans management system arguably needs to perform more of a proactive public trust function than the system on land, where it is hard to escape the sense that public intervention needs to be strongly justified against the status quo of personal liberty, market freedom and spatially defined property rights.

Because the marine space contains a wide variety of existing interests, a key question is: what is the extent to which a reformed system (no matter what its rationale) should be able to override or alter existing rights and interests?

6.6 Potential exceptions for intervention: Existing interests

The discussion above has been largely about what the *conceptual* rationale for having an oceans management system should be. Public interventions could be narrow (limited to the prevention of harm), or broad (whatever is best to achieve the public interest and te Tiriti obligations). However, reforms would not be starting from this blank slate. Therefore a more practically significant question might be: when should a new system be able to interfere with *existing* rights and interests? In other words, identifying a general rationale for intervention is a good starting point, but there might still be a need for exceptions.

Existing rights and interests in the marine context may be of a different nature to those on land, but they are still significant. Some private title over the seabed exists. Long-term resource consents have been granted. Customary title and protected customary rights are in the process of being recognised. Mining permits confer rights to take valuable petrochemicals and minerals. And fisheries quota provide the perpetual right to catch a proportion of the total allowable commercial catch (TACC). The ability to fish recreationally and access the sea are also often spoken of as “rights”.

Many of these existing interests could be impacted by a reformed system even if its general rationale for intervention was relatively narrow. For example, requiring externalities caused by trawling to be fully internalised (including the costs of benthic habitat damage) could significantly affect the value of some fishing quota. A broader rationale for intervention might see existing interests affected to an even greater extent, such as where the reallocation of a resource is deemed to be in the public interest, or necessary to meet te Tiriti obligations.

So should existing rights be protected, even if changing them would fall firmly within the scope of what the system might otherwise do? It is interesting that more targeted reform efforts in the past have been clear that interventions should *not unduly* affect existing rights. For example, this is often said in relation to the creation of new MPAs (where existing fishing rights are not to be undermined) and when it comes to creating new controls on land use (where existing use rights are protected under the RMA).²³

The first thing to note here is that not all interests are created equal. In particular, a distinction can be drawn between legal rights and expectations. Some things claimed as rights (such as the ability to continue clear-felling of pines, discharging wastewater into the sea, or bottom trawling) may really be expectations about the status quo continuing. Other interests, such as access to te moana for navigation and recreation, may be legally recognised as rights under international or domestic law.²⁴ Still other interests, such as fisheries quota holdings and resource consents, may be legally recognised rights and have significant monetary value.²⁵

Some rights can even be regarded as *property*, which for the most part is sacrosanct in our capitalist society.²⁶ Fisheries quota is a form of property interest, not just a regulatory permit.²⁷ Other rights may have a different status due to being part of te Tiriti settlements, which imparts a quasi-constitutional status and importance not seen in other contexts. It raises the difficult issue of whether fisheries quota transferred to iwi as part of the fisheries te Tiriti settlement has a different status to quota purchased separately by Māori or owned by other parties.

One approach would be to allow new interventions that shift people's *expectations* but do not infringe their *rights*.²⁸ The system could attempt to treat *property* rights as sacrosanct but other rights as more flexible. For example, it might mean that property rights in quota could not be undermined²⁹ but controls *could* be placed through conditions on resource consents (which are, legally speaking, not a form of property). However, it is not always easy to distinguish “property” from other kinds of rights, as the concept is a surprisingly fluid one.³⁰ And some other forms of rights may be regarded as even *more* sacrosanct than property (eg human rights

and te Tiriti settlement rights).³¹ The system could allow some rights to be impacted while protecting others more absolutely.

Even if a right clearly exists, and it can continue to be exercised, it can be unclear whether an intervention is actually an *infringement* of that right or not. For example, while establishing no-take MPAs (or recreational fishing parks or mātaihai reserves) may be in the public interest, they may comprise large areas of productive fishing grounds (as these are often the areas that need the most protection) and so reduce the value of some quota. At what point does this reduction in value amount to the erosion or regulatory “taking” of a right? This is by no means clear.³² On land, the RMA’s solution to this dilemma is to invalidate planning controls that render property rights incapable of reasonable use.³³ That involves an assessment of context and degree. A future system could also take this approach when it comes to marine property rights (eg invalidate interventions that would permanently prohibit fishing in an entire quota management area).³⁴

One approach could be to allow reform to alter people’s “expectations”, but not erode their property “rights”. However, it may prove difficult to draw a line between the two, as it is not always clear when an expectation becomes a right, and when an intervention amounts to an erosion of that right.



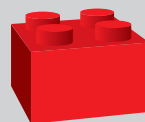
A future system might be permitted to alter people’s expectations but not erode explicitly recognised property rights.

Alternatively, a future system could focus less on the nature of the right (eg *property* rights) and more on the reason for which a right is eroded or removed. For example, on land there is a much higher bar for the compulsory acquisition of a property (transferring ownership) than for the imposition of public interest planning controls (restricting what an owner can do). Similarly, rights conferred by consent under the RMA cannot be extinguished in order to transfer them to another person (which would be a derogation of grant) but can be curtailed to deal with unanticipated adverse effects on the environment.

Thus, a future system could allow both expectations *and* rights to be eroded (or even extinguished) so long as the purpose of doing so was

legitimate. That could, for example, extend to the situation where a control was necessary to achieve targets for the protection of biodiversity and threatened species (including through the creation of MPAs) but not the reallocation of resources to achieve a more equitable distribution of wealth.

Reforms could be permitted to alter any existing rights and interests as long as there is a legitimate public purpose for doing so. That might, for example, allow regulatory intervention to protect the environment but not to reallocate resources to a “better” use.



A future system could permit the erosion of property rights but only for particular reasons.



Kaiteriteri Beach

Raewyn Peart

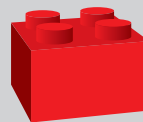
One could also make a distinction between different forms of property rights: those that are inherent features of our capitalist society (and cannot be extinguished), and those that have been brought into existence by the system (and are therefore amenable to change if they are no longer achieving their purpose). The QMS, for instance, was introduced not just because of the 1980s enthusiasm for privatisation per se, but also because it was seen as an effective way to achieve broader public policy goals around environmental sustainability and resource efficiency.³⁵ In that sense it is similar to the emissions trading scheme – it is a property-based tool.

But have some rights in the marine space become such an integral part of the fabric of society that, as with ownership of real estate on land, they should no longer be regarded as a tool to achieve public policy goals but rather a fundamental feature of capitalism that cannot realistically be undone? Consented rights under the RMA, although resembling property in many ways, are not clearly in this category. But private title to the seabed, small parcels of which exist due to historical reasons, more likely is.

If the QMS is still regarded as a tool within the system, to some the question may be a broader one: whether the system should continue to confer rights, not just whether it should be able to erode them. As explored in Chapter 8, some have floated alternatives to the QMS (although it is by no means clear that they would be *better*). And the door may be open to mechanisms by which coastal occupation rights (eg for aquaculture) could be moved to other sites, whether by mutual consent (where better conditions are found elsewhere) or by compulsion (where the impacts of inshore operations are no longer considered appropriate).

Perhaps the most useful way of thinking about the boundaries of a future system is that property rights themselves lie outside the system, and should not be extinguished except in the most exceptional of circumstances (as on land). But they *can* be linked to responsibilities. After all, a property right is not a freedom from obligation to society; it is simply a bundle of legal rights defensible against others.

Some rights – such as the Tiriti settlement rights or property rights that have become part of the fabric of capitalist society rather than a tool to achieve public interest outcomes – could be “off limits” for reform. However, it is not always clear when a right is within this category. A useful approach may be to not allow such rights to be extinguished, but to link them to public interest responsibilities.



A future system could permit the alteration of some property rights but not others.

Finally, we note that the biggest question when it comes to the defence of existing rights is usually not whether *some* public intervention is justified, but rather what *form* it takes. Command and control regulation is a very different way to influence behaviour than something more passive like an education campaign. It is often *regulatory* overreach that is the concern. Thus, for example, while it might be beyond the pale for the system to compulsorily acquire fisheries quota (or to effectively extinguish it by creating large no-take areas across an entire QMA), it might be more acceptable for the Crown to purchase quota within the market and retire it.³⁶ The provision of compensation or other assistance, in parallel, might also make some measures – even regulatory ones – more palatable. In short, debates about the proper scope of the system might be framed more usefully in terms of which *tools* should and should not be used, not whether the system should have any role at all. We discuss various tools in Chapters 8 to 10.

It may be that the system could legitimately interfere with existing rights and interests if “softer” mechanisms were used to do so instead of regulatory ones. Regulation may also be more palatable if accompanied by compensation or other assistance.



Commercial fishing vessels, Motueka

Raewyn Peart

6.7 Concluding comments

In this chapter we have considered what the outer boundaries of the system should be: where public intervention amounts to “overreach” and private choice or market freedom should be left to determine outcomes. The rationale for having a system can be interpreted narrowly (eg to prevent harm and provide public goods) or broadly (eg to achieve the public interest). The context in the marine environment may suggest that a broader scope for intervention is appropriate, although there remains a rich debate to be had, especially when it comes to the extent to which public intervention is able to alter or remove existing rights and interests. It may come down to which kinds of *tools* are used to intervene. Regulatory interference may, in some cases, be a step too far, and softer mechanisms may be seen as more legitimate.

Considering the rationale for having a system, and how it relates to existing interests, helps identify its outer boundaries – where it should not have a role. However, the discussion is equally useful in giving a sense of what specific roles a future system *should* be performing. There is a big difference between a system that *can* do things and a system that is *required* or *encouraged* to do them. We explored this area extensively in our previous work on resource management reform (and we refer readers

to our previous report for more detail),³⁷ where we identified seven core roles a future system could be expected to perform:

1. Setting environmental limits.
2. Making trade-offs above limits.
3. Providing public goods and services.
4. Pursuing positive outcomes.
5. Protecting the interests of mana whenua.
6. Allocating resources.
7. Resolving disputes.

It is important to note that these roles might look different in the marine context. Because the roles a system is expected to perform are closely linked to its purposes (for example, *why* environmental limits should be set in a particular place, or *why* we might allocate resources to some and not others), we explore them further in Chapter 7 when we look at what the objectives of a future system might be.

Summary of options for reform: The rationale for the system

- A future system’s ability to intervene could be narrow, based on internalising externalities or addressing other market failures.
- A future system could be designed to intervene whenever the public interest is at stake, providing more flexibility, but also less certainty, about scope creep and overreach.
- A future system could be able to intervene where necessary to meet the Tiriti o Waitangi obligations, even where those go beyond (or are different to) the public interest.
- A future system might be permitted to alter people’s expectations but not erode explicitly recognised property rights (eg quota).
- A future system could be allowed to erode property rights but only for particular reasons.
- A future system could be allowed to alter some property rights but not others.

Endnotes

- 1 This shift has occurred quite rapidly, given that the oceans were, until recently, seen largely as an inexhaustible resource basket and sink for waste.
- 2 If this is not addressed early in the reform process, opposing perspectives on what the system “is” can resurface later on in more specific situations. For example, in the terrestrial context the system cannot simply reallocate private land from one person to another, even if it would produce more equitable or sustainable outcomes. This is not just a question about the system’s objectives. It is more fundamental: the allocation of land is not considered to be within the scope of the resource management system at all.
- 3 For example, changing the purpose of the RMA/NBA to something like *te ora* or *te taiao*.
- 4 For example, proposals for new legislation to manage issues associated with carbon capture and storage.
- 5 For example, the response to Covid-19.
- 6 Greg Severinsen and Raewyn Peart *Reform of the Resource Management System: The Next Generation Synthesis Report* (Environmental Defence Society, Auckland, December 2018) at 80; Greg Severinsen *Reform of the Resource Management System: A model for the future: Synthesis report* (Environmental Defence Society, Auckland, December 2019) at 49.
- 7 Or, more accurately, where a person bears the cost of another’s actions that he or she has not agreed to bear. A positive externality is where those receiving the benefits of an action do not pay for it (allowing free-riding), such as a ship in the vicinity of a lighthouse.
- 8 It is generally accepted that it is impractical and undesirable to internalise all adverse effects. For example, changes in plans do not trigger compensation to existing residents: see Resource Management Act 1991, s 85.
- 9 Raewyn Peart *Farming the sea* (Environmental Defence Society, Auckland, 2019) at 93.
- 10 See Chapters 2 and 3.
- 11 Where there is a need to provide network infrastructure like roads, water pipes, public transport as well as social infrastructure like libraries and some housing.
- 12 See Greg Severinsen and Raewyn Peart *Reform of the Resource Management System: The Next Generation Synthesis Report* (Environmental Defence Society, Auckland, December 2018) at 85.
- 13 For example, retaining the natural amenity values associated with particular areas might be regarded as a public interest consideration, but could sometimes be regarded as *nimbyism* (protecting the value of private properties).
- 14 Greg Severinsen and Raewyn Peart *Reform of the Resource Management System: The Next Generation Synthesis Report* (Environmental Defence Society, Auckland, December 2018) at 84.
- 15 Adam Smith *An inquiry into the nature and causes of the wealth of nations* (1776).
- 16 There are risks associated with both: see Catherine Knight *Beyond Manapouri: 50 years of environmental politics in New Zealand* (Canterbury University Press, Christchurch, 2018) for an exposition of the risks of excessive government management, and Richard Thaler and Cass Sunstein *Nudge: Improving decisions about health, wealth and happiness* (Penguin, 2009) on the risks of leaving private choices uninfluenced.
- 17 See *Ngāi Tai ki Tāmaki Tribal Trust v Minister of Conservation* [2018] NZSC 122.
- 18 Jean Jaques Rousseau *Discourse on inequality* (1755).
- 19 See Marine and Coastal Area (Takutai Moana) Act 2011, s 11.
- 20 There is a patchwork of other rights in the marine area, including over structures such as wharves, jetties and marinas and for activities such as aquaculture (which are authorised under fixed-term coastal permits under the RMA).
- 21 Marine and Coastal Area (Takutai Moana) Act 2011, s 4(2)(e).
- 22 Of course, opposition to farming can be for reasons other than amenity, and not all farming operations – notably those involving intensive winter grazing – are “picturesque”.
- 23 Although the Randerson Panel’s report has more recently floated the idea that existing use rights might need to be extinguished where needed to defend environmental limits or adapt to climate change.
- 24 Although expectations around access are not unfettered; for example, there can be exclusion to protect nesting birds, to address biosecurity risks, and to impose safety zones for petroleum mining.
- 25 Consents are in the nature of property rights even though, legally, they are not. See D David Grinlinton “The nature of property rights in resource consents” (2007) 7 BRMB 37; Laura Fraser “Property rights in environmental management” (2008) 12 NZJEL 145; Barry Barton “The nature of resource consents” in *Environmental law: National issues intensive* (New Zealand Law Society, 2009).
- 26 For example, a fee simple estate in land.
- 27 See OECD *Using market mechanisms to manage fisheries: Smoothing the path* (OECD Publishing, 2006).
- 28 Whether it *should* shift expectations would still need to be justified (and there would be debates about compensation), but the point is that it is conceptually on the table as something the system *could* do.
- 29 It would not be clear what “undermined” meant. Effective extinguishment would be one line in the sand, but it is not clear whether any measure reducing the value of quota should be prevented. Value can already vary depending on market conditions and biophysical factors (eg collapse of a stock, fish moving due to climate change or habitat degradation) and no compensation is forthcoming for those things.
- 30 For example, some rights can exhibit features of property (eg tradeable consents) without *being* property, while property rights can be so heavily regulated that they start to resemble other forms of right. A property right is a general term used to describe a bundle of interests and rights people have that are generally tradeable in a market, but it is not a straightforward distinction; “implicit in a property right, generally, are all or some of the following rights: the right to use or enjoy the property, the right to exclude others, and the right to sell or give away”: see Australian Government “Definitions of property” (31 July 2015) <www.alrc.gov.au/publication/definitions-of-property>; Kenneth Palmer and others *New Zealand land law* (3rd ed, Thomson Reuters, New Zealand, 2017).
- 31 It might sound disingenuous to suggest that human rights could be altered by the system, but it might be a question of degree. For example, if the system were to give strong rights to nature, then these could conceivably clash with human rights.
- 32 This issue can be seen in the context of the proposed Rangitāhua/Kermadec Ocean Sanctuary. Here, proposed legislation would effectively set a catch limit to zero for the entire the QMA which surrounds the islands. We look at this in Chapter 7 in the context of principles (procedural justice).
- 33 See Resource Management Act 1991, s 85. That is quite different to where there is a desire to use land for a different *purpose* (or to extinguish a specific existing land use), in which case compensation is forthcoming through Public Works Act processes or on a willing seller/ willing buyer basis. This can be seen in the government’s recent purchase of dairy farming land around Lake Horowhenua and conversion to wetland, Radio New Zealand “Bid to restore Lake Horowhenua with new wetland project” (27 May 2021) *RNZ* (online ed, 27 May 2021) <www.rnz.co.nz>
- 34 Overturning planning controls on land requires a high bar, and there are much stronger property rights in land (ownership) than in quota (a right to a proportion of a stock once sustainability measures are taken).
- 35 See Raewyn Peart *Voices from the Sea* (Environmental Defence Society, Auckland, 2018).
- 36 In tandem with proportionate reductions in the TACC.
- 37 Greg Severinsen and Raewyn Peart *Reform of the Resource Management System: The Next Generation Synthesis Report* (Environmental Defence Society, Auckland, December 2018), ch 7.

7 Ethics, principles and objectives



Whangaroa harbour

7.1 Introduction

We concluded Chapter 6 by identifying seven distinct roles a future system might be expected to perform. For some of these, it is implicit what general outcomes are being sought. For example, the point of setting environmental limits is to protect the natural environment. What we are seeking by protecting the interests of mana whenua is, at least at a high level, self-evident. However, there is still a lot of room for debate. What *specifically* should environmental limits be seeking? To prevent ecosystem collapse, to prevent people being hospitalised by pollution, or to defend the mauri of a resource? And what are we seeking by allocating resources? Economic efficiency, social equity or environmental improvements? There is an even wider range of things the system might be seeking when it comes to making trade-offs or pursuing positive outcomes, ranging from general things, like “sustainability”, to quite specific things, like growing revenue from the aquaculture industry to \$3 billion by 2035.¹

In this chapter, we consider what we might want a future system to achieve, both generally and when performing its different roles. This involves looking at three things in turn: (1) what *ethics and worldviews* might underpin the system; (2) what legal and ethical *principles* could

operationalise those worldviews; and (3) what more specific *objectives* the system might have.

Why we might want to achieve something is just as important as *what* we want to achieve. That requires us to consider the ethics or worldviews underpinning the system. For example, seeking to safeguard the mana or mauri of the moana cannot be understood without engaging with the value system underpinning it (te ao Māori). And the ethical assumptions about *why* we might want MPAs can vary wildly between people and over time (eg scientific research, cultural connection, scenic values, biodiversity protection).² Worldviews and ethics need to be operationalised within a formal system of laws, and that can be done by developing legal and ethical *principles* such as various incarnations of “sustainability” and concepts like kaitiakitanga.

While principles are useful to guide decisions, it is also worth considering whether a future system would benefit from clearer, more directive and measurable *objectives*. It is one thing to say that the system should set environmental limits to reflect an eco-centric worldview, guided by principles like sustainability. It is quite another to say that the system must phase out all single use plastics by 2035, establish a 30 percent coverage of protected areas by 2030, or impose an immediate moratorium on deep sea mining.

Raewyn Peart



Boil up, Bay of Islands

7.2 Ethics and worldviews

Worldviews are about the basic ways in which we see ourselves in relation to our surroundings and the natural world. They are about what is right and what is wrong. We explored different worldviews in our previous work on resource management reform, but the topic is worth exploring further in the context of the marine environment. For example: why do people care so much about marine mammals like dolphins but not about other marine species? Is pollution prevention about human health and wellbeing, or the rights of ecosystems to thrive? Should living things have a dollar value when humans do not?³ Is our ethical obligation to nature simply to prevent extinctions, or does it go further than that? Is access to the oceans and its resources a “birth right” for all New Zealanders, or just those who can afford to pay (like the case for access to most land)? And what is the nature of our moral obligation to future generations? Below, we explore te ao Māori, welfare economics, anthropocentrism and ecocentrism.

Different worldviews seek to strike a balance between how people use/ impact the moana and how much they protect it. All require the benefits and costs of use and protection to be weighed, but built into the weighing process are inevitable value judgements that some benefits and costs are more significant than others.

None of this is about crunching numbers. Worldviews and ethics are really about storytelling. It is easy to see that in the context of te ao Māori, where oral tradition passes mātauranga down through the generations (see below). But it is no less true in Western and other traditions, where moral messages and narratives (eg the value of capitalism and markets and the centrality of human welfare) are often hidden within technocratic language. Neither science nor nature tells us what is right or wrong. It is something we tell ourselves. In our resource management project we said that:⁴

Both development and environmental health are thought of as “good”, so it makes sense that both underdevelopment and environmental degradation are thought of as “bad”. Yet one often must come at the expense of the other. This is different from an area like criminal ethics, where (despite arguments over definitions and exceptions) it is fairly simple to characterise crimes like murder as bad. There is no point at which there is too little murder.

Yet in the marine context the law is quite clear that too little death *can* be a bad thing; the concept of MSY is about harvesting natural fish stocks down significantly in order to shift the population dynamics (to produce lots of small, fast-growing fish rather than retaining older mature fish) for

human benefit. In that case, leaving too many fish in the sea can be seen as “waste”, because it lowers the overall stock “productivity”. That is not to suggest that maximising such natural yields is a wrong thing to do. It is simply to say that how we approach the life and death of species in the oceans is quite different to how we think about human life. Policy makers will need to consider which mix of worldviews should underpin a new generation of marine laws.

Different worldviews weigh the benefits and costs of use and protection of the oceans, but built into the weighing process are value judgements that some benefits and costs are more significant than others.

Te ao Māori

Te ao Māori is one way of seeing and living in the world, which has existed over centuries, incubated within an integrated social and cultural setting. It remains a powerful worldview amongst Māori in modern Aotearoa New Zealand, but it is not one that forms the foundations of our current oceans management system (even though components of it are present).⁵

Māori values are not homogenous. There is considerable diversity, just as there is within a “Western” worldview. That said, te ao Māori has strong common cultural roots. Hirini Moko Mead emphasised that “culture provides the general template of what Māori society was and is about. Through time the people developed systems that covered all aspects of life” led by the ancestors and then evolved through the generations down to contemporary Māori of today.⁶ All the activities that members of iwi, hapū and whānau engage in, is the ethical system and common law that is referred to as tikanga Māori.

Through the lens of te ao Māori, the environment is not seen as a collection of resources to exploit for human benefit, nor as a separate entity to protect; rather, people are seen as part of a cosmological system based on kinship, respect and reciprocity.⁷ Every aspect of corporeal and incorporeal life is connected. Dr Robert Joseph explained that:⁸

A traditional Māori cultural worldview ... was based on the Māori cosmogony (creation stories) that provided a blueprint for life setting down innumerable precedents by which communities were guided in the governance and regulation of their day-to-day existence. Māori worldviews generally acknowledged the natural order of living things

and the kaitiakitanga (stewardship) relationship to one another and to the environment. The overarching principle of balance underpinned all aspects of life and each person was an essential part of the collective. Māori worldviews are therefore ones of holism and physical and metaphysical realities where the past, the present and the future are forever interacting. The maintenance of the worldviews of life are dependent upon the maintenance of the culture and its many traditions, practices and rituals.

It has also been pointed out that the essence or philosophy that informs tikanga Māori is based on relational and genealogical connection to all facets of the environment:⁹

Whakapapa is not only a genealogical construct of who we are and where we are from, but it narrates our life through the people and places we come from. More importantly, it is a cultural tool used in connecting us to the environment which in turn, spiritually denotes a relationship to the atua (gods) who personify and represent these realms. Whakapapa therefore dictates our genealogical link back to our natural environment and therefore the atua that reside within these domains.

Māori atua exist as personifications of the natural world. [They] are derived from Māori mythology, theology, knowledge and history passed down as tradition through generations. The domains they represent are inherited by Māori and these environments become not only part of our culture, but part of our whakapapa.

Te moana has a central place in Māori worldviews. Māori have always been a seafaring people, tracing their histories back to Polynesia. The well-known origin story of Aotearoa sees the hero Maui catch the great fish of the North Island – te Ika a Maui – from his waka. According to some te ao Māori traditions, the oceans were the place from which life itself first appeared (according with Western scientific understanding). Te moana continues to have great significance, not just as a source of kai and resources, but as a powerful element in a tightly woven tapestry of existence. According to some:¹⁰

In the most well-known version of the Māori creation story, Tangaroa is the son of Papatūānuku, the earth mother, and Ranginui, the sky father. He is one of the 70 children who, when earth and sky were separated, went to live in the world that was created.

This is the story that tells us how the atua became the family entrusted to take care of specific areas of the environment. Oral tradition also says that

Tāne Mahuta was the one who created the female element from the clay of Papatūānuku, named Hineahuone. She then coupled with Tāne to produce the first human being, named Hinetītama, and she then coupled with Tane to produce the first human being named Hinetitama. It is this whakapapa that connects/links and infuses Māori to the environment. Moreover:¹¹

In some genealogies human history is traced from fish to amphibian, before finally taking human form. Perhaps the most well-known expression of this idea can be found in the whakairo (wood carvings) which adorn meeting houses throughout the country. The bulbous heads of the carved ancestors, their three fingers and serpentine bodies indicate the belief that humankind had marine origins.

Other commentators have noted other oral traditions relevant to the Māori creation story:¹²

Tangaroa was one of the many children involved and saw the repercussions of Tāne separating Ranginui and Papatūānuku. This separation caused conflict between the atua, more specifically to Tāwhirimātea who resented the thought of separating their parents. Tāwhiri expresses his rage through his control over the weather and how it continues to beat down on the domains of his siblings who remained with Papatūānuku. Some of this reflected on Tangaroa as the rage caused a separation of the sea creatures as some fled to land and the others fled deep into the ocean.

Tangaroa has whakapapa to connect the waters that rain down on our mountains and down through our valleys, fresh waters that run through our land, water that nourishes our soil, to the waters that we consume and comprise 80% of our human body.¹³

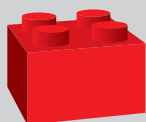
This account of atua linked to each other and to people through whakapapa – in tension and harmony – conceptualises the environment not as a series of separate domains and mapped boundaries, but rather as connected spheres, each with its own champion. In much Māori lore, the sea is something to be feared and respected – it takes on a human face and a human sense of injustice – not just a challenge to be conquered or a resource to be exploited.¹⁴ For Māori, marine protection is not about a preservationist, no-take approach.¹⁵ Nature is not conserved separately from humans, because what is important is maintaining relationships with the atua and tūpuna that are within. Failing to protect the mauri of the moana results in a diminishing of the mana (power, authority) of those responsible for its protection.

While the Māori worldview is a spiritual one, encapsulated in a rich oral storytelling tradition, it is also one that is translated into practice, and is ultimately designed to explain the world and guide human behaviour. Tikanga is central to the Māori outlook. Dr Joseph describes this as “values, principles, ethics or norms that determine appropriate conduct, the Māori way of doing things, and ways of doing and thinking held by Māori to be just and correct”.¹⁶ It is about *doing*, not just *explaining*. Further, Hirini Moko Mead has said of tikanga that:¹⁷

It is difficult to imagine any social situation where tikanga Māori has no place ... Tikanga Māori might be described as the Māori ethic ... Tika means “to be right” and thus tikanga Māori focusses on the correct way of doing things ... From this standpoint it is but a short step to seeing tikanga Māori generally as a normative system.

There is a question around whether te ao Māori is so intimately connected to Māori as a people that recognition of that world view would also need to bring with it a recognition of Māori stewardship of the oceans management system. Can we have a system where te ao Māori is administered by both Crown and Māori in partnership? Or a dual system whereby te ao Māori is administered or at least overseen by Māori? *He Puapua* has sparked an interesting and much broader-ranging debate in this area (see Chapter 4), and the ocean is one arena where it is particularly relevant. Power sharing is intimately related to a system founded at least partly in te ao Māori, and we explore institutional arrangements in Chapter 12.

Te ao Māori is a complex world view in which the moana plays an important role, and where the relationship between humans and nature is perceived and experienced as one of whakapapa and whanaungatanga rather than separation and hierarchy. It forms the normative foundation of tikanga – the right way of doing things. In many places the current system does not reflect te Māori.



The normative foundation of a future system could be based on te ao Māori and its concepts and principles.

A welfare economics view of the world

Anthropocentric worldviews put people at the centre of marine management. Within that broad church, some economic approaches construe human interests relatively narrowly – the overall aim is said to be the maximisation of social welfare.¹⁸ Social welfare in this context is generally seen as the product of two things: efficiency and equity, although the former is usually dominant:

an efficient allocation of resources means there is no way to increase one person's welfare without reducing another person's welfare...

but because many outcomes can be efficient, economic approaches generally determine which one is optimal by choosing the most equitable. Some may effectively see equity as unimportant (any efficient allocation is acceptable), but others may require that the welfare of those with the lowest welfare be enhanced.

Here, value and efficiency tend to be measured primarily in dollar figures. Traditionally, this has attracted the label of environmental economics, which has developed from the neoclassical school of economics.¹⁹

This ethic might be held by those who see the rationale for the oceans management system as one of internalising externalities (see Chapter 6). But even a system based on a broader rationale like the public interest might still adopt a worldview that seeks to maximise overall welfare (eg when allocating resources to their “best” use or when determining which aspects of the environment to enhance).²⁰ In short, this worldview might contemplate management measures necessary to achieve the public interest, but define it narrowly (as maximising overall welfare). That would be the overriding aim when the system was performing goals as diverse as setting limits, making trade-offs or allocating resources.

In the environmental context, the most famous (or infamous) expression of an economic approach is that attributed to Ronald Coase, in what has since become known as the Coase theorem.²¹ This echoes Hardin's seminal work on the tragedy of the commons, which sees environmental problems through an economic lens, and stemming from market failures.²² In other words, if the negative impacts of a person's resource use can be shared between many people, but its benefits can be individualised, any person acting rationally (in an economic sense) will cause overall environmental harm. Divergent solutions to this problem have been suggested: public regulation is one, taxation is another. Those writing in the tradition of the Coase theorem have proposed greater enclosure of resources – in other

words, the allocation of defined, divisible and defensible property rights.²³ We look at the use of property rights as an environmental management tool in Chapter 8.

Strict economic approaches to environmental ethics can be criticised in a number of ways.

- They do not allow the possibility that equity may be better enhanced by an economically inefficient outcome.²⁴ From one perspective, recreational fishing is very inefficient compared to commercial methods. But that does not necessarily mean the former has no or less value.
- They assume the existence of markets with no externalities or transaction costs,²⁵ and of economically rational actors.²⁶ Especially in the marine environment, we almost always have less than perfect knowledge of the environmental effects of an activity, and even in a world of full privatisation, markets may fail to internalise what are later realised to be the actual environmental costs of activities.²⁷ The continued collapse of fish stocks attests to that.

- Future generations cannot participate in markets, and so market transactions do not reflect their interests.
- Not all human values can be reduced to transactional or monetary terms. Human wellbeing derived from the environment is not just about services, and cannot necessarily be traded off against the same “dollar’s worth” of other forms of capital. There are also ethical questions about commodifying living parts of the environment.

Some innovative thinking has occurred in the field of environmental economics. The division between economists and ecocentrists (see further below) has also blurred, with many quite different framings now available for concepts like natural capital, ecosystem valuation, discounting, and green trading. Economically valuable ecological services are recognised, as opposed to just the protection of biodiversity. Many now recognise that the natural world has intrinsic value as well as its instrumental value as a source of resources. Kate Raworth’s concept of doughnut economics is a particularly prominent way in which neoclassical economic theory is being fundamentally transformed.²⁸

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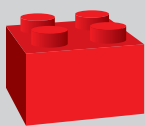


Recreational fishing, Wairau Bar

However environmental economics still has a methodological tendency to reduce the natural world to monetary or instrumental terms for ease of measurement, even if the end goal is broader than efficiency or an increase in human welfare. This is not to imply that such an approach is bad; imputing a monetary value to nature is much better than assuming it has no value at all or is “free” as a limitless sink for pollution. It is simply to say that there are other lenses through which we can think about what is right or wrong.

Generally speaking, there is a strong case that an economic approach to marine environmental matters is an inappropriate ethical foundation for the imposition of environmental limits or recognition of Te Tiriti o Waitangi, as it tends to treat different forms of “capital” (eg built and natural) as interchangeable. However, some may see it as more suitable to underpin other roles, such as making trade-offs (eg by undertaking cost-benefit analyses) or allocating resources (which should go to their most efficient use).

A narrow approach to economics sees the oceans as a source of instrumentally valued resources to be managed for the benefit of people. It seeks to measure value in monetary terms. That said, “green” approaches to economics have been developed in more recent times where the morality of decisions is not defined by solely economic factors.



The normative foundation of a future system could be based on a welfare economics view of the world, in which instrumental value is placed on the natural world as a collection of resources.

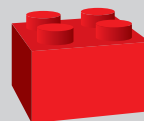
Anthropocentrism

An anthropocentric approach does not mean an economic lens needs to be used. What we want as consumers is not necessarily who we are as citizens of a society; we are moral agents, not rational automatons.²⁹ One pervasive line of thinking in Western thought, utilitarianism, has suggested a “good decision” is that which produces the most pleasure (utility or wellbeing) and the least pain for the most people – the greatest good for the greatest number.³⁰ The morality of this has been questioned over the centuries, as it can lead to great suffering for those in the minority and it is difficult to measure the pleasure of one against the pain of another.

Another approach focuses on deliberative democracy, whereby whether something is good or bad depends – to put it simply – on whether people say it is good or bad. A system designed on this basis would be flexible enough to accommodate shifting ethics. Indeed, some might argue that a bare bones framework like the EEZ Act already provides for that approach, and it has enabled a large degree of creative thinking (eg when it comes to the role of te Tiriti o Waitangi and a strict approach to pollution prevention).³¹ The RMA, in its broad concept of sustainable management, has also been capable of accommodating all sorts of changing values since 1991; te mana o te wai is a far cry from the more technocratic policy frameworks in first generation regional plans and national direction. And while something like the Marine Reserves Act admittedly allows very little normative flexibility (it is focused on research and not much else), the MACA Act provides much freedom for mana whenua to determine the values underpinning management in customary marine title areas.

Embracing the chaos and immeasurability of conflicting human values means that it is extremely important to get the process of decision-making right. Therefore, this kind of approach generally stresses the importance of participation, transparency, and rational discourse leading to decisions.³² Frameworks like the RMA reflect that strongly. Yet anthropocentrism is very much a concept serving human interests and measured by human values, which to some may no longer be an appropriate basis for a system. And weighing up such a large number of conflicting values, rather than focusing on a single metric like social welfare, can cause uncertainty, subjectivity and argument.

Anthropocentrism puts human interests at the forefront of decision-making, although that does not mean that value and morality need to be measured solely in economic terms. Some versions of anthropocentrism focus on the importance of democracy, which recognises that right and wrong can change according to the shifting values of society.



A future system could be based on anthropocentrism, where the multifaceted interests and values of society are put at the forefront of decisions.

Marine ecocentrism and biocentrism

In general terms, ecocentric approaches conceive of nature as a separate entity, with interests or rights that should be separately recognised and defended. Humans are not seen as inherently superior beings³³ but rather as part of a complex web of natural relationships that need to be respected.

Biocentric theories focus on the dignity and rights of the living world. Some animal rights theorists see the ability to feel physical or psychological pain as reflecting intrinsic value, giving rise to some interesting scientific debates about some fish species and their degree of consciousness or sentience.³⁴ Others have emphasised the wider value of individual plants and animals as centres of life capable, even without consciousness, of having some idea of their own good.³⁵

Broader nature-focused approaches extend rights and dignity to non-living aspects of the natural world, including geological features (eg rocky reefs).³⁶ Few ecocentric theories demand that people abandon all activities that exploit the natural world. To do so would be inconsistent with our own moral rights as part of an ecological community. Arguably, the current oceans management system contains elements of ecocentrism or biocentrism, in its recognition of animal rights and intrinsic value and providing strict protections for whales and dolphins.

A spotlight on valuing marine species

If one looks up while walking down the northern end of Wellington's Lambton Quay, one sign is particularly prominent: it has been posted by controversial businessman Sir Bob Jones, and reads "save the krill, kill the whales". One may assume it is there for shock value. But its author's point, he says, is genuine – to highlight the inordinate moral value that humans place on large creatures relative to small ones.³⁷ Whether or not one agrees with the specific message, it does highlight the interesting relationship between the different things we are trying to achieve through the oceans management system, not all of which are well defined or even deeply questioned.

Are we concerned about overall ecosystem health, and its productive capacity (food and other ecosystems service)? Preventing the extinction of threatened species? Preventing cruelty to animals? Protecting the lives of individual animals that we, as humans, tend to value above others for whatever reason? And why

do we do any of these things? For us, or for "nature"? A te ao Māori lens may point out that the compartmentalisation of such questions is itself the wrong approach. Aren't we instead concerned with the mauri and ora (wellbeing) of the moana, and the connected mana of the kaitiaki?

Some marine decision-making is based on the premise that it is an ethically bad thing to make a species extinct. That is arguably an ecocentric way of thinking; it is unlikely to undermine ecosystem services or impact on people's everyday lives if the fairy tern were no longer with us, just as we do not tend to notice any tangible effects from the demise of the moa. Yet it undeniably *matters*. A recent Cabinet paper has said, for example, that the rollout of cameras on boats will be "targeted to those fisheries that pose the greatest risk to protected species"³⁸ and a pilot project was focused on fishing in areas known to be frequented by the threatened Māui dolphin.

But is it wrong to kill a wild animal if the population of that species is healthy? We commonly kill wild fish, for example, and manage that harvest on the basis of MSY. We rarely frame that activity as a moral choice other than as an animal welfare issue (that the method of capture should be humane). But we treat marine mammals differently.

The Marine Mammals Protection Act makes it illegal to hunt (or otherwise harass) a marine mammal without a permit. The legislation followed that of the United States (the Marine Mammals Protection Act 1972) and came in the wake of the antiwhaling movement, which argued that whales had a right to life. The New Zealand Government has since been a strong supporter of the moratorium on whaling and has opposed the resumption of so-called scientific whaling by Japan.³⁹ Although this position has been partly based on the unsustainability of whaling, which decimated wild stocks, it also reflects the value placed on the intrinsic values of whales as well as abhorrence at the inhumane nature of whale harvesting practices.⁴⁰

Dolphins have been the subject of a more interesting legislative history in New Zealand. There have been regulations designed to protect individual animals (as opposed to a species in general). Pelorus Jack, the Risso's dolphin that followed ferries in Pelorus Sound during the late 1880s, prompted special regulations

under the Sea Fisheries Act 1894 that prohibited the harvest of Risso's dolphins in Cook Strait and were designed to protect them from hunting (which was legal at the time). In 1956, the Fisheries (Dolphin Protection) Regulations made it unlawful for anyone to take or molest a dolphin in the Hokianga Harbour, designed specifically to protect Opo, a bottlenose dolphin that had befriended humans there.⁴¹

Dolphins are highly intelligent animals capable of abstract thought and altruistic behaviour. Some have argued that dolphins have such impressive cognitive and social capabilities that they should be given a different legal status from other animals – that of a “non-human” person.⁴² So if they are highly intelligent creatures, capable of strong social bonding and suffering (recall the recent story of the orca mother who carried around the carcass of her dead calf for 17 days),⁴³ is it wrong to kill dolphins? The prohibition on (unpermitted) hunting in the Marine Mammals

Protection Act implies this is the case; it equally applies to threatened and non-threatened marine mammal species. But on the other hand, around 100 common dolphins are killed each year in trawl fisheries,⁴⁴ and this is enabled through a provision in the Act that provides a defence to prosecution if dolphin bycatch is reported. Common dolphins are not generally thought to be threatened. But equally, they are not regarded as a “pest” species where numbers need to be controlled. So if one accepts it is wrong to hunt them,⁴⁵ why is it okay for them to be entangled and killed in fishing nets?

Going even further, why is it ethically defensible to carve out a special place for marine mammals like dolphins and whales, but to allow other non-threatened species to be killed? Arguably, our perception of marine mammals is less an ecocentric ethic than a projection of our anthropocentric bias; the more a creature behaves or thinks like us, the more it seems to matter.



Memorial to Pelorus Jack, French Pass

Raewyn Peart

Ecocentrism can be criticised on a number of grounds. In practice, it must reflect human values, rather than the values of nature itself.⁴⁶ After all, limits to resource use are imposed by humans on humans. Fish can't speak, and we must attribute words to them. Furthermore, ecocentrism has little to say about thoroughly human concerns like the allocation of resources between people.

Generally speaking, ecocentric approaches conceive nature as a separate entity, with interests or rights that should be separately recognised and defended.



A future system could be based on ecocentrism, where nature is recognised as having intrinsic value alongside humans, not just as a set of resources or serving human needs.

A spotlight on ecosystem services and nature-based infrastructure

In Chapter 2, we touched upon the ecosystem services that our marine environment provides to people, from kai moana to nutrient cycling to carbon sequestration. Our point there was that when we protect the ability of the environment to perform those services we are still “using” it. Protection can be a use, and use can achieve protection; the distinction is not a binary one.

An interesting thing to ponder, however, is some people's tendency to think of these functions in an anthropocentric way – as *services* (also noting that this thinking is in contrast to te ao Māori/tikanga Māori). It is one thing to recognise the value of our oceans in providing us with sustenance, opportunities for recreation, and transport. However, there are numerous references in the literature to other “services” like the containment of sediment, the cycling of nutrients, the diffusion of pollutants, and the filtering of water. For example, some have pointed to the cleaning prowess of shellfish:⁴⁷

It has been estimated that with the historic coverage of mussel beds, the volume of the Firth [of Thames] could have been filtered in a single day. Current estimates are that remnant mussel beds take nearly two years to filter the same amount of water.

Those mussel beds – and a string of other habitats – would have done wonderful things for us if we had been more foresighted or ambitious in protecting them. But the reality is that, in 2022, we are expecting the natural world to perform an increasingly onerous range of services for us. The “water” that these mussels would now be required to filter is quite different to the water that existed 200 years ago, and much more polluted. Yet the message often seems to be that true problems only arise when the natural world is no longer capable of dealing with our rubbish – that we should hover somewhere around “maximum sustainable pollution” in the same way that in fishing we aim for “maximum sustainable yield”.

To put it another way, the danger is that we perpetuate the attitude of “if only we hadn't destroyed our ecosystem services, we wouldn't have to deal with the consequences of our increasingly unsustainable way of life”. Perhaps we need



Leather jacket, Cavalli Islands

to see the living world – including the mangroves, mussels and kelp forests that have to choke on our pollution – as having an existence that is about more than just servicing our needs and demands. The fact they are, increasingly, no longer doing so can be seen as nature’s last available form of protest, rather than just a breakdown of a human production line. These are living things that can lead a precarious existence and are fighting to survive.

Our outlook here has tangible consequences, not just for how we view the natural environment, but for the tools we use in a future system (see Chapter 8). For example, should we engineer entirely new kelp forests or mangroves in our estuaries where none existed before, to “put them to work” for us? Or should we seek to remove mangroves in an attempt to restore marine environments to their previous state (sandy beach), even if we lose the services they provide? Should we establish colonies of filter-feeding shellfish at stormwater outfalls, so that a continuation of excesses on land does not send our biogenic marine habitats over a tipping point? Should we instead focus on the cause of our pollution, and just leave the marine environment alone? Or is there an ethically sound middle ground to aim for?

The oceans provide people with many services, and pollution and other stressors can lead to those being threatened. A future system could strive to restore elements of the environment that provide those services, to reduce harm being felt by humans. That could arguably lead to a concept like “maximum sustainable pollution”.

7.3 Evaluating worldviews: a hybrid approach?

The foundations of the current oceans management system arguably rest on a tripartite marriage between economic rationalism, a strong sense of environmental activism, and a growing recognition of te Tiriti o Waitangi. Particularly in the marine space, there is an interesting mix of worldviews, where a highly instrumentalist view of fisheries exists alongside strong recognition of intrinsic value (eg for marine mammals) and a highly spiritual understanding of freshwater and its impact on estuaries (te mana o te wai). One statute, the RMA, has been described simply as “a complex set of values enshrined in law”.⁴⁸ Beyond that Act, there is even more ethical complexity.



Kelp forest, Mokohinau Islands

A future system is also unlikely to be founded upon any single worldview. That is the nature of a diverse society committed to liberal democracy, recognition of te Tiriti, and capitalism. Even people’s individual values are complex and hard to reduce to a simple list or label. The question is therefore not so much about which ethic(s) to adopt or reject, but more about the direction in which the system should head and where synergies can be found. There is no “right” answer when it comes to questions of right and wrong, only questions to ponder.

Yet they are worth pondering. While it would be unusual for a statute to proclaim its ethical underpinnings (eg to say it is “ecocentric” or “anthropocentric” in a purpose statement), a system that contains a jumble of inconsistent ethics is one likely to generate conflict, uncertainty and inefficiency (especially where incompatible ethics are split across different statutory frameworks). On the other hand, a system in which significant parts of the population cannot see their own basic outlooks reflected is unlikely to be durable. The normative integrity of the system therefore depends on how much normative unity there is in society.

Te ao Māori will need to stand alongside Western worldviews. Yet this shift poses challenges. Some may see anything less than full recognition of tikanga as fatally flawed, given that the Māori worldview is so interconnected, and may resist elements of it being “cherry-picked”. After all, tikanga is a normative system in its own right, not just a handful of principles to be added to a mix of others, and it may well be that “a culture cannot be understood fully in terms of the worldview of another”.⁴⁹ The existing system has struggled with such things. For example, while it is not the place of the oceans management system to resolve spiritual debates, spirituality is hard to divorce from the more concrete aspects of tikanga because the former is often used to explain the latter. Some may also object to the “co-opting” of Māori perspectives to apply to all New Zealanders, or resist the segregation of Māori values from Māori decision-makers.⁵⁰ On the other hand, we can build upon synergies to recognise that, for example, the environment is not comprised of resources, but rather taonga to be treasured; our institutions are not regulators and policy makers, but rather kaitiaki and stewards; and our water and living creatures are not there just to be used and owned, but have their own mana, mauri and dignity.

Raewyn Peart



Pou, Tāpapakanga regional park

A spotlight on te ao Māori and ecocentrism

There are potential nodes of agreement between te ao Māori and ecocentric. In the former, natural features are ancestors and relate to Māori people through ancestral relationships (whakapapa). Humans are not above nature; they are just one part of the intricate web of relationships that make up our world. Everything in this world is imbued with mauri, a life principle. It is an obligation for humans to uphold this mauri. The obligation stems from the principle of reciprocity: nature looks after us, thus we must look after it. This worldview is based on kinship, respect and reciprocity.⁵¹

The ecocentric perspective shares many values but comes from a Western point of view. It seeks to de-centre humanity from our assumed position of being above and in control of nature. Instead, it emphasises that we too are a part of nature, and that we rely on ecosystems to survive. It highlights that nature is inherently valuable, rather than only being valuable to the extent it serves human interests. For example, the Earth Law framework recognises that marine ecosystems: own themselves and have intrinsic value apart from human uses; have the right to perform all of their natural functions; and have the right to have a voice in decisions that may affect their health, including the right to legally defend themselves against damage (through human representation). The most significant differences between the ecocentric and te ao Māori perspectives is that the former does not include the metaphysical plane or account for the *relationship* between humans and the environment in the same way.

An interesting overlap between the ecocentric and Māori worldviews, is the idea of humans having a special obligation to advocate for nature, due to our sentience and language. For example, Christopher Stone, from an ecocentric perspective, notes that:⁵²

I do not think it too remote that we may come to regard the Earth, as some have suggested, as one organism, of which [humankind] is a functional part – the mind, perhaps: different from the rest of nature, but different as a man’s brain is from his lungs.

...

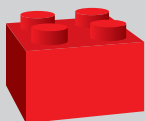
What is needed is a radical new theory or myth — felt as well as intellectualised — of man’s relationships to the rest of nature.

Comparably, Rev Māori Marsden, in explaining Māori cosmology, describes humankind as the “conscious mind” or “conscious awareness” of Papatūānuku.⁵³ This idea is what justifies the practice of human advocates in various legal personhood frameworks, whether humans are acting as a defender of basic rights in the court or are forming a management board to give effect to an entity’s interests.

It is also worth keeping in mind that the mix of ethics in an oceans management system may not be quite the same as on land, just as the ethics of the workplace can differ from those of the home. The longstanding possessive or defensive tradition of “my home is my castle” is a thoroughly terrestrial concept. People relate in different ways to the moana, which are arguably less about ownership, control, partition and exclusion as they are about respect, access and sharing. This might open the door to a system with a more ecocentric bent, or at least one focused less intensely on resources and profit.

Whether through failures of implementation or something more intrinsic, a largely Western, neoliberal and anthropocentric ethic has failed to prevent the problems described in Chapter 2. That does not mean we need to reject the ethic entirely, or replace it. But it means that we should be open to other ideas if they can orient our outlook in a way that is more likely to get us to where we want to go. After all, ethics are not just things we adopt and forget. They matter in practice. They affect what tools we deploy (eg rāhui, taxes, participatory planning or legal personhood), and the institutions we adopt (eg councils, a Tikanga Commission or marine guardians). They also shape the principles we use to guide decisions.

A future system will likely need to contain multiple worldviews to reflect Aotearoa New Zealand’s society. However, embracing synergies between ecocentrism and te ao Māori, within a plurality of ethics, may provide a positive direction of travel.



The normative basis of a future system could be one in which synergies between te ao Māori and ecocentrism are placed at the heart of decision-making.

7.4 Principles

Worldviews and ethics can be hard to pin down or encapsulate in words, and it is not immediately obvious how the discussion above might translate into the nuts and bolts of a new system. Yet an oceans management system needs specificity. One way to operationalise lofty ethics is through the creation or recognition of legal and ethical *principles*, such as sustainability, precaution, and ecosystem-based management. Many concepts, such as mauri (life force or essence), mana (honour and respect) and kaitiakitanga (caretaking/guardianship) are bound up in te ao Māori.⁵⁴ Below, we explore different principles that a future marine management system could adopt, drawing on the analysis in our resource management reform work. They include ecosystem-based management, sustainability, Māori-based principles, equity and justice, procedural justice, precaution and subsidiarity.

While some principles may produce binary choices, and different variations of principles are possible (eg “sustainable management” or “sustainable development”), the key thing to determine from a system design perspective will be how different principles interact with each other. Some principles can be regarded as substantive – they guide decision-makers in choices that will produce a tangible outcome. But equally important are procedural principles which guide *how* decisions are made.

Ecosystem-based management

Ecosystem-based management is a particularly important principle in the context of our oceans, given how connected they are. In theory it is simple: it is a management approach that recognises the full array of interactions within an ecosystem, including humans, rather than considering single issues (eg pollution), species (eg fish), or ecosystem services (eg food production) in isolation.⁵⁵ It resembles integrated management, but with the core concern that integration needs to focus on ecosystems rather than other things (eg the connection between land use and infrastructure).

The National Science Challenge ‘Sustainable Seas – Te Komata o Te Tonga’ has produced a wealth of interdisciplinary research exploring the development and application of ecosystem-based management in New Zealand, defining it as “a holistic and inclusive way to manage marine environments and the competing uses for, demands on, and ways that New Zealanders value them.”⁵⁶ Many of the features of this unifying principle can in fact be described as principles in their own right,⁵⁷ and some are explored below.

Ecosystem-based management is not well reflected in the existing system. Part of that is due to it not being expressly included in key legislation (eg the Marine Reserves Act) and only being implicit in others (eg the RMA and Fisheries Act). But a larger reason is that legislative and institutional frameworks are siloed in ways that do not manage ecosystems in a holistic way. That is noticeable when it comes to a fragmented approach to MPAs and the lack of connection between the management of fisheries and impacts from land-based activities on fish habitats. Rationalising legislation (see Chapter 11), modernising its normative foundations (eg in purpose statements), and improving connections between statutory processes (eg fisheries and catchment planning) could help give life to the principle.⁵⁸

There are challenges, however. Some have said that ecosystem-based management is unlikely to be successful where “ownership” and power sharing arrangements have not yet been established for Māori.⁵⁹ There can also be tensions between a Western conception of ecosystem-based management and Māori worldviews that conceptualise the human-nature relationship in different ways, although there are areas of overlap.

Despite ecosystem-based management’s international popularity in recent years, there is still a lack of consensus on its definition. It may be of most use as a general framing that colours other principles, such as sustainability and ecological justice. It is also equally about how people work together as it is about the marine environment itself. Collaboration and strong relationships are key to managing things in an integrated way.⁶⁰



Ecosystem-based management could form a core principle in a future system, expanding upon that of integrated management observable in frameworks like the RMA.

Sustainability and related principles

Sustainability provides a framework within which other more detailed substantive principles can be applied. To some, it may be a principle that is essentially about trade-offs – balancing the value of resource use with the value of environmental protection in a way that can be maintained over time. To others, it might involve setting hard environmental limits to safeguard the interests of future generations and the ability of the natural environment to support life. “Sustainable development” is, by contrast, concerned not only with ensuring environmental protection in the face of development pressures, but also with active recognition of the need to drive socio-economic development. “Sustainable management”, in the New Zealand experience of the term, has a narrower tradition of protecting

the environment but not addressing issues of equity or active pursuit of resource development.

A future system could potentially embrace a more active principle of sustainability in which there are duties to undertake sustainable development, as has been done in Wales.⁶¹ That might include, for example, not just facilitating new uses of the marine and coastal space (eg through green infrastructure, renewable energy, regenerative aquaculture and restoration of ecosystems) but actively driving them. Indeed, one could recognise a “social and economic development” principle more broadly – to recognise that the system should place value on, and incentivise or mandate, some resource uses that are in the public interest. This is already explicit in the Crown Minerals Act and Fisheries Act, where utilisation is expressly recognised as a good thing. The exposure draft of the NBA indicates a broader take on sustainability through the new concept of *te ora ngā o te taiao*, which is to be accompanied by a list of positive outcomes to be pursued (including the “protection and sustainable use” of the marine environment).⁶²

Other principles under the rubric of sustainability could include resilience and risk management. Resilience may be particularly important to ensure that a proactive approach is taken to minimising risks in the future (eg to guard against climate impacts like changing temperatures and storms on ecosystems, fish stocks and marine operations). Without it, the idea of sustainability may be too passive to meet the challenges of an unstable environment and rapid change, and may falsely assume we can control rather than being forced to respond to environmental change.⁶³



Recreational fishing, Raglan harbour

Raewyn Peart

A spotlight on sustainability and maximum sustainable yield

It is notable that the oceans management system adopts different versions of sustainability depending on what it is dealing with. Relatively minor differences are found between the principles underpinning the RMA and EEZ Act, which both focus on the concept of “sustainable management”. More notable is the purpose of the Fisheries Act, which is often described as “sustainable utilisation” (when what it really has is two potentially competing purposes).⁶⁴ First, there is an imperative to provide for the “utilisation” of a stock, which can be contrasted with the RMA’s softer requirement to “enable” people to provide for their own wellbeing, whether that is through resource utilisation or not. Secondly, there is the requirement to ensure “the potential of fisheries resources to meet the reasonably foreseeable needs of future generations” and to avoid, remedy or mitigate adverse effects of fishing on the aquatic environment, both concepts also reflected in the RMA and EEZ Act.

Fisheries could, of course, meet the needs of future generations if stocks were not fished *at all* by the current generation. Even a direction to utilise does not necessarily mean that stocks must be fished to the greatest extent possible. In fact the definition of “utilisation” in the Fisheries Act includes “conserving” as well as “using, enhancing and developing” fisheries resources. However, a conservation approach is not well reflected in other provisions.

When setting a TAC, the Minister generally seeks to maintain the stock at a level that can produce the “maximum sustainable yield” and to rebuild – or fish down – stocks to reach this Goldilocks objective.⁶⁵ This target partly stems from international law, where under UNCLOS, management measures must have the core focus of maintaining or restoring populations of harvested species at levels which can produce the MSY.⁶⁶ And states are required to grant surplus catch to other states if they cannot harvest the TAC set in any given year.⁶⁷

We have explored this principle and its history in previous work, noting that the underlying anthropocentric idea is that failing to fish down stocks is wasteful.⁶⁸ A strict theoretical approach to MSY has, however, proven untenable, due to assumptions about stocks and their environments that do not hold in the real world (eg annual variability in recruitment, growth rates, habitat change and

land-based impacts).⁶⁹ Other concepts, like maximum constant yield and current annual yield, have sought to iron out some flaws (eg avoiding the peaks and troughs of attempting to achieve MSY when stock sizes change year to year, by setting a more consistent and conservative TAC). The Fisheries Act direction has proven flexible enough to accommodate such innovations.

The direction in the Fisheries Act is to maintain stocks at or *above* a level that would achieve MSY, begging the question: what should justify TACs being set *above* this level? A precautionary buffer, or other value-based considerations? We might also take this further and ask how flexible the objectives of the Fisheries Act actually are, and the extent to which normative change away from MSY altogether could (if we wanted to) be achieved without legislative reform. The Act’s purpose is clearly much broader than this concept, extending to addressing adverse effects on the aquatic environment. In other words, sustainability in the sense of biological fish production (making sure enough fish are still there tomorrow and maximising their yield) by no means exhausts what the Act is there to do. It is equally about protecting the marine environment from the impacts of fishing, and not just from the removal of *fish*.

In 2009, the Supreme Court considered the application of MSY within the context of the purpose of the Fisheries Act in the *Kahawai* case, explaining that “the total allowable catch is the principal sustainability measure to maintain the fish stock at maximum sustainable yield or above it. But such maintenance does not exhaust sustainability or utilisation ends, which are concerned also with social, economic, and cultural well-being and the reasonably foreseeable needs of future generations.”⁷⁰

Even when it comes to stock management, the Act is broader than it might appear, and some have suggested that MSY is no longer an appropriate goal from an ethical standpoint even if technical challenges could be worked out.⁷¹ Even guidelines for the non-statutory Harvest Strategy Standard have recognised that “estimates of MSY-compatible points are only one of the inputs into the setting of targets. Other relevant inputs include economic, social, cultural and ecosystem considerations, which will generally result in targets equal to the MSY-compatible reference points or better.”⁷² However, specific policy on how

to operationalise these considerations has yet to emerge (see Chapter 8), and TAC setting is still treated as more of a technocratic exercise than an RMA-style value-based one despite the broad foundations of the Fisheries Act.

Some have described MSY-based catch limits as walking on a “razor edge” in an information poor environment.⁷³ While most assessed stocks are thought to be healthy,⁷⁴ this has not prevented collapses, the social and economic instability of closures, arguments about periods for rebuild, localised depletion and overfishing in practice.⁷⁵ It also remains that many stocks are not assessed (or assessed very infrequently), including ones that form an integral part of trophic webs.

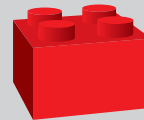
An objective based on a different principle, such as the retention of a representative range of age classes or abundance, could arguably avoid booms and busts, require less fishing effort (including for recreational and customary fishing), and improve the climate resilience of not only stocks but also the ecosystems of which they are a part. More specifically, some have suggested that management measures could aim to retain a higher proportion of original unfished biomass (eg no less than 50 percent) rather than seeking to maximise long-term yield per se.⁷⁶

Indeed, the Fisheries Act does not *always* require TACs to be set to achieve MSY. That includes for highly migratory species and – notably – where it is not possible to estimate it due to the biological characteristics of the species. Although a stock has to be explicitly included in a schedule to the Act for this specific exception to apply, it is interesting to consider the fact that TACs are frequently rolled over year on year in the absence of new information. Should this absence of new information, in light of the precautionary principle, effectively mean that it is not really possible to estimate MSY? Should stocks for which assessments are out of date or uncertain (or have never been undertaken) therefore be automatically subject to an obligation to set a TAC in a more cautious, or at least more nuanced, manner?⁷⁷

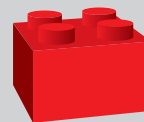
Finally, if the Fisheries Act allows consideration of broad social, cultural and economic factors – as highlighted by the Supreme Court – that begs the question as to whether the concept of sustainable utilisation is really that different from that of sustainable management under the RMA and EEZ Act. Indeed,

sustainable management was originally mooted as the purpose of the Fisheries Act.⁷⁸ The two concepts may in fact be converging, given the increasing focus on achieving “utilisation” outcomes for some things (such as urban land) under the RMA and a proposed focus on marine use as one outcome sought under the NBA.⁷⁹ That, in turn, begs the question whether continued statutory silos are justified for fishing and other aspects of marine management (see Chapter 11).

Sustainable management, in the New Zealand experience of the term, has a reasonably narrow focus of protecting the environment. It seeks only to enable socioeconomic wellbeing, not to drive it. Sustainable utilisation in the context of fisheries is broader, seeking to optimise the benefits of resource use.



Sustainable management could be recast as a broader concept of sustainability (eg *te oranga o te taiao*, or *te mana o te moana*), potentially embracing the social and economic dimensions of resource use and protection.



The principle at the heart of fisheries management could be reframed from one of sustainable utilisation to one more like sustainable management or *te oranga o te taiao* at the core of frameworks like the RMA/NBA.

Māori-based principles

Māori worldviews incorporate not just an ethical and relational understanding of the marine environment, but also various values and principles embedded in tikanga. The central Māori idea is often kaitiakitanga (stewardship, see spotlight below), but other important concepts are mauri (life-force or essence) and mātauranga Māori. This is more than just the knowledge of scientific facts, and it may be said to encompass broader ideas like wisdom (having an ethical and spiritual component – knowing what *ought* to be done) and systems of knowing. A future system will need to provide recognition, not only for these

concepts, but also for the ability of Māori to exercise them (see Chapter 12). Other important values relevant to marine management include the following.⁸⁰

- Whanaungatanga – “the centrality of relationships to Māori life”;
- Manaakitanga (and kaitiakitanga) – “nurturing relationships, looking after people, and being very careful how others are treated” and an ethic of guardianship;
- Mana – “the importance of spiritually sanctioned authority and the limits on Māori leadership”;
- Tapu/noa – “respect for the spiritual character of all things”;
- Utu – “the principle of balance and reciprocity”.

These values are intertwined with intangible or spiritual relationships as an absolute foundation to Māori society as well as the key institutions of te ao Māori.⁸¹ Dr Robert Joseph has added the concepts of koha (gift exchange); aroha (charity and generosity); and hau (respect for the vital essence of a person, place or object).⁸² Tikanga is also bound up with customary rights and practices, including with respect to the use of marine areas and the cultural harvest of marine species. In other words, Māori values concerning the environment do not make a binary distinction between use and protection. Instead, it is about mediating a relationship through connection with resources.

There are general concepts of tikanga that Māori embrace, and these could be used to anchor a future oceans management system. However, pinning down such concepts or defining them in a Western framework of formal legal mechanisms like legislation, regulation and institutions can prove challenging. For example, commentators have pointed out the challenge of defining tikanga Māori through a Western judicial system:

metaphysical concepts do not fit well within this objective framework, which depends on the presence of physical facts that can be quantified by science in order to render them more or less probative...⁸³

there is danger in assigning a Pākehā term to a Māori concept, as it isolates that concept from the Māori worldview it is born from.⁸⁴

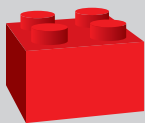
A spotlight on kaitiakitanga

Attempts to pin down complex ethical concepts, like kaitiakitanga, in bite sized statutory provisions have fallen short. For instance, the former Aquaculture Steering Group described kaitiakitanga as having the following elements (in the context of aquaculture reform), noting that it is wider and more complex than existing legal definition:⁸⁵

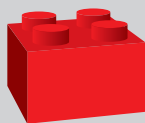
- founded in whakapapa – the relationship between everything and everybody in the natural world – there is no distinction between people and their environment
- a set of inalienable responsibilities, duties and obligations that are not able to be delegated or abrogated
- a web of obligations: to the taonga, to the atua and to ourselves and our uri. Kaitiaki have a responsibility to provide for everyone and ensure everyone benefits
- independent of “ownership” in a European sense
- seamless and all encompassing – making no distinction between moana and whenua
- expressed in ways that are appropriate to the place and to the circumstances
- given practical effect by exercising control over access to resources and sharing the benefits of the use of those resources (ie it has a strong social dimension that “environmentalism” under the RMA does not)
- enabled through rangatiratanga, which includes the authority that is needed to control access to and use of resources, and to determine how the benefits will be shared.

Further challenges arise with the interpretation of Māori terms within a statute, and the question of who interprets them. This issue is playing out in resource management reforms, in that it is unclear who will get to interpret the broad and novel concept of “te oranga o te taiao”.⁸⁶ Some have criticised the inclusion of te ao Māori concepts in legislation as a way to “co-opt” Māori values into the political process without corresponding Māori involvement.⁸⁷

Alongside concepts central to te ao Māori, a separate set of principles relating to te Tiriti o Waitangi have evolved in the courts and have become increasingly important as “part of the fabric of Aotearoa New Zealand law”.⁸⁸ There is a rich history of jurisprudence here, which other writers have laid out at length.⁸⁹ In all legislation, the Treaty must be considered as an aid to statutory interpretation.⁹⁰ More specific principles that will need to inform a future system include a duty on both Māori and the Crown to act reasonably and in good faith (including a duty to consult);⁹¹ the active protection of Māori interests by the Crown;⁹² the making of informed decisions by the Crown; the remediation of past grievances;⁹³ and the right of the Crown to govern by pursuing its policies in the interests of the whole community. The Waitangi Tribunal has also identified an underlying principle of reciprocity.



A future system could seek to give effect to the principles of te Tiriti o Waitangi that have been developed in the courts, or recognise and adhere to te Tiriti itself.



The normative core of a future system could be based on te ao Māori concepts such as kaitiakitanga, mana and mauri.

Equity and justice

Principles based on justice and equity go beyond the concerns of overall environmental sustainability. They are about fairness, and the distribution of costs and benefits of resource use and protection. We explored these principles in more depth in the project’s working paper.

Distributional equity (sometimes referred to as intragenerational equity) is about making sure that the benefit of resource use rights are distributed in an equitable way.⁹⁴ For that reason, it is intimately related to the system’s allocative role. However, it is not just about allocating resources themselves, but also distributing the benefits that flow *from* resource use and protection.

Within Aotearoa New Zealand’s current system, the concept of distributional equity has been fairly weak.⁹⁵ For instance, the concept of sustainable development promoted in the Brundtland report⁹⁶ was intentionally avoided in the RMA due to its association with the distribution of wealth and rights to socio-economic development.⁹⁷ As such, there are

few allocative principles to be found in the RMA. Under the Fisheries Act, after the initial allocation of quota, the distribution of commercial rights to fish is left to the market.

Questions about distributional equity abound in the context of management of a shared space like the oceans. Is it equitable, for instance, that the allocation of coastal space is still largely achieved through a reactive, first-in-first-served process under the RMA?⁹⁸ If not, who should receive these “rights” and on what basis (and for what activities)? Should the market decide, or should that be the job of a well-intentioned public authority? Should communities and their representatives get a say? And should such rights be given away for free (on a cost recovery basis), or should there be a return to the public and Māori (by imposing a resource rental or koha)?

Furthermore, is it fair that the grant of new aquaculture rights is, essentially, dependent on not having an undue adverse effect on wild fishing interests?⁹⁹ And is it fair that, albeit in a fairly unconscious fashion, the interests of some fishers, aquaculture proponents and recreationalists are effectively subservient to the “rights” of landowners who discharge nutrients and sediments into harbours, impacting the productivity of the marine environment? And finally, is it fair that the financial benefits of harvesting wild fish – a common resource – accrue to quota holders without a portion being returned to the public through a tax or resource rental? (On a deeper level, does society still even regard fish as a “common” resource of New Zealanders, or is it rather a “shared” resource between commercial, customary and recreational fishers?)

Similarly, if one accepts that the public *should* receive some financial benefit from the use of a public resource one can also ask whether it would be fairest to characterise that as a cost recovery levy type arrangement, a tax, a koha, or a resource rental, and what such revenue should be used for (eg marine conservation efforts, investing in the development of a fishery, assisting kaitiaki, or a general pot of government money). All of these questions are far from settled in the marine context.



Distributional equity or intra-generational equity could be expressly recognised as a principle in a future system, particularly to guide decisions about allocation.

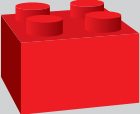
Environmental justice is about who bears the cost of environmental degradation. The principle is closely related to indigenous environmental

justice, which, in Aotearoa New Zealand, is often framed around obligations and redress under te Tiriti o Waitangi.¹⁰⁰ It is not about protecting the environment per se, only about the ways in which the costs and benefits arising from use are distributed among people.

At present, a lot of the environmental costs of fishing, land-based discharges and other activities are borne disproportionately by New Zealanders as a whole. Coastal communities and Māori – many of whom are advocating for greater involvement in decision-making around fisheries and marine protection – are particularly impacted by the damage that occurs in their watery backyards.

For example, it is arguably unjust that some people in Aotearoa New Zealand cannot use and enjoy their coastal environment (at least without the risk of illness) because of nutrient discharges from land-based activities, chemical contamination from stormwater (much of the impacts which remain unknown), microplastic and other waste, or discharges from public wastewater systems. Because of urban growth pressures and historical infrastructure underinvestment in some urban parts of Aotearoa New Zealand, these impacts are not felt evenly across the country.¹⁰¹ To Māori, this harm has a spiritual or metaphysical component. And fishers (whether customary, recreational or commercial) are impacted disproportionately by contaminants entering the sea from land-based activities.

Environmental justice and distributional equity seek to distribute the costs and benefits of resource use and protection between groups in present-day society, according to equity or sensitivity to harm.

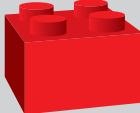
 **The principle of environmental justice could be strengthened in a future system, reflecting a broader understanding of the social elements of sustainability.**

Ecological justice sees the natural world as an actor within, not an object outside, the human community of justice.¹⁰² The inclusion of humans as part of the environment aligns with te ao Māori (eg in oral traditions like the creation story of Ranginui and Papatūānuku). The concept is not unfamiliar to the existing system – the existing prohibition on hunting marine mammals is not just because some are threatened, but also because hunting them is seen as the “wrong” thing to do (see the spotlight earlier).

Current laws see dolphins, whales, seals and sea lions as different or special, and deserving of a kind of justice closer to that which humans enjoy.¹⁰³

But if we took the concept further, should society build institutions that give the oceans a voice of their own? Can this build on the innovative legal personhood developed as part of the settlement processes for Te Urewera and Te Awa Tupua/Whanganui River,¹⁰⁴ and what would be the challenges in giving the oceans as a whole legal personhood (eg through recognition as Tangaroa or Hinemoana, or concepts like te mana o te moana)? Instead of a resource rental going back into the public purse, should that be treated as “payment” or koha to nature for its services (or compensation for past harm) and be invested in regeneration projects? We explore legal personhood in Chapter 8.

The principle of ecological justice sees the natural world as an actor within, not an object outside, the human community of justice. This has implications for our choice of tools, as well as institutional arrangements – including the idea of conferring legal personhood on aspects of the environment.

 **There could be express recognition of ecological justice in a future system, embracing an ecocentric ethic and welcoming nature into human systems of justice.**



Hector's dolphins, Akaroa harbour

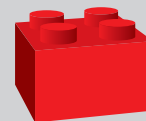
Raewyn Peart

The principle of intergenerational equity is about the relative value we give to the interests of current and future generations.¹⁰⁵ It tells us that the latter should be recognised and safeguarded to at least some extent.¹⁰⁶ It is a morally charged obligation to future generations to act prudently; a direction not to consider cumulative, long-term effects as inconsequential;¹⁰⁷ and an exhortation to live on the dividends of, rather than erode, our natural capital.¹⁰⁸ Edith Brown Weiss has identified three key components in this concept: conserving the diversity of our resource base; maintaining the quality of our planet; and providing equitable access to our legacy.¹⁰⁹ It invites into the system of justice those who are not yet born.

But different worldviews can give a different flavour to the principle; te ao Māori approaches future generations as a continuity between atua, ancestors and mokopuna, from the perspective of whakapapa. The appropriate Māori term may be taonga tuku iho, meaning a gift passed down through the generations.¹¹⁰ Ecocentrism looks at it from the perspective of future generations of *nature*.

In particular, intergenerational equity may point to the need to actively *enhance* the marine environment to restore its productive potential where it has been degraded (or where people have benefited from its past degradation), and to set firm environmental limits to prevent (at a very minimum) marine ecosystem collapse and exceeding ecological tipping points.

Intergenerational equity is about maintaining the ability of current people to meet their needs while not compromising the needs of future generations. However, what the needs and interests of future generations are with respect to the oceans is not always clear.



Intergenerational equity could be strengthened in a future system by defining more specifically what the relative interests of current and future generations are, including with respect to restoring, enhancing and developing the marine space.

Procedural justice

The literature generally draws a distinction between issues of substantive justice and procedural justice.¹¹¹ Even if an outcome is fair, it does not mean that the process to get there has been. The significance of this has been seen in the case of Rangitāhua/Kermadec Islands, where the substance of a proposal for protection is arguably less of an issue than the way in which (and by whom) the proposal has been developed and communicated (see the spotlight below).¹¹²

The Waitangi Tribunal has also been scathing about procedural justice for Māori under the RMA, which was “seen as a beacon of hope for Māori ... Nearly 20 years after the RMA was enacted, it is fair to say that the legislation has delivered Māori scarcely a shadow of its original promise.”¹¹³



Recreational fishing, Firth of Thames

The same can be said of the EEZ Act. The MACA Act at least provides an opportunity for statutory processes to be coloured more by tikanga, but there have been some criticisms that Western style courts are still responsible for conferring rights and resolving disputes.

Procedural justice is also closely connected to what we might call the “participatory” principle: who gets to be notified and involved in processes leading to key decisions, and the importance of deliberative democracy. Wide participation in oceans management decision-making allows choices to be informed by local knowledge and the values of communities, and to provide catharsis for genuinely held views.¹¹⁴ The principle has generally operated on the assumption that people can participate to the extent that their interests are affected.¹¹⁵

Broad participatory rights are particularly noticeable when producing plans, policy statements and regulatory tools that are values-based. They are more targeted when they only involve impacts on defined interests (eg property rights).¹¹⁶ As Barton and colleagues have observed, robust participatory rights at least under the RMA “reflect a social consensus that goes back long before 1991”.¹¹⁷ That is particularly relevant in the marine space, where there are fewer fixed private property rights and arguably a greater “public” interest in management. Everyone is a stakeholder, not just those who use the marine area directly. That has implications for the fisheries system, where the provision and use of information in stock assessments, fisheries planning and sustainability measures is focused on the science of fish stocks rather than the values of communities or mātauranga Māori.¹¹⁸

Mana whenua involvement across the whole system is an important end in its own right under te Tiriti o Waitangi, separate to broader notions of public participation.¹¹⁹ It is also concerned with the sharing of power and institutional arrangements (see Chapter 12), not just how Māori are consulted.

That said, participation cannot be absolute, and needs careful constraints. It can be costly and lead to delays in achieving urgent outcomes. In other words, what is procedurally just might be unjust when it comes to the substantive results. That has arguably been the case in the past when it comes to the delays faced in protections for the Māui dolphin due to litigation. But procedural justice goes to other system design features too, not just participation. It supports broad access, and openness, to information (including mātauranga Māori),¹²⁰ as well as access to judicial redress for breaches of the law or improper use of public powers.

A spotlight on the proposed Rangitāhua/Kermadec Ocean Sanctuary

The lack of a clear framework for MPAs beyond the coastal marine area, has resulted in ad hoc and hotly contested mechanisms for creating protections on a case by case basis, which can have significant implications for the fairness of the process. Nowhere is this more obvious than in the context of the proposed Rangitāhua/Kermadec Islands Ocean Sanctuary. On 29 September 2015, at the United Nations General Assembly in New York, Prime Minister John Key announced that Aotearoa New Zealand would create an oceans sanctuary in the EEZ around the islands. At 620,000 km², and twice the area of the country's landmass, this was to be “one of the world's largest and most significant fully protected ocean areas”.¹²¹ The area is renowned for its high biodiversity, with 32 percent of all fish species known in Aotearoa New Zealand being from the region. The Kermadec Islands themselves are a nature reserve managed by the Department of Conservation. A marine reserve protecting the territorial sea surrounding the islands was put in place in 1990. In addition, a benthic protection area established under the Fisheries Act in 2007 protects the EEZ around the islands from bottom impacting fishing methods. Ngāti Kuri and Te Aupōuri have mana whenua status over the area.

The announcement of the Sanctuary followed a long campaign, beginning in 2008, led by the USA-based Pew Foundation and supported by WWF New Zealand and Forest and Bird. The Kermadec Islands had been one of nine focus areas worldwide supported by Pew as part of its Global Ocean Legacy Project. This aimed to establish the world's first generation of great marine parks, with others in places such as the Pitcairn Islands, Easter Island and Palau.¹²² The departmental disclosure statement for the Bill indicated that the decision to create the sanctuary was largely informed by a number of reports prepared by the Pew Foundation.¹²³

The National-led Government's decision to create the sanctuary was kept secret until just before the United Nations announcement. Key affected parties such as Māori fisheries trust Te Ohu Kaimoana, Ngāti Kuri, Te Aupōuri, deep sea mining company Nautilus Minerals NZ Limited and the Pew Foundation were only informed by telephone the night prior to the announcement.¹²⁴

The Kermadec Ocean Sanctuary Bill was introduced into Parliament on 8 March 2016, with its first reading on 15 March when it was referred to the Local Government and Environment Select Committee. The purpose of the Bill was to “preserve the Kermadec Ocean Sanctuary in its natural state”.¹²⁵ Shortly afterwards, Te Ohu Kaimoana announced that it would be launching judicial review proceedings against the Government. This was soon followed by the New Zealand Fishing Industry Association which also launched proceedings. Meanwhile public submissions were sought on the Bill and the Select Committee reported back on 22 July 2016 with some minor recommended changes. The legal proceedings are currently stayed pending passage of the Bill through Parliament.¹²⁶ The Bill is stalled while the parties seek a negotiated settlement and is still awaiting its second reading.

The legal challenges against the Bill raise three key allegations:¹²⁷

1. The Bill effectively confiscates fisheries quota. The legislation would set the TAC (and also the TACC, which cannot legally exceed the TAC), for the QMA which surrounds the islands, to zero.¹²⁸
2. The establishment of the Sanctuary is a breach of the Crown’s duty of good faith to Māori (through the failure to undertake fully informed consultation, and to proceed without consent of Te Ohu Kaimoana or iwi and without compensation).
3. The actions of the Crown are contrary to the 1992 Māori fisheries settlement.

The situation is a complex one. The quota rights in question have never been exercised, as there is currently no fishery at the Kermadec Islands.¹²⁹ However, all the quota that has been issued by the Crown so far in the area (apart from for migratory species which can also be caught elsewhere) was provided to Māori as part of the Treaty fisheries settlement.¹³⁰ Te Ohu Kaimoana has argued that the proposed measures undermine not just property rights per se, but also te Tiriti rights under the fisheries settlement by unilaterally changing the redress provided after the fact. This is through reducing the value of the quota held by Māori in FMA10 and also undermining the ability of Māori to exercise rangatiratanga through managing and utilising the fish stock.¹³¹

Te Ohu Kaimoana’s claims have yet to be determined by the Courts, and this seems unlikely to occur in the future, due to the Crown’s focus on negotiating a settlement to the proceedings. The controversy over the Sanctuary may have had a chilling effect on other marine protection initiatives, including the development of new MPA legislation. This has yet to progress despite the release of a discussion document on a new Marine Protected Areas Act in January 2016 by the National-led government.¹³² Nevertheless, both a legislated solution to the Rangitāhua/Kermadec Sanctuary, and to MPA legislation more generally, still appear to be on the table for the present government.

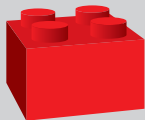
The Kermadec’s experience is a cautionary tale about how the reform process is managed, but also a reminder that a continued ad hoc approach to marine issues reflects a system that does not deal with issues of procedural fairness and justice in a coherent or effective way. There is no transparent standardised process provided for Māori or stakeholder engagement,¹³³ other than the select committee process to review new legislation. There is no predictable framing for what the purpose of bespoke legislation would be, including Māori principles, or a consistent approach to how customary or commercial rights are to be treated.



Te Whanganui-A-Hei (Cathedral Cove) marine reserve

Raewyn Peart

Procedural justice, which is closely linked to participatory rights, is as important as the justice of substantive outcomes. In particular, the involvement of *mana whenua* can be regarded as an important element of indigenous justice in its own right, not just a means to an end.



A principle of procedural justice could be included or reflected more strongly in a future system, outlining common elements of all processes to ensure they are fair, including for *mana whenua*.

Precaution

The precautionary principle tells us to take care where we face environmental risk or uncertainty. Where it is unclear whether an adverse effect will occur, that does not excuse a lack of action to address the effect.¹³⁴ In short, it is better to be safe than sorry. This goes further than the principle of prevention, which simply holds that it is better to prevent harm than to respond to it after the fact.¹³⁵

The precautionary principle has a rich history in international law, with its most famous formulation being in Principle 15 of the Rio Declaration on Environment and Development (1992). This proclaims that “where there are threats of serious or irreversible damage, lack of full scientific certainty shall not be used as a reason for postponing cost-effective measures to prevent environmental degradation”. Generally, the greater a risk’s probability and magnitude, the more precautionary the response needs to be. Whether remaining risks are acceptable or not cannot be answered in the abstract; it depends largely on value-based policies relating to the kinds of impacts.

In some contexts, residual risks can be managed through the principle of adaptive management (changing controls as more information comes to light), rather than being avoided altogether.¹³⁶ Yet it can be arguable whether adaptive management enhances precaution (by imposing more controls over time) or weakens it (by allowing risky activities to proceed).

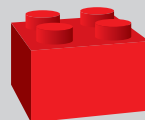
Precaution is already a significant part of our existing system. Case law has developed a great deal of nuance around this under the RMA, and marine dumping regulations are highly precautionary (they assume dumping is prohibited unless expressly authorised). More explicit “information” principles about the treatment of uncertainty exist under the Fisheries Act and the EEZ Act.¹³⁷ In the latter, the courts have relied heavily on precaution as a reason

to decline or impose conditions on consent for mining activities, showing the value of a clearly defined statutory principle, not just a general ethos or approach.¹³⁸ And precaution is proposed to be codified in the NBA both as a general principle and in the form of a mandatory buffer when setting environmental limits above a point that threatens ecological integrity.¹³⁹

Overall, the precautionary principle (or “approach”) is particularly important in the marine environment, given the difficulty of obtaining information.¹⁴⁰ Yet while the current system recognises precaution in many ways, in others it is more questionable. For example, despite a precautionary principle in the Fisheries Act, most catch limits are rolled over on an annual basis without a stock assessment. Many related and dependent stocks are not assessed at all. There is relatively little attention paid to the land-based impacts that could impact on stocks in the future. This is understandable given the cost and time involved in the process, and the number of interrelated stocks, yet it does beg the question as to whether we should be fishing at all if we don’t know the true status of the stock (and its habitats). Precaution does not demand *perfect* information, but it does guide the nature of the response if information is lacking. Some have argued that the precautionary principle has been used in ways that actually undermines its intent.¹⁴¹

Moreover, it is arguable that precaution is not baked into the broader fabric of the system when caution is needed across statutory boundaries. For example, cumulative risks from catchments, fishing, mining and pests pose threats of serious and potentially irreversible damage, yet uncertainty about how they relate has still led to siloed approaches to individual pressures (eg tackling fisheries, biosecurity and sedimentation separately) and a lack of cost effective measures being taken across them (eg MPAs and marine spatial planning). Precaution might not be achieved by simply considering whether or not to allow new activities; it may require proactive measures to improve information or to enhance the environment.

The precautionary principle states that where there is uncertainty as to the adverse effects of an activity, this is not a reason to fail to take action to address them. It includes approaches to risk identification, risk assessment, and risk management.



A broader precautionary principle could be adopted at a more systemic and proactive level, including obligations to take positive action to enhance the resilience of the environment where future cumulative impacts are uncertain.

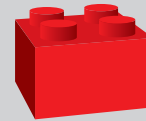
Subsidiarity

Decisions can be made by a variety of institutions at a variety of levels of governance. Subsidiarity provides that decisions should be made closest to, and in line with the values of, those most affected by them (the relevant community of interest).¹⁴² It is not the same thing as localism.

It can be difficult to determine where the appropriate community of interest lies, because many communities have legitimate and conflicting interests.¹⁴³ This is particularly evident in the marine context. For example, it is by no means clear that residents of a region (those who vote for regional councillors) are the appropriate community of interest when it comes to impacts on the moana at the outer edge of the coastal marine area, far from where most people live and work. Nor is it clear that centralised authority should be dominant when it comes to making decisions about highly localised depletion of inshore fish stocks that affect the ability of marae and communities to feed themselves and sustain the ecosystems in their backyards. Efforts to lay down a series of localised rāhui (leading to fisheries closures), and to pursue small-scale MPAs under the RMA in the Bay of Plenty, highlight this tension. There are also unresolved questions about “devolving” decision-making to particular communities of interest (eg mana whenua, sector groups, or collaborative groups), which is not about centralisation or localisation per se; instead, this is about how the system identifies and defines dominant communities of interest.

For its part, the RMA provides for political decision-making power at the planning level to shift as the relevant community of interest changes,¹⁴⁴ and the national level Environment Court has wide-ranging jurisdiction on appeal (even for local matters).¹⁴⁵ But aside from the NZCPS, and the approval role of the Minister of Conservation for regional coastal plans, central government has not taken a proactive role in managing the coastal marine area. Conversely, there are few meaningful mechanisms in the Fisheries Act for *local* level influence, only input.¹⁴⁶ Holders of a customary marine title recognised under the MACA Act are given the power to grant RMA “permission rights” within a title area, without which an activity requiring a coastal permit cannot proceed.¹⁴⁷ But these do not directly influence fisheries decisions, which remain centralised. And while some frameworks provide for devolution of decisions to mana whenua and industry or other stakeholder groups,¹⁴⁸ the existing system provides little by way of guidance as to when or why such things should happen.

Subsidiarity seeks to locate decision-making responsibilities closest to (and according to the values of) the relevant community of interest. There is a separate Māori community of interest alongside national, regional and local ones.



A future system could provide more clarity as to what subsidiarity means in the marine environment, and when it is appropriate for decisions to be centralised or devolved to councils, mana whenua, or stakeholder groups.

Other principles

A number of other principles can be noted more briefly. Some of these are not spoken of as “principles”, in the sense that it is hard to define them in a way that can be included in legislation. Yet they often lie behind design choices and decisions.

- *Efficiency*: In terms of process, decisions should be streamlined but must be balanced against the need for good information and public participation. It must also be remembered that efficiency is only about the most efficient way to achieve a desired outcome, and not a substitute for debate about what that outcome should be. Sometimes the best outcomes will require time and cost. Efficiency is important both in terms of administrative process and the use of resources themselves, such as how scarce things like fisheries and marine space are used as well as the kinds of tools used to manage that use (eg market-based ones versus regulatory ones).¹⁴⁹ It has implications for broader design features too; a highly fragmented system or one that is unclear is bound to be inefficient. Yet a costly system can still be an efficient one if it is good at delivering what we want.
- The *conservation* principle: This recognises that protection and enhancement of the environment must be relatively absolute in some geographical areas, for some species, and for the functioning of the natural world as a whole. It is a useful label for the idea that underpins the moral imperative to prevent extinctions and to impose environmental limits focused on ecosystems. It encompasses the principle of *non-regression*, which states that measures beneficial for the environment should not subsequently be removed or eroded, and the *public trust doctrine*, under which the state acts as trustee of the

ecological health of “public” areas.¹⁵⁰ It is not just about preventing extinctions or saving threatened species from further decline; conservation is equally about regeneration and restoration.

- *Property rights and security of investment* are not usually considered to be a principle per se, but recognition of these is a core tenet of capitalism that generally goes unspoken. It is implicit, for example, in provisions that require decision-makers to have regard to the level of investment made when deciding whether to renew an existing consent. This does not mean that property rights have to be *granted* (ie that marine resources are privatised), but it does mean there is a high threshold before they can be extinguished, where they already exist. The defence of property rights can be regarded as a principle (to be balanced against other principles), but an alternative is that property *rights* are regarded as something that lies beyond the scope of the system entirely (see Chapter 6).
- Similarly, *economic growth* is a principle that lies behind parts of the system, but is seldom invoked specifically. Instead, often the broader notion of *wellbeing* is enshrined in legislation (eg the RMA and the Local Government Act) or in other mechanisms (eg the Treasury's wellbeing framework for funding decisions). The concept of wellbeing is an even more slippery one than sustainability, because

often different types of wellbeing (social, cultural, economic and environmental) have no clear hierarchy.

- The principle of *polluter-/user-pays* seeks to distribute the costs of resource use between private and public interests. It generally places costs on the polluter or user (or those who benefit from it) unless there is good reason not to, and is one expression of *environmental justice*. A future system may need to be much clearer and more transparent about who actually bears the cost of polluting activities, and provide measures to transition towards a more equitable distribution of costs. For instance, activities generating sediment on land are not generally held accountable to those who suffer from its impacts (eg through lower productivity in estuaries and food-producing ecosystems). This does not mean that people should have a right to pollute or use resources simply by paying for them. It also has implications for how acceptable offsetting mechanisms are in the marine environment (degrading one environment to enhance another), and the deployment of tools like marine biobanking (see Chapter 8).

Many other principles may be important in designing a future system, including efficiency, conservation, non-regression, property rights, economic growth and the polluter/user-pays principle.



Forestry harvesting, Havelock

Raewyn Peart

7.5 How principles are used

When it comes to system design, there are two distinct ways to think about the principles described above. First, principles are norms that underpin and guide a reform exercise itself. These might be called “foundational” principles. They operate in the “backroom” of reform. For example, a principle of non-regression might result in any attempt to weaken environmental limits in national direction being subject to review or even appeal in the Environment Court. Ecosystem-based management might see a tighter relationship between the RMA and Fisheries Act, or the merging of those frameworks into one statute. And a participatory principle might tell policy-makers that the public has a right to be involved in any decision involving value-based decisions or ones that affect them personally. Even assumptions about the sanctity of property rights, capitalism and liberal democracy can be seen in this light. Foundational principles are not necessarily spelt out in a statute. Instead, they guide choices when reforming the system.

Secondly, principles can themselves be specifically defined in legislation or developed through case law. These might be called “operational” principles. Operational principles have life and influence through the ongoing operation of the system. They are spelt out meticulously as a way to guide the exercise of discretion and as an aid to interpretation of other provisions. For example, a statute like the RMA contains principles like inter-generational equity and sustainability that are subject to constant interpretation when making decisions on plans and consents. The Fisheries Act and EEZ Act have information principles that can be the subject of litigation.

In reform exercises, more importance is often placed on how operational principles like “sustainable management” and “te ora o te taiao” are drafted in legislation than on how the foundational principles of reform itself are articulated (if at all). The former are seen as legal, whereas the latter are in the realm of politics. However, it is important to give due consideration to both. Foundational principles can be just as significant as operational ones, as they drive all sorts of system features (eg the tools that are used and the institutions that support them) – not just the purpose and principles of legislative silos.

Principles can be used in two key ways: (1) as foundational principles, where they guide other system design choices without being included in legislation, and (2) as operational principles, which are included in legislation and are used as guides to ongoing statutory decision-making.

7.6 Choices on principles for the future

In a future system, we face a number of choices when it comes to principles. We can adopt some (or a particular variation of them, such as sustainable management) and reject others (such as sustainable development). We can carefully articulate the relationships between different principles, either within a broad framing (eg linking use and protection within the principle of sustainability through a term like “while” or “subject to”) or by establishing a hierarchy or synergistic relationship between different principles (eg making clear there is a hierarchy between the utilisation of fisheries and conservation of marine mammals).

General principles like sustainability, precaution and justice can also be worded very differently depending on the worldviews underpinning them. For example, if something is “sustainable”, what are we actually trying to “sustain”? Is it the ability for a resource to keep us alive? Its ability to support economic growth? The ability for a species to be kept from the verge of extinction? The ability for mana whenua to exercise kaitiaki responsibilities and protect (and use) taonga? This difference is most obvious when we look at the conceptual gulf between sustainability that is underpinned by concepts like kaitiakitanga, whakapapa and whanaungatanga; or framings like te mana o te moana and te ora o te taiao; and notions like MSY or maximum economic yield.

We can also think about when principles *bite* – whether a principle gives a reasonably clear indication of an outcome in primary legislation, or whether more generally worded principles are left to be interpreted and elaborated in more specific contexts through policies, regulations and permits. For example, the RMA in Part 2 gives only a general sense of what marine outcomes are envisaged from particular activities, and even in a lot of plans and national direction (eg the NZCPS) policies do not have clear hierarchies or directive language.¹⁵¹ The real meaning or impact of principles is only felt when they are balanced or crystallised in regulatory mechanisms like rules, standards and – most often – in consent conditions (or refusals).

The principles outlined above (and potentially many others) could be adopted, rejected or modified. They might be expressed differently in different legislative contexts.



Principles in a future system could be made more specific and directive in legislation, giving greater clarity as to what outcomes are expected and less room for interpretation by policy makers and the courts.

7.7 Objectives and the system's core roles

It is possible for principles to be quite specific and directive. At some point, these are better described as objectives rather than principles. For example, while one could say that the system is aiming for “resilience” or “ecological justice”, it is hard to measure success for such things; a future system could be aiming for much more specific and measurable objectives than that.

For the most part, the current oceans management system is defined by principles and shies away from specific, time-bound objectives. Even provisions specially *called* objectives in RMA plans are often vague to the point of confusion. There are some exceptions – the principle of sustainable utilisation in the fisheries context, for example, has arguably crystallised into a measurable objective – MSY (see the spotlight earlier). Indeed, it can be plotted on a graph, something which is well out of reach for a principle like sustainable management under the RMA. Yet even this objective, like the more general ones of the RMA and EEZ Act, is still essentially seeking to maintain a static state. As pointed out in Chapter 3, such things are about management, not change. They are not *strategic* objectives.

Many specific objectives could be sought in the future, and we offer some thoughts below as to what they might be. But the key design question for the future is whether the system should consciously drive towards a new future, rather than leaving big-picture objective setting to the vagaries of politics of the day. To put it another way: if one core role for the system is to pursue positive outcomes, should marine management frameworks become more like climate change (see the spotlight below), where the statute contains a clear imperative for measurable change¹⁵² and powerful legal mechanisms to get there? We have said previously that:¹⁵³

If the system lacks clear goals for the future,¹⁵⁴ we need to give it some – and ways to get there ... If it does not, we may become stuck in a politicised cycle focused on the pros and cons of the status quo, not the opportunities and risks we face in the future.

A spotlight on climate change

As New Zealand Inc, we are developing a general vision for land use change (driven by climate change imperatives). That is illustrated in incentives for afforestation (through the emissions trading scheme and One Billion Trees programme), national direction for the protection of productive land, and proposals for a formal framework for regional spatial planning. We are also looking to transform a whole sector – transport – to make it carbon neutral, including through quite direct interventions like banning the importation of conventional vehicles and subsidising the uptake of electric vehicles. All of this is in the service of a clear statutory direction to meet Aotearoa New Zealand's nationally determined contribution under the Paris Agreement and to limit global temperature rise to below 1.5 degrees Celsius. This is far from the laissez faire approach to resource management – avoiding, remedying and mitigating – that we have become used to over the past three decades. Our objectives are becoming more specific, and more geared to forcing change. And we are pursuing an overhaul of the way society operates to get there.



Meola Creek, Auckland

Raewyn Peart

This begs the question: should we not also have a similarly (or more) ambitious strategy for what would be the best use of ocean space? For example, should we proactively plan for the roll out of offshore renewable energy projects, in order to mitigate climate change, provide for energy security, and soften the impact on workers in the oil and gas sector? Do we need dedicated zones earmarked for desalination in anticipation of meeting new freshwater drinking needs in Auckland (see Chapter 2), or do we wait for the market to take the lead and deal with any spatial conflicts later on? Should we seek to utilise new resources like rare minerals, or existing resources in new ways (like deriving pharmaceuticals from fish and shellfish), to bring economic opportunities?

The current system has relatively few specific objectives for change baked into its legislative fabric, instead relying on general principles like sustainability. Climate change is one exception where objectives are clear and direct.



A future system could legislate for a much more specific set of objectives, including timeframes or milestones for achieving change.

Rather than offering a laundry list of possible objectives, it is useful to consider them in the context of the more specific roles the system might be expected to perform. For example, although there may be some overlap, what we are seeking to achieve by imposing environmental limits is likely to be quite different to our aims for allocating resources. We consider a range of potential objectives below.

Environmental limit setting

The existing system arguably does not impose true environmental “limits” (see Chapter 3). Limits, sometimes referred to as bottom lines, are about identifying minimum acceptable outcomes – the points at which harm is to be prevented no matter what the trade-off. These can be expressed in different ways, such as a minimum state for the environment (eg minimum coastal water quality, or minimum populations of species) or the maximum amount of pressure allowed (eg maximum concentration of a contaminant or maximum catch of fish). Limits can be set using different tools, such as a cap and trade system, prohibited activity regulations, or a fisheries bag limit (see Chapter 8). But our concern here is not *how* they

are set, but whether, and why, limit setting is a distinct role a future system should be expected to perform at all.

That is not a foregone conclusion. We have previously said that on land this is “one of the least controversial ways in which the system plays a role”.¹⁵⁵ It can be seen in strict minimum noise standards, controls on air quality, or nitrogen caps in a catchment. Such things are measurable and there is a clear link between the activity being restricted and the minimum public good outcome being defended. Swimmable rivers might, for instance, be regarded as a limit or bottom line at which point no more agricultural intensification can occur in a catchment. But it is arguably more difficult to put the concept into practice at sea, where the environment is more fluid, connected and fluctuating, things are harder to measure and observe, and our understanding of ecological processes and the impacts of cumulative impacts are poorer. It is also harder to ascertain when limits have been breached (due to monitoring constraints).¹⁵⁶ In workshops, some instead expressed support for concepts like continuous improvement and adaptive management, seeing firm limits as too difficult in practice.

That said, some things might be amenable to limit setting in the marine environment. For example, point source discharges (eg through wastewater standards for outfalls) can be controlled. Fisheries are already familiar with the concept of a catch limit (including, in practice, a “hard” limit on stock size) even where there is incomplete information.¹⁵⁷ Marine biosecurity has very strict limits on the introduction of pest organisms (although less so once they are established and financial trade-offs are finely balanced).¹⁵⁸ And limits can be placed on mortality when it comes to protected species (eg maximum bycatch before closure of a fishery).

Furthermore, while they can be hard to measure and attribute causation to, overall precautionary limits could still be required for diffuse discharges, such as sediment from urban development or forestry harvesting operations, by targeting limits spatially at particular land uses (ie identifying where those activities cannot under any circumstances occur). Limits could also be imposed “upstream” by prohibiting the creation or manufacture of products containing pollutants that we know end up in the marine environment (eg microbeads and heavy metals). It depends what one means by a “limit” – they do not necessarily require specific numbers to be set. Indeed, limits recognised by the Supreme Court under the NZCPS are non-regulatory policies that contain strong language like the broad requirement to “avoid” adverse effects.

The issue is therefore not *always* that environmental limits are hard to set and monitor. Rather, the current system does not necessarily *want* to set them if that would prevent trade-offs being considered on a case by case basis. For example, imposing an inflexible limit on sedimentation in the Marlborough Sounds might require plantation forests in some catchments to be left completely unharvested, undermining significant investments in forestry and expectations that go back decades. Do we want to set limits here, or instead allow trade-offs depending on the extent of investment in any given catchment? Slowing the speed at which a fishery is reopened after closure, which remains largely a discretionary decision,¹⁵⁹ would also affect short term profitability in the fishery. Should we take into account economic factors in deciding how fast a stock rebuild should be, or consider only biophysical factors in how we get back to healthy stocks?¹⁶⁰ A lot comes down to whether we want to set hard and ambitious limits or not.

There might also be strong *social* resistance to imposing environmental limits. For example, restrictions on where cats can be owned, and treating some as a biosecurity risk to be eradicated, may ease the toxoplasmosis threat to the Māui dolphin (see Chapter 2). But controls on domestic cats would upset many pet lovers; 44 percent of homes in the country have one.¹⁶¹ So do we set some kind of limit on cat ownership, or balance the interests of pet owners against those of dolphins? The former would come at a cost.

Yet it is interesting that, for many non-environmental matters, the current system is not shy about setting strict limits. Such a role has been strongly embraced where immediate social and economic impacts would be felt. For example, there is little opposition to protecting sensitive infrastructure like submarine cables and pipelines (even from economically and socially valuable activities involving trawling and anchoring)¹⁶² or excluding people from areas around mining operations or dangerous shipwrecks. An uncompromising approach to biosecurity (which closed down an entire bluff oyster aquaculture industry in the Marlborough Sounds) can at least partly be explained by its potentially catastrophic economic impacts for other sectors (in that case the wild bluff oyster fishery). The Building Code is understandably uncompromising in its requirements that buildings will not cause loss of life in an earthquake. And many “environmental” success stories (eg air quality improvements) can be put down to the imperative to prevent direct human health impacts rather than protect the environment per se. In all this we can see that limit setting is primarily a product of the political will at any given time, not a core feature of the current system. Should protecting something like a reef or seamount through *environmental* limits be so different?

Perhaps. Yet strict limit setting in some contexts may be an uneasy fit with tikanga. That may be the case where limits prevent important cultural uses by prioritising *other* uses (eg commercial fishing which causes local depletion), or where no-take spatial protections sever relationships with the moana (which rely on mana whenua interacting with it as kaitiaki). Limits can be culturally problematic where they are permanent (eg in the creation of a marine reserve) rather than temporary (eg through rāhui to allow stocks and habitats to replenish). In isolation, limits can also be seen as encouraging a race to the bottom. That is inconsistent with the need to enhance already degraded environments as well as te ao Māori, which is about maintaining a reciprocal relationship with nature rather than just preventing collapse.

A key choice is whether the concept of an environmental limit (a non-negotiable and absolute minimum outcome to be defended), as opposed to making trade-offs by balancing competing concerns, is a distinct role a future system should be expected to perform.



A future system could specifically define what an environmental limit is, and require such limits to be set for a defined list of things in the marine environment.



Sandager's wrasse, Mokohinau Islands

Tanya Peart

When it comes to something like submarine pipelines, it is reasonably easy to articulate why we have limits – a pipeline is either broken or not, and any activity that could damage (eg anchoring) it is duly prohibited. It is, however, harder to articulate exactly why environmental limits are needed and therefore where they should be placed.

One option would be for limits to be set in places that would prevent the collapse of marine ecosystems. That would be an outcome that must, at all costs, be prevented. Alternatively, limits might be set to achieve anthropocentric objectives such as the protection of a basic level of human health (eg when setting pollution standards). Indeed, both of those things are proposed to be point of limit setting in the new NBA, which provides:¹⁶³

The purpose of environmental limits is to protect either or both of the following:

- (a) The ecological integrity of the natural environment:
- (b) Human health.

However, limits could aim to achieve a range of other outcomes, such as protecting the pristine state of areas that are valuable to recreation or scientific research (eg marine reserves). They could aim to prevent the extinction of species. Or they could be set at points that maintain the mana or mauri of a resource or ecosystem, which may well be a much higher minimum standard than the prevention of ecological collapse. The goal could be more specific still: for instance, remove all forms of marine life from threatened status by a particular date.

The purpose of limit setting might also be phrased in a relative or an absolute way. The former would essentially treat the status quo as being broadly acceptable as a minimum standard, but any *further* degradation would be strictly prevented. This kind of approach is evident in how wetlands are being treated under the government's essential freshwater reforms (where national direction mandates "no further loss" in the extent of wetlands).¹⁶⁴ Alternatively, the purpose could be based on a more absolute measure of environmental quality, which might mean, for some things (eg where entire ecosystems have tipped over into a new, less productive state), limits would already have been infringed.



Saltmarsh and mangroves, Miranda

What we are hoping to achieve by setting limits also affects the scale at which they are imposed. If we are wanting to preserve an *overall* minimum measure of biodiversity or representative ecosystems, we might be happy to trade-off harm in one place for compensation or net gain elsewhere. In that case, limits could still be maintained while providing for environmental offsetting or compensation. That becomes very important when considering the design of MPAs (see Chapter 9), given that absolute protection in one place can concentrate fishing pressures (and cause limits to be breached) in others.¹⁶⁵ It is also relevant when considering the boundaries of QMAAs (see Chapter 8), where overall catch limits can be adhered to but still cause localised depletion of significant value to marae, recreational fishers and local communities.

Environmental limits might also be imposed to protect the values of specific locations – values that could vary markedly from place to place – which is quite a different purpose to those outlined above. For example, many existing marine reserves are arguably more about protecting the unique values of specific places (eg for research, tourism, particular uses like diving, or distinctive features or life forms) rather than defending biodiversity outcomes per se. Should the system have a role in imposing inflexible and permanent limits to achieve the former, or only the latter? What makes one marine space more important than others? And would we contemplate spatial limits that moved – such as protected areas that shifted from one place to another as the climate changed – to achieve overall biodiversity gains? Or are protections inherently tied to place?

Moreover, what we are seeking to achieve by imposing environmental limits may depend on what we include in the concept of the “environment” and how far that goes. There may not be consensus about which *types* of things require uncompromising limits and which do not. For example, would we want to put inflexible prohibitions on particular activities (eg ports, aquaculture or commercial fishing) on the grounds that they would affect the natural character or “amenity” of marine environments?¹⁶⁶ Or prohibit activities to protect marine heritage (eg shipwrecks) or culturally or spiritually significant sites (like wāhi tapu?) Or to strictly limit the impacts of one activity (eg aquaculture) on another (eg fishing)? Those are all “environmental” impacts under the broad definition in the RMA, but they may not necessarily reflect the kind of objectives for which limits are required. Then there are even more complex and controversial things like limits to economic growth or limits on human population. Are we hoping to pursue such things by setting uncompromising environmental limits? Some may be intrigued by the prospect. But core “limits” in the marine context might have a distinct purpose relating to strict protection of basic

ecological function and structure, human health, and harm to threatened or protected species.

It is also interesting to consider whether the purpose of limit setting in a framework like the RMA or NBA could be more *marine* focused than the RMA is currently. For example, the EEZ Act has a dual purpose (essentially, sustainability *and* the prevention of pollution), and its strong second pillar has been interpreted as being significant for how tools are used, such as when determining applications for consent.¹⁶⁷ That could be reflected in the NBA as well, given the strong imperative to reduce or phase out pollution in coastal areas.

Trade-offs and dispute resolution

Many activities can have impacts on the moana, but that does not mean we simply ban them all. Instead, the system often has to figure out if harm is a price worth paying. And although trade-offs should not be conflated with limit setting (as it arguably has been in practice when considering the recovery of some fish stocks),¹⁶⁸ it is not enough for the system to just set limits and walk away. It needs a mechanism to balance various competing matters. Indeed, one commentator has pointed out, “in the absence of [any] trade-offs, decision criteria can become paralysed”.¹⁶⁹

Often a conversation about trade-offs is framed around how the system strikes a balance between environmental harm and economic and social benefit. Some have suggested that existing statutes like the RMA (at least on their face) fail to recognise the benefits of the latter.¹⁷⁰ There is something in this reasoning. Arguably, the RMA was never intended to have such an “active” planning role, as some hoped that market forces and personal choices would resolve any trade-offs as long as environmental limits (which were, at least in theory, contemplated by section 5 of the Act) were not infringed.¹⁷¹ Yet value-based trade-offs were inevitable in practice, as decision-makers were specifically directed to consider the positive effects of activities, not just negative ones. Much consent authority time has gone into weighing up the pros and cons of activities in light of various policies and objectives. The Act’s silence as to the benefits of resource use did not mean they could be ignored – it just meant the process of weighing them against other things was left almost bereft of statutory guidance.

We have previously described this as the Act’s inbuilt identity crisis – that it even has a role in guiding choices above environmental bottom lines was never made particularly clear.¹⁷² At least in the marine environment, the mandatory NZCPS has assisted somewhat (it talks about the value

of things like infrastructure alongside the importance of protecting the natural environment), but its policies have still required extensive litigation – including all the way to the Supreme Court – to determine what their relationships mean in practice. Things are even less clear under the EEZ Act, which has been left entirely devoid of such policy instruments. As described in Chapter 3, this can leave novel activities like offshore aquaculture and deep-sea mining in policy limbo, wasting applicants’ time and effort and causing angst for other stakeholder groups engaged in costly litigation.¹⁷³

It would be possible for such things to be rectified by “rebalancing” the list of outcomes sought by such statutes (as long as they were clearly subject to environmental limits) and by clarifying the relationships between the outcomes sought. At least at a high level, key trade-offs (eg do we want to pursue offshore wind or allow deep seabed mining? Where and where not, and subject to what standards?) could be made up front for all to see. Indeed, the exposure draft of the NBA has included the broad outcome of “sustainable use” of the marine environment, although due to its generality and the fact it is located next to the term “protection”, that is hardly a meaningful guide to how trade-offs should be made on the ground.¹⁷⁴ It does, however, signal intent about what kinds of things the Act will be expected to do differently in the future, including taking a more active approach to balancing the pros and cons of use and protection.

In a future system, this role could also be reframed as one in which synergies are pursued. For instance, the rollout of more nuanced MPAs could be framed not just around where harm (eg from bottom trawling on the benthos) is no longer worth the benefit, but also as an opportunity to protect areas of value as nurseries for stock recruitment (from both fishing pressure and land-based sedimentation). The system could be reoriented to seek win-win situations where the social and economic benefits of use and environmental wellbeing can be enhanced. Indeed, the Randerson Panel has proposed to transform the “trade-offs” implicit in the RMA into the synergistic “outcomes” focus of the NBA, and we have previously said that:¹⁷⁵

We need no longer accept that a project that enhances economic wellbeing must come at an environmental cost, or simply be as environmentally neutral as possible. The system should therefore think not just about “trade-offs” but also about “convergence” and a “race to the top”.

It is also important to remember that making trade-offs is not just about environment versus development. It is also about making choices between

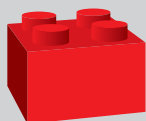
different forms of use and development. Although an assessment of trade-offs between the costs and benefits of various marine activities is a well-worn path under a consenting framework like the RMA... a key question for the future is whether the system should deal with more strategic trade-offs in advance.

At present, a lot of tensions between marine activities span different statutory frameworks and can be brushed under the carpet, because they do not need to be addressed within a single decision. For example, a consent authority considering an application to occupy the seabed under the RMA is not required to consider the opportunity cost (whether a different kind of use might be better).¹⁷⁶ And it is at least possible for mining operations to be consented in benthic protection areas even though bottom contact fishing methods are not permitted.¹⁷⁷ The system¹⁷⁸ could have a more proactive role in making trade-offs across a range of such things, including the following.

- Whether sustainable offshore aquaculture operations should be supported irrespective of their potential to impact adversely on wild fishing interests.
- Whether the development of sustainable inshore forms of aquaculture like seaweed and some shellfish is more important than the harvesting of plantation forests or coastal urban development (which could have impacts on it).
- Whether fish and other resources are better used as a source of food, or whether other uses should be supported instead/as well (eg nutraceuticals).
- Whether, even with the emissions trading scheme, climate change mitigation is more important than exploration for petroleum.
- Whether sub-seabed space should be used for oil and gas extraction or a reservoir for compressed carbon dioxide captured from industrial emissions.
- Whether the economic benefits of commercial fishing outweigh its impacts on recreational fishing opportunities.
- Whether the eco-tourism and recreational benefits of protected areas outweigh the impacts on other sectors like fishing and mining.

The point here is not to suggest which choices the system should make, and we invite readers to consider that for themselves. Instead, it is to ask whether a future system should be involved in making such choices at all. Such things could, for example, be treated as matters for inter-sectoral dispute resolution and facilitated through mediation or collaborative spatial planning when conflicts arise. Or they could be lent considerable policy support and strategic direction through legislation. In the marine context, there is almost always a public interest element in the resolution of disputes, which sets it apart from land where some disputes (eg between neighbours) are almost wholly private in nature.

The current system has something of an identity crisis when it comes to making trade-offs above environmental limits, because it provides little guidance as to when the pros should outweigh the cons of particular activities, or whether this is really something the system should be doing at all. A future system could be clearer about why this role is being performed, and tackle some difficult strategic trade-offs between types of activities in the marine space.



A future system could be expected to guide more clearly how trade-offs are made between different forms of wellbeing above environmental limits. It could also reconceptualise the role from one of balancing things against each other to seeking win-win situations.

Allocation

The system's role in making trade-offs is closely connected to another role: allocation. This is because higher level choices about *kinds* of desirable activity are sometimes linked to very individual issues about who gets to use a resource and – if there is a limited amount of it – who does not. For example, strategic policy choices can be made by allocating space to some activities in advance (eg aquaculture, recreational fishing parks, wind farm zones). In practice, this excludes not just some uses, but also some groups of people.

However, the system could have a much broader role in making allocative choices. Not only could it determine to what kind of *use* marine resources

are put, but also who specifically gets to undertake the use (individuals, companies or groups). An allocative role is complex in the following ways.

- Different resources (eg fish, minerals, occupation of coastal space) can be, and currently are, treated differently from each other. They are allocated for different purposes and by different mechanisms.
- It is not just tangible “things” that require allocation, it is any right that is scarce (eg the right to conduct tourism operations in a marine reserve, or the right to catch a proportion of a TAC).
- There are three distinct kinds of allocative decision – (1) an initial allocation, (2) reallocation (the transfer of existing or previous rights or use from one person to another), and (3) deallocation (in what relative measure people give up their rights when scarcity increases).

Aside from some private land titles (traded via the market) and fisheries quota (again, bought and sold on the market with some economic regulation), marine resources rely heavily on regulatory mechanisms for allocation.¹⁷⁹ However, aside from Crown owned minerals, it is questionable whether the existing system has really performed an allocative role other than by a sidewind. Rights – especially under the RMA – are recognised largely on a first in, first served basis:¹⁸⁰

Although the RMA does not really care how resources are allocated, it still goes ahead and does it anyway. A resource consent decision creates a legally defensible and sometimes exclusive right to use scarce resources (and sometimes for a long time – up to 35 years).

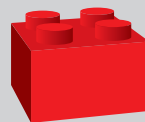
Concessions and permits under conservation legislation are also usually granted on a first in, first served basis. One of the growing criticisms of the system is that, while it deals with sustainable management and conservation of resources, it does not provide sufficient direction as to the allocation of many of those resources between different classes of activities. Should allocation be towards the highest and best use, or to the most deserving applicants? One commentator has noted the “ingenuity” of councils in trying to use a framework like the RMA to “make the allocation of resources a necessary means of controlling effects”.¹⁸¹ Deficiencies in the RMA are particularly obvious in the case of aquaculture, and they are equally present under the EEZ Act (under which aquaculture might be possible in future). A future system could be designed to perform an allocative role more proactively in the future, and we explore options for doing so in Chapter 8 when we look at resource use rights.

An even bigger question is the extent to which the system should have a role in reallocating *existing* rights, such as fisheries quota and rights to occupy marine space. The legitimacy of doing so might depend on several things – the nature of the right (eg property vs other rights), the mechanism by which it is done (eg regulatory taking, extinguishing consent, or voluntary buy back) and the purpose for which reallocation is pursued (eg to improve the environment by removing the worst performing rights holders from the system, or to achieve greater social equity). Whether the system should have a reallocative role is therefore not a simple yes or no answer.

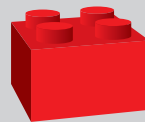
Finally, there is role of *deallocation* to consider. If the system is to reduce the allocation of something where it is the overall impact, not a particular user, that is important, then there is the question as to whose rights should be removed and in what measure. That is reasonably straightforward when it comes to reducing commercial harvesting levels, as quota rights are already expressed as a proportion of the TACC and decrease accordingly when the TACC is reduced.

However, the situation is more complex when fishing effort needs to be “deallocated” by spatially excluding fishing from particular areas (eg through MPAs or rāhui). Why should quota holders in one QMA give up more space than those in a neighbouring one? Should MPAs be distributed fairly across all QMAs so as to equitably share the cost amongst the sector, rather than being determined according to ecological criteria? This “spatial” deallocation of rights is perhaps the thorniest “deallocative” role the system will have to grapple with in future. Other situations are challenging too, such as when it comes to the proportions in which recreational and commercial take should be reduced when a TAC is lowered.

The current system performs a crucial allocative role – it determines who gets to use what and for how long – but in many contexts it does not do so in a proactive way or consider what use would be best. A future system will need to engage more directly with its allocative role, as the marine space becomes more congested and contested, and potentially consider how rights could be reallocated or even deallocated (and in what shares people would have to give up rights and interests).



The system could be expected to guide how rights to use or benefit from different resources are distributed.



The system could be expected to guide how some existing rights might be reallocated to “better” uses or users over time.

If the system were to play a more proactive role in allocating or reallocating resources, the purposes of doing so would need to be made clearer. At a general level, the system might seek fairness, efficiency and/or certainty. But there are many options for what that could look like.

On the one hand, the system could seek to simply allocate marine resources to those who can pay for them. That could, like the QMS, be primarily about economic efficiency. It could also be about generating an ongoing return to the public, as is already the case for royalties for Crown owned minerals.

Raewyn Peart



Harvested gurnard in slurry

Alternatively, the objective behind allocation could be to provide certainty of rights and a clear pathway for resolving disputes. Under this approach, the outcome (who gets to use a resource, and who benefits from it) is less important, as long as there is clarity as to who benefits and a process that all can use to get to it. For example, a first in, first served approach to allocation gives some clarity (in that we all know we need to get in before someone else). So too does giving a preferential right to an incumbent user (eg an existing consent holder) based on the amount they have invested, or creating fixed proportions for commercial and recreational portions of a TAC. To some, that certainty may be all that is needed. But should an incumbent be preferred when it comes to reconsenting operations? Should that depend on how much investment has been made, whether a new entrant would produce better outcomes, or whether there is some other driver (eg to return rights to mana whenua or a more environmentally friendly use)?

Although there is now opportunity under the RMA and Conservation Act to make more value-based allocative decisions (where applications can be compared on their merits rather than assessed in isolation)¹⁸² it is by no means clear why one activity or operator should be preferred over another and what the objective of allocation (if any) is. This dilemma can be seen vividly in the lack of normative guidance around allocation in the RMA and EEZ Act (or instruments made under them). It can also be seen under the Fisheries Act in the respective allocation of harvest shares to the recreational and commercial sectors when a TAC is changed. Here, the courts have held that:¹⁸³

The sequential nature of the method of allocation provided for in s 21 does not indicate that non-commercial fishing interests are to be given any substantive priority over commercial interests... It leaves that to the judgment of the Minister.

However, there is very little in the Act or elsewhere to guide the Minister's judgement. Similar difficulties with allocative questions can be seen in conservation legislation, where there is little direction as to who should receive concessions to undertake tourism, research or other activities in protected areas or in relation to protected species.

In a future system, the purpose of allocation or reallocation might be more normatively rich than just providing certainty of rights. It could tackle these difficult questions head-on. Resources (or a proportion of them) could, for example, be allocated in ways that would achieve greater social equity or improve the socio-economic status of the most

vulnerable, or to uses/users that would provide the most wellbeing for society or the environment.

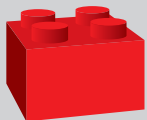
Instead of arguing interminably over who got an application in first, we could ask: which proposal would *best* promote sustainable management (or *te ora* or *te taiao*)? Which method of fishing would produce the *most* sustainable or efficient outcome?¹⁸⁴ That could produce a race to the top, where the system's allocative role also performs quite a different role – environmental enhancement. A particularly interesting case is Victoria, Australia where, because of the potential for the use of carbon capture and storage technology, subsurface space (including offshore) has to be allocated between existing petroleum interests and new carbon storage interests. A targeted law introduces a test partly based on the “public interest”.¹⁸⁵

Importantly, allocative principles would need to include reference to *te Tiriti o Waitangi*, both in the sense of upholding specific settlements (for fisheries and aquaculture) and the influence of the principles of *te Tiriti* more generally. The latter is just as important as the former; with case law highlighting the need for the Minister to consider whether *iwi* should be given preference, or whether no rights should be granted at all.¹⁸⁶

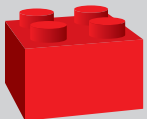
A degree of specificity would be needed around what allocative principles meant in practice (eg through national direction), to provide a reasonable degree of certainty to those seeking to use resources. Principles might also look different depending on the resource in question – for example, efficiency might be regarded as more important than equity when it comes to the allocation (or reallocation) of commercial fishing rights.

In developing allocation objectives, we also need to decide whether the system should go further and actively “pick winners” by allocating scarce resources to (or away from) specific kinds of uses or users in advance. For example, marine spatial planning (see Chapter 10) might have a specific legislative direction to facilitate offshore renewable energy or to secure recreational fishing experiences by designating suitable space for such things (and preventing conflicting uses, as we do with submarine cables and pipelines). Some have also suggested giving explicit priority to recreational fishing over commercial fishing when setting a TACC.¹⁸⁷ This more directive and activity-based approach to allocation goes much further than a principled-based approach. It is also intimately linked to the system's role in pursuing positive outcomes, because rights can be allocated to particular people or uses that would provide the most overall benefit for people or the environment.

The current system tends to shy away from principles relating to the allocation (and reallocation) of resources. A future system might be expected to determine more clearly why different resources should be allocated to different uses or users. This could be about creating a clear and certain pathway for rights to be conferred.



A future system could include an overarching set of allocative principles, which might look different for different resources. That could relate to the use to which resources are put, or which users can benefit from them. In particular, how the Tiriti provisions are worded will have implications for how resources and rights are allocated.



The system might take a more directive approach to allocation, where resources are reserved for uses or users that are seen as more deserving than others.

Pursuing positive outcomes

To some, the system may need to be a vehicle for driving positive change. Arguably we need to improve things, not just set bottom lines or allow harm when the benefits are “worth it”. That is especially the case if, as explored in Chapter 6, the basic rationale for having a system is to pursue the public interest rather than just internalising externalities. Allocating resources to the “best” uses could help, but the pursuit of positive outcomes could go well beyond an allocative role.

Part of this might involve proactively providing public goods and services – tangible things that contribute to people’s wellbeing. In the marine space, the system currently plays less of a role than on land, given that there are no roads, water connections or railways at sea. Although ports need to be provided for, blue highways themselves are provided by nature. That said, some marine infrastructure must be delivered, maintained and protected (eg for navigation, public transport and submarine cables), and that will remain an essential role in the future.

However, there may be potential for this role to be expanded. For instance, a much broader range of infrastructure might be considered important to achieve the public interest, such as offshore wind or tidal energy, justifying more active intervention (eg subsidies, the delivery of publicly funded pilot projects, or even direct provision of this kind of infrastructure as there is with roads, prisons and schools). The system may also have a stronger role to play when it comes to the provision, design and location of infrastructure on land, to reduce the environmental impacts of things like wastewater facilities and desalination plants on the moana.

Extending this role even further, there is an opportunity to rethink how a future system conceives of public services and goods. Ecosystems providing services (eg shellfish beds, reefs, carbon sinks) could be thought of as a form of infrastructure not dissimilar to concrete pipes and electricity lines. Indeed, things like MPA corridors could be regarded as ecological *network* infrastructure like the national grid, which needs to be proactively maintained and improved. Similarly, public authorities frequently provide museums, education facilities and research centres on land. MPAs could perform similar functions at sea by acting as living laboratories and classrooms.¹⁸⁸ In fact, when reconceptualising marine ecology as infrastructure, it is intriguing to consider close similarities with publicly provided land-based infrastructure in the more traditional sense.¹⁸⁹

Public goods usually exhibit particular features that means the market does not provide them (or may provide them at excessive cost).¹⁹⁰ They are often non-excludable (it is hard to prevent people using them, and therefore to require people to pay for them), non-rival (one person’s use does not prevent another person’s use),¹⁹¹ or have network monopoly characteristics (it is not practical or efficient to provide competing networks...

Those features apply equally to marine habitats as they do to things like roads or lighthouses, suggesting that ecological services are something the system needs to “provide”, not just “protect”. It is also interesting to compare the importance of *coordinating* things like roads, pipes and public transport with land use (eg residential rezoning), which is a significant theme in current resource management reforms. But human uses like fishing, tourism and recreation also need to “plug in” to ecological services to work. These services need to be present at the right time and the right place. If we were to reconceptualise the natural environment as a form of ecological infrastructure, it may transform the role the system is expected to perform and people’s perception of environmental enhancement.

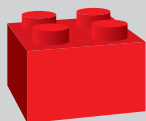
Of course, to many people environmental enhancement is important for its own sake, not just to provide services to people. That is already recognised in conservation laws to some extent, and enhancement is mentioned in the Fisheries Act and the RMA (although the latter does not really provide mechanisms to achieve it). Yet a future system could have a clearer role to play in pursuing positive environmental change rather than just dealing with the impacts of private activities as they arise. This is, for example, made explicit in Wales.

A spotlight on the Welsh approach

The Welsh Well-being of Future Generations Act places positive obligations on public bodies which “must” carry out sustainable development.¹⁹² This is a stronger and more active requirement than to simply “promote” in the RMA. The definition of “sustainable development” refers to the process of “improving” economic, social, environment and cultural wellbeing¹⁹³ rather than “enabling” people and communities to do this for themselves under section 5(2) of the RMA. In Wales, public bodies are required to set and publish objectives designed to maximise their contribution to achieving each of the seven wellbeing goals, and to take “all reasonable steps” to meet the objectives.

Indicative drafting of the proposed NBA suggests there will be a new focus on achieving positive outcomes in the marine environment, not just on the management of adverse effects. But it remains to be seen whether that is an aspiration or whether it will actually provide the toolkit to achieve it.¹⁹⁴

The current system seeks to pursue some positive outcomes in the marine environment, but it is largely defined by passive management. A future system could change its orientation to drive positive change. If it does so, it will need to provide the actual tools to enable change to happen, not just aspirational statements.



The system could be reoriented to drive positive change more than it does at present. The concept of providing public goods and services could be broadened to include the active provision and protection of *ecosystem services*.

The system could also clarify what its aims for “environmental enhancement” are in the marine context. It is not always clear how far we should go here. For example, to some, restoration might mean returning the oceans to the state they were in prior to human or Pākehā settlement. To others, it might simply mean returning to a safe space above environmental limits. And some might aim for a state of continual improvement rather than an end point. This is how the NBA proposes to treat enhancement, where there are many references to “improving” and “restoring” but no answer to the obvious question – restoring to *what*? And how will we know if things have been improved *enough*?

This issue is highlighted by the case of mangroves, which have expanded in some estuarine environments as a natural response to environmental change (eg more sediment coming down catchments). These are “new” anthropogenically driven ecosystems, but they are not necessarily *bad* ones. So should we aim to restore such environments to states that preceded human activity, such as sandy shorelines? Or embrace nature’s response to human activity?

There are many possible objectives for environmental enhancement. The proposed NBA has an extensive list. These include improving significant habitats of indigenous fauna, enhancing public access, increasing the removal of greenhouse gases from the atmosphere and improving the resilience of the environment to climate change. However, those outcomes are still fairly general, and others (eg the protection and sustainable use of the marine environment) are potentially conflicting.

There are clearer objectives set out in international documents. For example, the FAO Code of Conduct for Responsible Fishers (a voluntary document) includes targets which could, conceivably, be incorporated into fisheries legislation, such as (for example):

pollution, waste, discards, catch by lost or abandoned gear, catch of nontarget species, both fish and non-fish species, and impacts on associated or dependent species are minimised, through measures including, to the extent practicable, the development and use of selective, environmentally safe and cost-effective fishing gear and techniques.

The Aichi biodiversity targets also provide examples of how the objectives of a future system could be made more specific, including by incorporating timeframes for achieving them. For example, those targets include the following.

- By 2020 all fish and invertebrate stocks and aquatic plants are managed and harvested sustainably, legally and applying ecosystem-

based approaches, so that overfishing is avoided, recovery plans and measures are in place for all depleted species, fisheries have no significant adverse impacts on threatened species and vulnerable ecosystems and the impacts of fisheries on stocks, species and ecosystems are within safe ecological limits.

- By 2020, pollution, including from excess nutrients, has been brought to levels that are not detrimental to ecosystem function and biodiversity.
- By 2020, invasive alien species and pathways are identified and prioritised, priority species are controlled or eradicated, and measures are in place to manage pathways to prevent their introduction and establishment.
- By 2015, the multiple anthropogenic pressures on coral reefs, and other vulnerable ecosystems impacted by climate change or ocean acidification are minimised, so as to maintain their integrity and functioning.
- By 2020, at least 17 percent of terrestrial and inland water areas, and 10 percent of coastal and marine areas, especially areas of particular importance for biodiversity and ecosystem services, are conserved through effectively and equitably managed, ecologically representative and well-connected systems of protected areas and other effective area-based conservation measures, and integrated into the wider landscapes and seascapes.

This final target raises the possibility of a future system using legislated objectives for the deployment of specific tools (MPAs), not just for general outcomes (such as the NBA's "improving ecological integrity").

Raewyn Peart



Cape Rodney-Okakari Point marine reserve

A spotlight on objectives for marine protection

In 2014, following concerns that the Aichi targets were inadequate,¹⁹⁵ the IUCN World Parks Congress adopted a recommendation to “urgently increase the ocean area that is effectively and equitably managed in ecologically representative and well-connected systems of MPAs or other effective conservation measures”.¹⁹⁶ A target of 30 percent of each marine habitat was recommended for marine protection purposes.¹⁹⁷ The recommendation was endorsed by the IUCN World Conservation Congress in 2016, with the Congress issuing a resolution encouraging governments to designate and implement at least 30 percent of their national waters as MPAs and other effective area-based conservation measures by 2030.¹⁹⁸ In 2021, the High Ambition Coalition for Nature and People (an intergovernmental group of more than 57 countries) was established to advance a global deal for nature and people at the Conference of the parties to the Convention on Biological Diversity, with the central goal of protecting at least 30 percent of the world’s land and ocean by 2030 (the 30x30 target).¹⁹⁹ The target has gained widespread support.²⁰⁰ The draft Post-2020 Global Biodiversity Framework sets action targets for 2030, including the following.

- Retain and restore freshwater, marine and terrestrial ecosystems, increasing by at least [50 percent] the land and sea area under comprehensive spatial planning addressing land/sea use change, achieving by 2030 a net increase in area, connectivity and integrity and retaining existing intact areas and wilderness.
- Protect sites of particular importance for biodiversity through protected areas and other effective area-based conservation measures, by 2030 covering at least [60 percent] of such sites and at least [30 percent] of land and sea areas with at least [10 percent] under strict protection.
- Control all pathways for the introduction of invasive alien species, achieving by 2030 a [50 percent] reduction in the rate of new introductions, and eradicate or control invasive alien species to eliminate or reduce their impacts by 2030 in at least [50 percent] of priority sites.
- Reduce by 2030 pollution from excess nutrients, biocides, plastic waste and other sources by at least [50 percent].

Such targets relate to the overall coverage of MPAs, but equally important are objectives about *where* they should, and should not, be located. Take marine reserves, for example. These have been shown to effectively restore biodiversity and enhance ecosystem resilience: a meta-analysis of studies around the world showed the biomass of fish was 670 percent greater than in adjacent unprotected areas, and 343 percent greater than in partially protected areas.²⁰¹ Do we want to establish them only when and where they don't disrupt existing uses and rights? Or is that precisely their purpose? Should we “go hard” and establish no-take marine reserves where it would be scientifically and ecologically beneficial, or do we need to trade that off against other uses and cultural interests?

It is interesting to consider the approach of Natura 2000 areas in Europe, which adopt ecologically based criteria for the establishment of protected areas, rather than weighing up environmental benefits against property rights and economic value.²⁰² However, as has been seen in the context of the Rangatāhua Kermadec Island Sanctuary (see the spotlight earlier), tying objectives to the deployment of a particular brand of no-take MPA (where cultural connections are severed) may lack the nuance required of a system under te Tiriti o Waitangi.

For its part, the non-statutory biodiversity strategy *Te mana o te taiao – Aotearoa New Zealand biodiversity strategy* has three sets of goals for MPAs. The first is that by 2025 “a protection standard for coastal and marine ecosystems will be established with implementation underway”. Next, by 2030, that “significant progress is made in establishing an effective network of marine protected areas and other protection tools”; and by 2035 (on the way to 2050 goals) that “an effective network of marine protected areas and other tools, including marine and coastal ecosystems of high biodiversity value is established and is meeting the agreed protection standard”.²⁰³ However, MPA planning processes “will be underpinned by a commitment to minimise the adverse impacts of new MPAs on existing users of the marine environment and Treaty settlement obligations”.²⁰⁴



St Clair seawall, Dunedin

contaminants before they reach the sea) while providing habitat for marine species like seabirds. Demand side tools for freshwater use (especially in Auckland) such as water meters and onsite urban rainwater tanks may reduce the need to consider (or frequently use) solutions like desalination that can potentially put pressure on marine environments (see Chapter 2). And supporting the managed retreat of coastal infrastructure, rather than building defensive structures, could have benefits for marine ecosystems (eg by preserving inter-tidal habitats) as well as public access to beaches.



A future system could legislate for measurable environmental enhancement objectives. These could reflect those contained in international fora (eg the Aichi biodiversity targets) and could relate to the deployment of particular tools, such as MPAs.

It is also worth considering what objectives an oceans management system might have for land-based management. For example, the system might have a role in increasing the use of nature-based solutions to address land-based pollution (eg use of urban wetlands to filter stormwater

The moana is a valuable space for people, not just nature, and it is worth contemplating what social and economic objectives a future system might have to improve people's overall wellbeing. Indeed, social and economic goals can sometimes have potential benefits for environmental wellbeing as well. For example, commercial fishers operating more profitable businesses are better able to invest in new, more environmentally benign equipment.

But there is a significant question mark over the extent to which the formal system should be expected to set purely social or economic objectives – especially if they are legislated – or simply enable people (including markets) to provide for their own wellbeing.²⁰⁵ It might, for example, raise the uncomfortable prospects of market regulation and economic planning. If this is a legitimate role, however, possible objectives might encompass the following (and we invite readers to consider others).

- Creating marine-related jobs and economic or export opportunities, including for Māori business.
- Supporting the growth of coastal tourism (eg by supporting artisanal fishers or establishing recreational fishing parks and MPAs).
- Supporting the economic health of small coastal communities (eg by planning the location of licensed fish receivers and new infrastructure like wind energy).
- Diversifying the fishing industry by encouraging new entrants.
- Connecting future generations to the sea by supporting young people to obtain fishing rights or coastal permits.
- Supporting the deployment of high value industries, such as offshore aquaculture or mining for rare earth minerals.
- Ensuring high value uses of marine resources (eg fresh rather than frozen fish products; nutraceuticals).



A future system could provide formal mechanisms by which legally influential objectives could be set for achieving social and economic outcomes.

Some sectors might be actively encouraged in a future system by being offered policy or financial support, while others might be discouraged or even stopped. For example, a timeline could be established for phasing out oil and gas exploration or bottom contact fishing methods, a moratorium could be placed on deep seabed mining and novel activities like ocean fertilisation (see below), or a target set for ramping up offshore aquaculture. Some might be keen to pursue *diversification* of the blue economy. Still others might see value in utilising marine minerals that can both support the economy and be used in technologies designed to address the climate crisis (such as solar panels and electric vehicles).

A spotlight on ocean fertilisation

In 2008, parties to the London Convention and London Protocol adopted a Resolution in response to concerns over the potential adverse effects of large-scale ocean fertilisation on marine biodiversity. The Resolution defines ocean fertilisation as “any activity undertaken by humans with the principal intention of stimulating primary productivity by the oceans”.²⁰⁶

Ocean fertilisation has been promoted as a possible solution to the problem of increasing atmospheric carbon dioxide levels. It involves the dumping of iron compounds into the water column to stimulate phytoplankton growth in the surface layer of the oceans. Phytoplankton produces organic matter that absorbs carbon dioxide from the water column, establishing a gradient between the air and sea, that promotes uptake of atmospheric carbon dioxide by the ocean. The Resolution agrees that, given the present state of knowledge, ocean fertilisation should not be allowed other than for legitimate scientific research purposes.²⁰⁷ This demonstrates that some objectives may be constrained by international obligations, but it also highlights that even “environmental” objectives in the marine space can conflict.

Going even further, a future system might seek to transform entire industries (see spotlight below). For example, one objective might be to decarbonise coastal shipping by supporting biofuels and providing incentives for changes in ship design. Oil and gas could transition to offshore tidal or wind energy in a way that is planned and supported by the system over time. It is also interesting to consider the place of fishing and aquaculture.

A spotlight on the role of the system in transforming industries

One reasonably radical question may be whether we should be more assertive in shifting our reliance from inshore wild fisheries (where information is difficult to obtain and impacts are potentially severe) towards sustainable forms of aquaculture (in appropriate places and with appropriate controls). Would such a shift have the potential to restore and enhance the marine environment while creating high value products and supporting local communities near to where activities are located? Will people a millennium from now look back at our 21st century reliance on wild fisheries and compare it to a mechanised version of pre-agricultural hunter gatherer societies?²⁰⁸ After all, we do not rely on land-based hunting in that way. Is our obligation to future generations both to reverse marine degradation, and at the same time, to proactively invest in alternative food production methods that are less vulnerable to information gaps and collapse?

Such a shift is fairly speculative, of course, and our intention is by no means to recommend it. Aquaculture itself has a number of environmental impacts and risks, including with respect to biosecurity and climate change.²⁰⁹ There are also difficult questions of ethics around whether wild or farmed protein is preferable from an animal welfare perspective. “Hunting and gathering” might sound uncivilised, but our millennia long experiment with agriculture has hardly been an environmental success story either – would we be creating similar problems by extensively farming the sea? Commercial fishing provides many benefits and it can be done sustainably if controls and market incentives are in place. One fisher has put it elegantly, in that “if farming is the backbone of this country the fishing industry is what it floats on”.²¹⁰

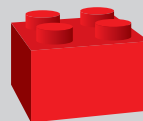
Yet the thought experiment serves to broaden the horizons of our thinking. It tests our worldviews. In particular, it requires us to think about how specific the system’s objectives should be. Is it the role of public authorities to use the system to plan industry transitions at all, or is this “economic” planning still as much an anathema as it was in the 1980s? Or does it depend on *how* public authorities do such things (either through regulation or softer incentives)? For example, a push to offshore energy could be pursued by providing additional financial incentives for projects, or alternatively by constraining the ability to get consent for land-based wind farms.

In the biosecurity context, regulation can already be very intrusive where there is an incursion triggering an emergency response (requiring, for example, the destruction of privately owned livestock

and crops, albeit with compensation). That can create immense hardship and apply to large areas, but it is an intervention that is seldom questioned. This is partly because it is for the overall *benefit* of the sector, not a way to transition *away* from the sector. It does beg the question, however: how deep does an *ecological* emergency need to become for similarly directive measures to be taken in the oceans?

Jared Diamond has pointed to an extreme example from the past, where there is evidence that tribal chiefs on the tiny Pacific island of Tikopia decided to slaughter every single pig on the island,²¹¹ because of the environmental impacts they were having and their inefficiency as a mode of food production. They did not bother assessing pig farm “consent applications”, or imposing conditions on them. Islanders simply shifted their source of protein elsewhere (fish and turtles); their objective morphed from environmental sustainability to a very specific course of action. While this relied on a system much more totalitarian than our own, and a population where each person likely knew each other individually, its small island context made it much more obvious to those living there that something needed to change. In our vast oceans, we cannot rely on the evidence of our own eyes and must instead use science.

Such questions about the proper role of the system are relevant to *any* objective that involves transitioning away from current industries, activities or practices, not just fishing. Deep seabed mining is one notable example where some are calling for an end to (or moratorium on) the activity, not just the management of its effects.²¹² That is a specific objective that goes beyond a general principle. Similarly, at least for now, we are well down the same path with respect to offshore petroleum extraction, where new permits will not be granted.²¹³ Climate change and Covid-19 are increasingly testing assumptions about the role of the system in setting quite specific objectives to be pursued through many means (including strict regulatory restrictions).²¹⁴ Is it appropriate for “planning”, rather than the market, to play a greater role in determining what the future uses of our marine space should be?



A future system could contain objectives relating to particular sectors or activities (eg whether to expand them or phase them out), reflecting a more interventionist approach to resource or economic planning in the marine environment.

Protecting the interests of mana whenua

Intimately linked to all of the above roles is a vitally important function: protecting the interests of mana whenua. This role is necessary to discharge te Tiriti obligations, but can also be seen as a broader role that is founded in the rights of indigenous peoples (eg under UNDRIP).²¹⁵ The role is about specifically recognising the Māori people as important actors in the management of marine resources, and a distinct community of interest within Aotearoa New Zealand.

An important objective of this role will be the defence of te Tiriti settlements. Notable in the marine space are settlements for fisheries and aquaculture, as well as the broader framework established under the MACA Act (although that is not a settlement, and not necessarily “full and final”). However, the objectives of the oceans management system here potentially extend well beyond redress for past grievances. The system must actively protect Māori, and (speaking generally) their taonga, from harm. That needs to encompass intangible impacts, such as effects on cultural wellbeing and ancestral connections. The system also needs to have effective mechanisms for applying traditional knowledge (mātauranga Māori).

At the same time, te Tiriti is as much about recognising Māori rights to manage and use resources as it is about protecting the environment from use (there is no binary distinction in te ao Māori). The system could be expected to enable Māori to undertake practices according to their own cultural and spiritual beliefs, and manage resources consistently with tikanga.

The system's role here colours the exercise of all its other roles, although there are also potential conflicts between them. For example, environmental limits that take spatial forms (eg permanent no-take protected areas) might run up against settlement rights and expectations (maintaining the value of fisheries quota) and the need in te ao Māori to retain ancestral connections to the moana through a degree of use. The pursuit of positive outcomes might be more targeted, such as in the support for a thriving *Māori* aquaculture industry rather than just an aquaculture industry per se. And – perhaps most significantly – the system's allocative role will be affected by te Tiriti obligations (see the discussion earlier).

Protecting the interests of mana whenua is not just about how the system's other roles are performed, however. It goes further, because it is not just about *what* the system must do (eg limit setting, allocation) but

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Ngāti Paoa waka, Karaka Bay, Auckland

also *how* it is done (eg *rāhui*) and, even more importantly, *who* should do it (eg *mana whenua* or co-governed entities). For instance, not only could the system be expected to promote *kaitiakitanga*, but also to respect Māori as *kaitiaki*. Such objectives are intimately linked to questions of institutional design, discussed in Chapter 12.

A particularly important question is how legislation articulates its objectives with respect to *te Tiriti o Waitangi*. As discussed in Chapter 3, current statutes refer to *te Tiriti* in many different ways. A future system could roll out more consistent *te Tiriti* clauses across all marine legislation, but exactly what they should say remains debatable. The Waitangi Tribunal, for instance, recommended that all persons acting under legislation “shall act in a manner that is consistent with the principles of the Treaty of Waitangi”.²¹⁶ The proposed NBA seems set to have an obligation to *give effect* to the principles of *te Tiriti*,²¹⁷ mirroring the existing provision in the Conservation Act. And it would be possible to refer to *te Tiriti* itself, rather than just principles that have been developed in the courts – although that may require the resolution of issues about which *version* or *interpretation* of *te Tiriti* is the right one, which has been neatly sidestepped by instead focusing on principles. All of these are potential options for the future.

Here, language matters. For instance, “giving effect” to the principles of *te Tiriti* has been held to require consideration to be given to priority access for *mana whenua* in the granting of concessions on conservation land.²¹⁸ This is because it is one way that *iwi* can strengthen ancestral connections with *te taiao*, and give practical effect to the principles of *te Tiriti*.²¹⁹

How a similarly strong direction might apply to the marine context in other legislation remains unclear but may be significant. It might, for example, have an impact on how coastal space is allocated (well beyond the context of aquaculture, where a settlement has been reached), and how the assimilative capacity of estuaries might be allocated (eg to allow development of Māori agricultural land in fully allocated catchments). It might influence the treatment of *mātauranga Māori* as evidence (eg in legislative requirements to use “best available information”), as well as how *tikanga* is used to interpret other legal provisions (as has been done under the MACA Act). If a broader range of MPAs were required to *give effect* to *te Tiriti* or its principles (as the Marine Reserves Act currently must),²²⁰ then that has significant implications for strategic decisions about where they go, what they can restrict, and whether they are temporary or permanent.²²¹ Stronger *te Tiriti* clauses might also have implications for how power sharing and co-governance tools are used. It is interesting to consider section 33 of the RMA here.

A spotlight on section 33 of the RMA

The power to transfer decision-making to *mana whenua* has existed since the inception of the RMA, but has been used only once.²²² The proposed NBA, and other marine legislation, could put more framing around when that type of mechanism *should* or *should not* be used in the marine context, rather than just leaving it in the toolbox to gather dust.

That is part of a bigger conversation about what the shift to “giving effect” to *te Tiriti* principles means under a new system. The Waitangi Tribunal has noted that control over *taonga* in the marine environment should shift from being a settlement-based grievance process to a proactive management process that occurs as of right.²²³ That does not mean that all powers should be transferred under something like section 33, but it does point to a need to clarify why the tool is there and how it will be used.

The Randerson Panel has envisaged that (in the context of the NBA) this would be addressed in the agreements negotiated between *iwi* and councils, and clear principles could be included in legislation to guide how that would work.²²⁴ There are a number of possibilities; for example, recognition of customary marine title under the MACA Act could bring with it some expectation of a transfer of powers under the NBA (eg to act as a consent authority, not just to interpose a veto power) or Fisheries Act (eg to impose binding *rāhui* to halt fishing), and ancestral connections with *taonga* species could see some powers transferred under conservation legislation.

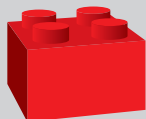


Umupuia marae

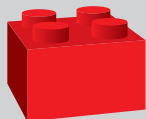
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An important lesson, perhaps, is that the system should tackle strategic issues about te Tiriti head-on and be more specific about what its objectives are (and how they will interact with other objectives). If it does not, it will likely be left to litigation in the courts to determine. It may not be immediately obvious what “giving effect to” or “complying with” te Tiriti means, but such general provisions could over time transform how a whole range of other parts of the system work.

An important cross-cutting role that a future system will need to perform is protecting the interests of mana whenua. This is not just about the outcomes that need to be achieved (eg protection of wāhi tapu sites) but also about *how* those outcomes are produced (eg processes that involve partnership or shared decision-making). The objectives the system has for te Tiriti o Waitangi, and how Treaty clauses are expressed, will be significant.



In performing all its other roles, the system could be expected to protect the interests of mana whenua.



Stronger and more consistent te Tiriti clauses could be deployed across marine legislation in the future. It could be made clearer what these clauses mean in practice and how te Tiriti objectives affect or interact with other objectives.

7.7 Concluding comments

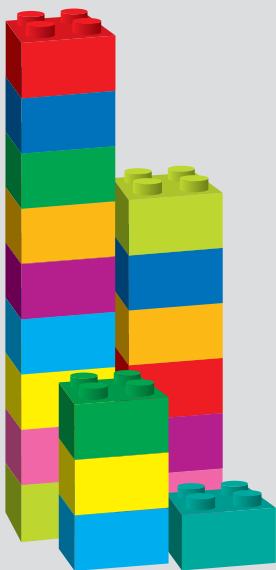
In this chapter we have considered what the normative foundations of a future system might be. The bedrock of a system will be the worldviews and ethics that underpin it. In the future, this is likely to be a mix of many different values, but there is potential for synergies between te ao Māori, ecocentrism and new approaches to green economics (eg the “doughnut” model) to steer us in a positive direction. That might see a softening of some of the neoliberal features that were established in the late 1980s and early 1990s.

Ethics need to be operationalised, and legal principles provide a way to do this. There are numerous choices on this front, not just in terms of the principles we adopt and how we express them in legislation, but also how we conceive of the relationships between them and how specific they are. At the more specific end of the scale, a future system could be more directive as to what it is seeking to achieve (and by when) for things like limit setting, making trade-offs, pursuing positive outcomes and allocating resources. Some objectives could, like in the context of climate change, be codified in legislation and an accountability framework established around them.

The recurrent theme in this chapter is that the current system is by no means clear about the different roles it is expected to perform and the objectives that should be driving them. In particular, it is not clear whether the current system adequately distinguishes limit setting from trade-offs, whether it has a legitimate role in determining how trade-offs are made above limits, and whether it is concerned with some forms of resource allocation.

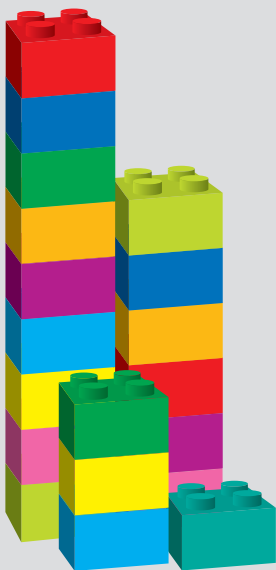
On the one hand, it is vital for accountability and action that reasonably specific objectives exist at the highest levels of legislation and are not treated just as politically driven policies. The biodiversity crisis, like the climate crisis, requires a predictable and planned pathway out of danger, not just management. And legislating for objectives gives them a stronger status and moral significance.²²⁵ That is equally the case when it comes to objectives relating to Māori, which in the past have tended to be relatively unclear and based on general clauses referencing the principles of te Tiriti. However, we also need to be wary of treating a particular solution as an objective in its own right, as it might create issues of path dependency and blind us to alternative options. MPAs are of particular interest here. Should we legislate a target for their deployment (eg for their coverage and location)? Or do we set general biodiversity objectives (eg maximum mortality of protected species) and allow all sorts of other tools (eg bycatch controls) to be deployed to achieve them?

Exactly what the system’s objectives might be is up for debate. So too are the tools we might use to achieve them, and that is what we now turn to in Chapter 8. As we do so, it is worth remembering the words of the late Bishop Manuhua Bennett, who described the Māori system of tikanga as “doing things right, doing things the right way, and doing things for the right reasons”.²²⁶ Our choice of tools – the right way of doing things – needs a strong ethical underpinning.



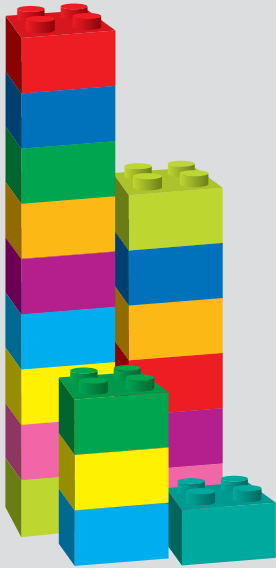
Summary of options for reform: Ethics, principles and objectives

- The normative foundation of a future system could be based on te ao Māori and its concepts and principles.
- The normative foundation of a future system could be based on a welfare economics view of the world, in which instrumental value is placed on the natural world as a collection of resources.
- A future system could be based on anthropocentrism, where the multifaceted interests and values of society are put at the forefront of decisions.
- A future system could be based on ecocentrism, where nature is recognised as having intrinsic value alongside humans, not just as a set of resources or serving human needs.
- The normative basis of a future system could be one in which synergies between te ao Māori and ecocentrism are placed at the heart of decision-making.
- Ecosystem-based management could form a core principle in a future system, expanding upon that of integrated management observable in frameworks like the RMA.
- Sustainable management could be recast as a broader concept of sustainability (eg te oranga o te taiao, or te mana o te moana), potentially embracing the social and economic dimensions of resource use and protection.
- The principle at the heart of fisheries management could be reframed from one of sustainable utilisation to one more like sustainable management or te oranga o te taiao at the core of frameworks like the RMA/NBA.
- A future system could seek to give effect to the principles of te Tiriti o Waitangi that have been developed in the courts, or recognise and adhere to te Tiriti itself.
- The normative core of a future system could be based on te ao Māori concepts such as kaitiakitanga, mana and mauri.
- Distributional equity or intra-generational equity could be expressly recognised as a principle in a future system, particularly to guide decisions about allocation.
- The principle of environmental justice could be strengthened in a future system, reflecting a broader understanding of the social elements of sustainability.
- There could be express recognition of ecological justice in a future system, embracing an ecocentric ethic and welcoming nature into human systems of justice.



Summary of options for reform: Ethics, principles and objectives *(continued)*

- Inter-generational equity could be strengthened in a future system by defining more specifically what the relative interests of current and future generations are, including with respect to restoring, enhancing and developing the marine space.
- A principle of procedural justice could be included or reflected more strongly in a future system, outlining common elements of all processes to ensure they are fair, including for mana whenua.
- A broader precautionary principle could be adopted at a more systemic and proactive level, including obligations to take positive action to enhance the resilience of the environment where future cumulative impacts are uncertain.
- A future system could provide more clarity as to what subsidiarity means in the marine environment, and when it is appropriate for decisions to be centralised or devolved to councils, mana whenua, or stakeholder groups.
- Principles in a future system could be made more specific and directive in legislation, giving greater clarity as to what outcomes are expected and less room for interpretation by policy makers and the courts.
- A future system could legislate for a much more specific set of objectives, including timeframes or milestones for achieving change.
- A future system could specifically define what an environmental limit is, and require such limits to be set for a defined list of things in the marine environment.
- A future system could be expected to guide more clearly how trade-offs are made between different forms of wellbeing above environmental limits. It could also reconceptualise the role from one of balancing things against each other to seeking win-win situations.
- The system could be expected to guide how rights to use or benefit from different resources are distributed.
- The system could be expected to guide how some existing rights might be reallocated to “better” uses or users over time.
- A future system could include an overarching set of allocative principles, which might look different for different resources. That could relate to the use to which resources are put, or which users can benefit from them. In particular, how the Tiriti provisions are worded will have implications for how resources and rights are allocated.



Summary of options for reform: Ethics, principles and objectives *(continued)*

- The system might take a more directive approach to allocation, where resources are reserved for uses or users that are seen as more deserving than others.
- The system could be reoriented to drive positive change more than it does at present. The concept of providing public goods and services could be broadened to include the active provision and protection of ecosystem services.
- A future system could legislate for measurable environmental enhancement objectives. These could reflect those contained in international fora (eg the Aichi biodiversity targets) and could relate to the deployment of particular tools, such as MPAs.
- A future system could provide formal mechanisms by which legally influential objectives could be set for achieving social and economic outcomes.
- A future system could contain objectives relating to particular sectors or activities (eg whether to expand them or phase them out), reflecting a more interventionist approach to resource or economic planning in the marine environment.
- In performing all its other roles, the system could be expected to protect the interests of mana whenua.
- Stronger and more consistent Te Tiriti clauses could be deployed across marine legislation in the future. It could be made clearer what these clauses mean in practice and how Te Tiriti objectives affect or interact with other objectives.

Endnotes

- 1 An aspiration contained in the government's aquaculture strategy.
- 2 For example, to protect specific places, safeguard particular species, enhance surrounding fisheries, or achieve overall biodiversity gains.
- 3 Dollar values for human life are possible, but they are generally descriptive (ie measuring value after the fact, eg by measuring traffic deaths against appetite for investment in safety measures) rather than normative (what a life ought to be worth).
- 4 Greg Severinsen and Raewyn Peart *Reform of the Resource Management System: The Next Generation Synthesis Report* (Environmental Defence Society, Auckland, December 2018) at 54.
- 5 See generally Robert Joseph and others *Stemming the Colonial Tide: Shared Māori Governance Jurisdiction and Ecosystem-Based Management over the Marine and Coastal Seascape in Aotearoa New Zealand – Possible Ways Forward* (Ko Ngā Moana Whakauka and Te Mata Hautū Taketake – the Māori and Indigenous Governance Centre, Waikato, 2020).
- 6 Hirini Moko Mead *Tikanga Māori: Living by Māori Values* (revised edition, Huia Publishers, Wellington, 2016) at 14.
- 7 Ulrich Klein, "Belief-Views on Nature – Western Environmental Ethics and Māori World Views", (2002) 4 NZJEL L 81 at 104-106.
- 8 Robert Joseph and others *Stemming the Colonial Tide: Shared Māori Governance Jurisdiction and Ecosystem-Based Management over the Marine and Coastal Seascape in Aotearoa New Zealand – Possible Ways Forward* (Ko Ngā Moana Whakauka and Te Mata Hautū Taketake – the Māori and Indigenous Governance Centre, Waikato, 2020) at 49.
- 9 Benjamin Dimitrios Hanara and Anne-Marie Jackson *Tangaroa Ara Rau: Tangaroa the Atua of Human Movement* (Ngā Pae o te Māramatanga, ngā Ākonga Report 18INT01, 2019) at 3 (citations omitted).
- 10 Te Ahukaramū Charles Royal "Tangaroa – the sea - The importance of the sea" (12 June 2006) Te Ara - the Encyclopedia of New Zealand <www.TeAra.govt.nz/en/tangaroa-the-sea/page-1>.
- 11 Te Ahukaramū Charles Royal "Tangaroa – the sea - The importance of the sea" (12 June 2006) Te Ara - the Encyclopedia of New Zealand <www.TeAra.govt.nz/en/tangaroa-the-sea/page-1>.
- 12 Benjamin Dimitrios Hanara and Anne-Marie Jackson *Tangaroa Ara Rau: Tangaroa the Atua of Human Movement* (Ngā Pae o te Māramatanga, ngā Ākonga Report 18INT01, 2019) at 7.
- 13 Benjamin Dimitrios Hanara and Anne-Marie Jackson *Tangaroa Ara Rau: Tangaroa the Atua of Human Movement* (Ngā Pae o te Māramatanga, ngā Ākonga Report 18INT01, 2019) at 5.
- 14 See the story of Tinirau at Te Ahukaramū Charles Royal "Tangaroa – the sea - Tinirau and Kae" (12 June 2006) Te Ara - the Encyclopedia of New Zealand <www.TeAra.govt.nz/en/tangaroa-the-sea/page-3>.
- 15 Gerard Hutching and Carl Walrond "Marine conservation - Protected areas" (12 June 2006, updated 1 September 2015) Te Ara - the Encyclopedia of New Zealand <www.teara.govt.nz>.
- 16 Robert Joseph and others *Stemming the Colonial Tide: Shared Maori Governance Jurisdiction and Ecosystem-Based Management over the Marine and Coastal Seascape in Aotearoa New Zealand – Possible Ways Forward* (Ko Ngā Moana Whakauka and Te Mata Hautū Taketake – the Māori and Indigenous Governance Centre, Waikato, 2020) at 50.
- 17 Hirini Moko Mead *Tikanga Māori: Living by Māori Values* (revised edition, Huia Publishers, Wellington, 2016) at 16.
- 18 Richard L Revesz *Foundations of Environmental Law and Policy* (Oxford University Press, New York 1997) at 3.
- 19 See Tim Hazledine "Economics and the resource management system" in Greg Severinsen and Raewyn Peart *Reform of the Resource Management System: The Next Generation – Working Paper 3* (Environmental Defence Society, Auckland, September 2018).
- 20 Enhancing some aspects of the environment (such as key fisheries habitat) might yield greater overall welfare for people than others (such as threatened species habitats).
- 21 RH Coase "The problem of social cost" (1960) 3 Journal of Law and Economics 1. The Coase theorem has since been developed mostly by other economists, based on a portion of Coase's article.
- 22 See Garrett Hardin "The Tragedy of the commons" (1968) 162 Science 1243.
- 23 See, for example, the description of property rights in Marilyn Bramley and Jeff McNeill "Up the creek and down the river: In-stream ecological values and property rights under the RMA" in Klaus Bosselmann and Vernon Tava (eds) *Water rights and sustainability* (New Zealand Centre for Environmental Law, Auckland, 2011) at 175.
- 24 See the economist's conception of wellbeing in Michael Pickford "Economic efficiency and the Resource Management Act" (2014) 18 NZJEL 149 at 153.
- 25 A limitation that Coase recognised explicitly.
- 26 On the myth of the rational actor, see David Orrell *Economyths: 11 Ways Economics Gets It Wrong* (Icon Books, London, 2010).
- 27 Lester B Lave "The Strategy of Social Regulation: Decision Frameworks for Policy" in Richard L Revesz *Foundations of Environmental Law and Policy* (Oxford University Press, New York 1997) at 85.
- 28 Kate Raworth *Doughnut Economics: Seven Ways to Think Like a 21st-Century Economist* (Random House, London, 2017).
- 29 Mark Sagoff "The Economy of the Earth: Philosophy, Law and the Environment" in Richard L Revesz *Foundations of Environmental Law and Policy* (Oxford University Press, New York 1997).
- 30 See Tim Denne "Resource management law reform and economics" in Greg Severinsen and Raewyn Peart *Reform of the Resource Management System: The Next Generation – Working Paper 3* (Environmental Defence Society, Auckland, September 2018). Mostly the developments are attributed to Jeremy Bentham and John Stuart Mill: see John Kenneth Galbraith *A History of Economics: The Past as the Present* (Penguin Books, London, 1987). Robert L Heilbroner *The Worldly Philosophers: The Lives, Times and Ideas of the Great Economic Thinkers* (7th ed, Simon & Schuster, New York 1953).
- 31 See the creative interpretations of the EEZ Act's te Tiriti clause in *Trans-Tasman Resources Ltd v Taranaki-Whanganui Conservation Board* [2020] NZCA 86.
- 32 See Klaus Bosselmann *The principle of sustainability: Transforming law and governance* (Ashgate Publishing, Surrey, 2008) at 49; Robyn Eckersley *The Green State: Rethinking Democracy and Sovereignty* (MIT Press, Cambridge, 2004). at 141.
- 33 See generally Christopher Stone "Should trees have standing? Towards legal rights for natural objects" (1972) 45 S Cal LR 450; and John Kleinig *Valuing Life* (Princeton University Press, Princeton, 1991).; Robyn Eckersley *The Green State: Rethinking Democracy and Sovereignty* (MIT Press, Cambridge, 2004).
- 34 Tamar Stelling "Do lobsters and other invertebrates feel pain? New research has some answers" *The Washington Post* (online ed, Washington DC, March 10 2014).
- 35 Paul W Taylor "Respect for nature: A theory of environmental ethics" (Princeton University Press, Princeton, 1986) in Richard L Revesz *Foundations of Environmental Law and Policy* (Oxford University Press, New York 1997) at 32.
- 36 See Christopher Stone "Should trees have standing? Towards legal rights for natural objects" (1972) 45 S Cal LR 450.
- 37 Bob Jones "Anti-whaling outcry simply sizeism" *NZ Herald* (online ed, 7 August 2012).
- 38 Minister for Oceans and Fisheries "On-board cameras across the inshore fishing fleet" (2 July 2021).
- 39 See Ministry of Foreign Affairs and Trade "Joint statement against whaling" (media statement, 18 December 2017)..
- 40 Department of Conservation *The conservation of whales in the 21st century* (2004).
- 41 Raewyn Peart *Dolphins of Aotearoa: Living with New Zealand dolphins* (Craig Potton Publishing, Nelson, 2013).
- 42 Thomas White *In defense of dolphins: The new moral frontier* (Blackwell Publishing, Malden, 2007).
- 43 See Lori Cuthbert and Douglas Main "Orca Mother Drops Calf, After Unprecedented 17 Days of Mourning" *National Geographic* (online ed, Washington DC, 13 August 2018).
- 44 The latest estimate was 104 common dolphin captures in trawl fisheries for the 2014-15 fishing year. See ER Abraham and K Berkenbusch "Estimated captures of New Zealand fur seal, New Zealand sea lion, common dolphin, and turtles in New Zealand commercial fisheries, 1995-96 to 2014-15" (Ministry for Primary Industries, New Zealand Aquatic Environment and Biodiversity Report No 188, November 2017).
- 45 It is not a universal view; for example, some iwi have expressed an interest in harvesting marine mammals if and when the populations are at suitable levels.
- 46 See Klaus Bosselmann "The Concept of Sustainable Development" in Klaus Bosselmann, David Grinlinton and Prue Taylor (eds) *Environmental Law for a Sustainable Society* (2nd ed, New Zealand Centre for Environmental Law, Auckland 2013). at 101; Mark Christensen "Valuation of natural assets under the Resource Management Act" (2013) 17 NZJEL 291 at 298.
- 47 Ministry for the Environment and Statistics New Zealand *Our marine environment 2019* (ME 1468, October 2019) at 22.
- 48 David Young *Values as law: The history and efficacy of the Resource Management Act* (Victoria University of Wellington Institute of Policy Studies, Wellington, 2001) at 85.
- 49 SRobert Joseph and others *The Treaty, Tikanga Māori, Ecosystem-based Management, Mainstream Law and Power Sharing for Environmental Integrity in Aotearoa New Zealand – Possible Ways Forward* (Ko Ngā Moana Whakauka and Te Mata Hautū Taketake – the Māori and Indigenous Governance Centre, Waikato, 2019).

- 50 Robert Joseph and others *The Treaty, Tikanga Māori, Ecosystem-based Management, Mainstream Law and Power Sharing for Environmental Integrity in Aotearoa New Zealand – Possible Ways Forward* (Ko Ngā Moana Whakauka and Te Mata Hautū Taketake – the Māori and Indigenous Governance Centre, Waikato, 2019). Joseph points out that we need to take care here, given that Māori decision-makers are not only driven by traditional tikanga values; Māori corporations are increasingly powerful and have economic drivers.
- 51 Ulrich Klein, “Belief-Views on Nature – Western Environmental Ethics and Māori World Views”, (2002) 4 NZJEL L 81 at 104-106.
- 52 Christopher Stone “Should trees have standing? Towards legal rights for natural objects” (1972) 45 S Cal LR 450.
- 53 Te Ahukaramū Charles Royal (ed) *The Woven Universe: Selected writings of Rev Māori Marsden* (The estate of Rev Māori Marsden, Ōtaki, 2003) at 46 and 69.
- 54 Joseph Williams “Lex Aotearoa: An Heroic Attempt to Map the Maori Dimension in Modern New Zealand Law” (2013) 21 Wai L Rev 1.
- 55 See National Oceanic and Atmospheric Administration (NOAA) (6 December 2021) “Ecosystem-Based Management” National Marine Ecosystem Status <www.ecowatch.noaa.gov/about/>
- 56 See Sustainable Seas National Science Challenge “Why do we need EBM?” <www.sustainableseaschallenge.co.nz>
- 57 Judi Hewitt and others “Proposed ecosystem-based management principles for New Zealand” (2018) November RM Journal 10.
- 58 Raewyn Peart, Alison Greenaway and Lara Taylor “Enabling Ecosystem-Based Management: Is Aotearoa New Zealand’s Legal Framework up to the Task?” (2019) 23 NZJEL 31 at 64.
- 59 Lara Taylor, Tania Te Whenua and Bonny Hatami *How current legislative frameworks enable customary management & ecosystem-based management in Aotearoa New Zealand – the contemporary practice of rāhui* (Sustainable Seas National Science Challenge, EBM Discussion Paper, April 2018).
- 60 Robert Joseph and others *Stemming the Colonial Tide: Shared Māori Governance Jurisdiction and Ecosystem-Based Management over the Marine and Coastal Seascape in Aotearoa New Zealand – Possible Ways Forward* (Ko Ngā Moana Whakauka and Te Mata Hautū Taketake – the Māori and Indigenous Governance Centre, Waikato, 2020) at 490-491. at 490-491.
- 61 Well-being of Future Generations (Wales) Act 2015 (UK).
- 62 See cl 8.
- 63 Nicholas Robinson “Re-conceptualising Sustainability: The Anthropocene Agenda” (2015) 11(7) RM Theory & Practice 99 at 108; Melinda Harm Benson and Robin Kundis Craig “The End of Sustainability” (2014) 27 Society & Natural Resources 777 at 778. at 778.
- 64 “The purpose of this Act is to provide for the utilisation of fisheries resources while ensuring sustainability”. See *Recreational Fishing Council Inc v Sanford Limited* [2009] NZSC 54 , [2009] 3 NZLR 438 at [30].
- 65 Fisheries Act 1996, s 13(2)(a).
- 66 UNCLOS, art 61(3). They must also consider cumulative effects on non-target species with a view to maintaining or restoring populations of such species to levels at which their reproduction is not seriously threatened.
- 67 UNCLOS, art 62(2).
- 68 Raewyn Peart *Voices from the Sea: Managing New Zealand’s Fisheries* (Environmental Defence Society, Auckland, 2018) at 33.
- 69 Raewyn Peart *Voices from the Sea: Managing New Zealand’s Fisheries* (Environmental Defence Society, Auckland, 2018).
- 70 *New Zealand Recreational Fishing Council & Ors v Sanford Limited* [2009] NZSC 45, [2009] 3 NZLR 438. at [21].
- 71 Raewyn Peart *Voices from the Sea: Managing New Zealand’s Fisheries* (Environmental Defence Society, Auckland, 2018) at 34.
- 72 Ministry of Fisheries *Operational Guidelines for New Zealand’s Harvest Strategy Standard* (2011) at 3.
- 73 Raewyn Peart *Voices from the Sea: Managing New Zealand’s Fisheries* (Environmental Defence Society, Auckland, 2018) at 37.
- 74 Fisheries New Zealand *The Status of New Zealand’s Fisheries 2020* (February 2021).
- 75 See *Royal Forest and Bird Protection Society v Minister of Fisheries* [2021] NZHC 1427.
- 76 LegaSea *Rescue Fish Ika Rauora: A pathway to fish abundance and marine ecosystem recovery* (May 2020) at 18.
- 77 That might not be desirable if a single stock is the focus. For example, sometimes the abundance of some species might be too high for ecosystem stability (eg kina in the Hauraki Gulf) or for effective operation of the market (eg where snapper is abundant and the inability to obtain ACE cover for it prevents fishers from obtaining species that are often caught with it).
- 78 Philip Wheeler and others *Sustainable fisheries: Tiakina nga taonga Tangaroa, Report of the Fisheries Task Force to the Minister of Fisheries on the review of fisheries legislation* (Fisheries Task Force, Wellington, 1992).
- 79 Ministry for the Environment *Natural and Built Environments Bill* (Exposure Draft, 2021), cl 8; Ministry for the Environment *Natural and Built Environments Bill Parliamentary Paper on the Exposure Draft* (June, 2021).
- 80 Joseph Williams “He Aha te Tikanga Māori?” (paper presented to Mai I Te Ata Hāpara Hui, Te Wānanga o Raukawa, Ōtaki, August 2000) at 9. Law Commission *Māori Custom and Values in New Zealand Law* (NZLC SP9, 2001) at 28-40.
- 81 Carwyn Jones “A Māori Constitutional Tradition” (2014) 12 NZJPI 187 at 190; Joseph Williams, “Lex Aotearoa: An Heroic Attempt to map the Maori Dimension in Modern New Zealand Law” [2013] 21 Wai L Rev 1 at 2-5; Linda Te Aho “Tikanga Maori, historical context and the interface with Pakeha law in Aotearoa/New Zealand” [2007] NZ Ybk NZ Jur 4 at 11.
- 82 Robert Joseph and others *Stemming the Colonial Tide: Shared Māori Governance Jurisdiction and Ecosystem-Based Management over the Marine and Coastal Seascape in Aotearoa New Zealand – Possible Ways Forward* (Ko Ngā Moana Whakauka and Te Mata Hautū Taketake – the Māori and Indigenous Governance Centre, Waikato, 2020) at 55—56.
- 83 Toni Love “Incorporating Māori Approaches to Ecosystem Management in Marine Management” (2018) July Māori LR, as cited in Robert Joseph and others *Stemming the Colonial Tide: Shared Māori Governance Jurisdiction and Ecosystem-Based Management over the Marine and Coastal Seascape in Aotearoa New Zealand – Possible Ways Forward* (Ko Ngā Moana Whakauka and Te Mata Hautū Taketake – the Māori and Indigenous Governance Centre, Waikato, 2020) at 207.
- 84 Mason Durie *Te Mana, Te Kawanatanga – The Politics of Māori Self-Determination* (Oxford University Press, Melbourne, 1998), as cited in Robert Joseph and others *Stemming the Colonial Tide: Shared Māori Governance Jurisdiction and Ecosystem-Based Management over the Marine and Coastal Seascape in Aotearoa New Zealand – Possible Ways Forward* (Ko Ngā Moana Whakauka and Te Mata Hautū Taketake – the Māori and Indigenous Governance Centre, Waikato, 2020) at 208.
- 85 Aquaculture Steering Group *He Maara Mataitai – Ngangaru Ana* (date unknown).
- 86 Ministry for the Environment *Natural and Built Environments Bill*, cl 5.
- 87 Greg Severinsen and Raewyn Peart *Reform of the Resource Management System: The Next Generation – Synthesis Report* (Environmental Defence Society, Auckland, December 2018) at 71.
- 88 *Huakina Development Trust v Waikato Valley Authority* [1987] 2 NZLR 188 (HC) at 210.
- 89 See, for example, Waitangi Tribunal *Ko Aotearoa Tenei: A Report into Claims Concerning New Zealand Law and Policy Affecting Māori Culture and Identity* (Wai 262, 2011).
- 90 *Trans-Tasman Resources Ltd v Taranaki-Whanganui Conservation Board* [2020] NZCA 86. See Matthew SR Palmer *The Treaty of Waitangi in New Zealand’s law and constitution* (Victoria University Press, Wellington, 2008). On the first detailed formulation of the principles, see *New Zealand Māori Council v Attorney General* [1987] 1 NZLR 641 (CA) [Lands Case].
- 91 Waitangi Tribunal *The Radio Spectrum Management and Development Final Report* (Wai 776, 1999) at 38-39.
- 92 Waitangi Tribunal *Te Maunga Railways Land Report* (Wai 315, 1994) at 67-68.
- 93 See Richard Boast “The Treaty of Waitangi and environmental law” in Rob Harris (ed) *Handbook of environmental law* (Royal Forest and Bird Protection Society of New Zealand, Wellington, 2004) at 517.
- 94 David R Keller (ed) *Environmental ethics: The big questions* (Wiley-Blackwell, Chichester (West Sussex), 2010) at 418.
- 95 Ezekiel Hudspeth “Freshwater Management in New Zealand: a Challenge for Ecology, Equity and Economic Efficiency” (2013) 16 NZJEL 277 at 285.
- 96 Kenneth Palmer “The Resource Management Act 1991” in Derek Nolan (ed) *Environmental and Resource Management Law* (5th ed, LexisNexis, Wellington, 2015). at 111-112; World Commission on Environment and Development *Our common future* (Oxford University Press, Oxford, 1987) at 43.
- 97 Ministry for the Environment *Report of the review group on the Resource Management Bill* (1991) at [3.3]; *St Columba’s Environmental House Group v Hawkes Bay Regional Council* [1994] NZRMA 560 (PT) at 528.
- 98 See Greg Severinsen and Raewyn Peart *Reform of the Resource Management System: The Next Generation – Synthesis Report* (Environmental Defence Society, Auckland, December 2018) at 114.
- 99 Fisheries New Zealand “Undue adverse effects test for marine farms” (5 July 2021) Ministry for Primary Industries <www.mpi.govt.nz>
- 100 On indigenous environmental justice, see Meg Parsons, Karen Fisher and Roa Petra Crease

- De-colonising blue spaces in the Anthropocene: Freshwater management in Aotearoa New Zealand* (Palgrave Macmillan, Cham (Switzerland), 2021).
- 101 On infrastructure failures, underinvestment and its impacts, see New Zealand Productivity Commission *Local government funding and financing* (November 2019); Cabinet Minute of Decision "Strengthening the Regulation of Drinking Water, Wastewater and Stormwater" (1 July 2019) CAB-19-MIN-0332.
- 102 See generally Paul W Taylor *Respect for nature: A theory of environmental ethics* (Princeton University Press, Princeton, 1986) in Richard L. Revesz *Foundations of environmental law and policy* (Oxford University Press, New York, 1997); Christopher Stone "Should trees have standing? Towards legal rights for natural objects" (1972) 45 S Cal LR 450; Robyn Eckersley *Environmentalism and political theory: Toward an ecocentric approach* (State University of New York Press, New York, 1992).
- 103 See Greg Severinsen and Raewyn Peart *Reform of the Resource Management System: The Next Generation – Synthesis Report* (Environmental Defence Society, Auckland, December 2018) at 58.
- 104 See Meg Parsons, Karen Fisher and Roa Petra Crease *De-colonising blue spaces in the Anthropocene: Freshwater management in Aotearoa New Zealand* (Palgrave Macmillan, Cham (Switzerland), 2021).
- 105 See generally Edith Brown Weiss *In Fairness to Future Generations* (United Nations University Press, Tokyo, 1989); Edith Brown Weiss "Intergenerational Justice and International Law" in Salvino Busuttil and others (eds) *Our Responsibilities to Future Generations* (Foundation for International Studies, Malta, 1990); Klaus Bosselmann *The principle of sustainability: Transforming law and governance* (Ashgate Publishing, Surrey, 2008).
- 106 Kenneth Palmer "Origins and Guiding Ideas of Environmental Law" in Klaus Bosselmann, David Grinlinton and Prue Taylor (eds) *Environmental Law for a Sustainable Society* (2nd ed, New Zealand Centre for Environmental Law, Auckland 2013); Resource Management Act 1991, s 5(2)(a); Conservation Act 1987, ss 2 and 6(c).
- 107 See Gerd Winter "A fundament and two pillars: The concept of sustainable development 20 years after the Brundtland Report" in Hans Christian Bugge and Christina Voigt (eds) *Sustainable development in international and national law* (Europe Law Publishing, Groningen (Neds), 2008) at 28.
- 108 Klaus Bosselmann *The principle of sustainability: Transforming law and governance* (Ashgate Publishing, Surrey, 2008) at 34.
- 109 Edith Brown Weiss "Intergenerational Justice and International Law" in Salvino Busuttil and others (eds) *Our Responsibilities to Future Generations* (Foundation for International Studies, Malta, 1990), from 98.
- 110 The gift can refer to any relevant oral tradition/customary practice.
- 111 See generally David Miller "Justice" (26 June 2017) Stanford Encyclopedia of Philosophy <<https://plato.stanford.edu/entries/justice/#ProcVersSubsJust>> at [2.3].
- 112 See, for example, Ben France-Hudson "The Kermadec/Rangitāhua Ocean Sanctuary: Expropriation-free but a breach of good faith" [2016] RM Theory & Practice 55.
- 113 Waitangi Tribunal *Ko Aotearoa Tenei: A Report into Claims Concerning New Zealand Law and Policy Affecting Māori Culture and Identity* (Wai 262, 2011).
- 114 *Murray v Whakatane District Council* [1999] 3 NZLR 276 (HC) at 46.
- 115 See D Sheppard "The Resource Management Act – from principles to practice" in Trevor Daya-Winterbottom (ed) *Frontiers of resource management law* (Thomson Reuters, Wellington, 2012) at 226.
- 116 For example, cross-boundary disputes that are non-notified.
- 117 Barry Barton, Kimberley Jane Jordan and Greg Severinsen *Carbon capture and storage: Designing the legal and regulatory framework for New Zealand* (Centre for Environmental, Energy and Resources Law, University of Waikato, 2013) at 78.
- 118 See Chapter 8 of the project's working paper on science and information in the fisheries system.
- 119 *Wellington International Airport v Air New Zealand* [1993] 1 NZLR 671 (CA) at 675.
- 120 See Meg Parsons and Lara Taylor "Why Indigenous knowledge should be an essential part of how we govern the world's oceans" *The Conversation* (online ed, 8 June 2021).
- 121 New Zealand Government "Kermadec Ocean Sanctuary Bill introduced" (press release, 9 March 2016)
- 122 PEW "Global Ocean Legacy" (6 July 2018) <<https://www.pewtrusts.org/en/projects/archived-projects/global-ocean-legacy>>
- 123 Ministry for the Environment *Departmental disclosure statement: Kermadec Oceans Sanctuary Bill* (26 February 2016).
- 124 Cabinet Paper "Establishment of a Kermadec Ocean Sanctuary" (1 September 2015, Cabinet Economic Growth and Infrastructure Committee).
- 125 Kermadec Ocean Sanctuary Bill 2016 (120-2), cl 3.
- 126 At the time of writing, we understand that an announcement on the future of the sanctuary is imminent.
- 127 Ben France-Hudson "The Kermadec/Rangitāhua Ocean Sanctuary: Expropriation-free but a breach of good faith" [2016] RM Theory & Practice 55 at 61.
- 128 The legal property right conferred by fisheries quota is a proportion of the TACC rather than a set amount of fish, so the quota right is not technically extinguished by this measure (it simply could not be utilised until the TACC was increased). But a proportion of zero remains zero, effectively rendering the quota of no value (and in perpetuity assuming the Ocean Sanctuary is a permanent feature). The practical effect is largely the same.
- 129 When the QMS was introduced, nominal quota was created for FMA10 in the event that a commercial fishery was established there. The reasons for the lack of development of a commercial fishery in FMA10 include the presence of the marine reserve in the territorial sea which prevents a coastal fishery, the benthic protection area which prevents bottom trawling within the EEZ and the distance of the islands from the mainland. A small part of the catch of high migratory species (4 percent) such as bigeye tuna, swordfish and moonfish was being taken from within FMA10, but as the QMA for these species includes the entire country's EEZ, they could also be caught outside the Sanctuary area by quota owners and would not be "extinguished" in the same way.
- 130 16 percent is held by Te Ohu Kaimoana, and has yet to be distributed to iwi: see Cabinet Paper "Establishment of a Kermadec Ocean Sanctuary" (1 September 2015, Cabinet Economic Growth and Infrastructure Committee); Kate Allan "A Kermadec/Rangitāhua Ocean Sanctuary: Issues and insights into marine protection processes" (LLB (Hons) Dissertation, Victoria University of Wellington, 2017).
- 131 Ben France-Hudson "The Kermadec/Rangitāhua Ocean Sanctuary: Expropriation-free but a breach of good faith" [2016] Resource Management Theory and Practice 55 at 78.
- 132 Ministry for the Environment *A new Marine Protected Areas Act: Consultation document* (ME 1224, 1 January 2015).
- 133 Toni Love "The Kermadecs conundrum: Marine protected areas and democratic process" (2017) 13(2) Policy Quarterly 17 at 20.
- 134 See United Nations Framework Convention on Climate Change 1771 UNTS 107 (signed 9 May 1992, entered into force 21 March 1994), art 3.3: "Where there are threats of serious or irreversible damage, lack of full scientific certainty should not be used as a reason for postponing such measures."
- 135 Which in turn emerged from the less ambitious principle of "no harm" and the associated duty to avoid "transboundary harm", which developed as interstate obligations rather than environmental protections per se.
- 136 *Sustain Our Sounds Inc v New Zealand King Salmon Co* [2014] NZSC 40, [2014] 1 NZLR 673.
- 137 Fisheries Act 1996, s 10; EEZ Act, ss 34, 61.
- 138 See *Trans-Tasman Resources Ltd v Taranaki-Whanganui Conservation Board* [2020] NZCA 86; Environmental Protection Authority *Decision on Marine Consent Application: Chatham Rock Phosphate Limited: to mine phosphorite nodules on the Chatham Rise* (February 2015).
- 139 See cl 15.
- 140 See generally Greg Severinsen "Letting our standards slip? Precaution and the standard of proof under the Resource Management Act 1991" (2014) 18 NZJEL 173; Catherine Iorns Magallanes and Greg Severinsen "Diving in the deep end: Precaution and seabed mining in New Zealand's exclusive economic zone" (2015) 13 NZJPL 201.
- 141 Catherine Iorns Magallanes "The precautionary principle in the New Zealand Fisheries Act: Challenges in the New Zealand Court of Appeal" (Victoria University of Wellington Legal Research Paper 59, May 2014).
- 142 Compare Allison Arthur-Young "Climate Change" in Derek Nolan (ed) *Environmental and Resource Management Law* (5th ed, LexisNexis, Wellington, 2015). at 1150.
- 143 See generally Marilyn Bramley "Institutional and governance structures of environmental law" in Peter Salmon and David Grinlinton (eds) *Environmental law in New Zealand* (1st ed, Thomson Reuters, Wellington, 2015) at 398.
- 144 See Resource Management Act 1991, pt 6A, ss 25A, 43-58A, 62(3), 67(3), 75(3).
- 145 Resource Management Act 1991, sch 1, cl 27.
- 146 In that regional councils do not have powers under it.
- 147 Marine and Coastal Area (Takutai Moana) Act 2011, s 68.
- 148 Such as under section 33 of the RMA, or fisheries plan making under the Fisheries Act.
- 149 Resource Management Act 1991, s 7(b); Ezekiel Hudspith "Freshwater management in New Zealand" (2013) 16 NZJEL 277 at 284.
- 150 See Joseph Sax "The Public Trust Doctrine in Natural Resource Law: Effective Judicial

- Intervention” (1970) 68 Mich L Rev 471.
- 151 Some, such as a direction to “avoid” effects, have a more directive approach and can operate as “limits”. See *Environmental Defence Society Inc v New Zealand King Salmon Co Ltd* [2014] NZSC 38, [2014] 1 NZLR 593.
- 152 Although litigation underway at the time of writing is testing whether the objectives in the Act are aspirations or duties.
- 153 Greg Severinsen and Raewyn Peart *Reform of the Resource Management System: The Next Generation – Synthesis Report and Next Steps (Summary for Policy Makers)* (Environmental Defence Society, Auckland, February 2019).
- 154 See Inga Carlman “The Resource Management Act 1991 Through External Eyes” (2007) 11 NZJEL 181.
- 155 Greg Severinsen and Raewyn Peart *Reform of the Resource Management System: The Next Generation Synthesis Report* (Environmental Defence Society, Auckland, December 2018) at 94.
- 156 See Raewyn Peart *Voices from the Sea: Managing New Zealand's Fisheries* (Environmental Defence Society, Auckland, 2018).
- 157 The “floor” for a stock, reflected in the concept of soft and hard limits in the Harvest Strategy Standard, does not have a statutory basis and its legal influence could be clearer (see Chapter 8).
- 158 For example, the Auckland Council decided against taking measures to control cats in areas containing threatened species vulnerable to cats, which may be relevant in the future when it comes to measures to prevent toxoplasmosis in the Māui dolphin.
- 159 See Chapter 8 on the role of the Harvest Strategy Standard in determining rebuild times under the Fisheries Act.
- 160 See *Royal Forest and Bird Protection Society v Minister of Fisheries* [2021] NZHC 1427.
- 161 New Zealand Companion Animal Council *Companion Animals in New Zealand 2016* (June 2016).
- 162 See Submarine Cables and Pipelines Protection Act.
- 163 See cl 7.
- 164 Partly reflecting that the tools in the RMA to achieve that limit rely on preventing activities rather than requiring restoration. Other mechanisms, including government funding, are intended to restore and enhance degraded wetlands.
- 165 Unless overall pressure is reduced by amending the TAC.
- 166 The NZCPS essentially imposes “limits” on natural character through strong “avoidance” policies, which were recognised by the Supreme Court in *King Salmon*.
- 167 *Trans-Tasman Resources Ltd v Taranaki-Whanganui Conservation Board* [2020] NZCA 86 at [221].
- 168 See the spotlight on the case of Tarakihi and the Harvest Strategy Standard in Chapter 8.
- 169 Tim Denne “Resource management law reform and economics” in Greg Severinsen and Raewyn Peart *Reform of the Resource Management System: The Next Generation – Working Paper 3* (Environmental Defence Society, Auckland, September 2018).
- 170 See New Zealand Productivity Commission *Better Urban Planning: Final Report* (February 2017) at 258, citing Caroline Miller *Culture and Capability within the New Zealand Planning System* (New Zealand Productivity Commission, December 2015); *Report of the Minister for the Environment's Resource Management Act 1991 Principles Technical Advisory Group* (CR 129, 1 July 2012) from 35.
- 171 Compare the oft cited words of Hon Simon Upton upon the second reading of the Bill that would become the RMA: (4 July 1991) 516 NZPD 3019–3020.
- 172 See New Zealand Productivity Commission *Better urban planning: Final Report* (February 2017) at 122, citing the submission of Caroline Miller *Culture and Capability within the New Zealand Planning System* (New Zealand Productivity Commission, December 2015).
- 173 How value-based trade-offs are to be made under the Fisheries Act has also been left unclear; the concept of utilisation in the purpose of the Act embraces social, economic and cultural wellbeing, but in practice controls on catch are driven by biological considerations rather than the broader values of communities.
- 174 Ministry for the Environment *Natural and Built Environments Bill* (Exposure Draft, 2021), cl 8; Ministry for the Environment *Natural and Built Environments Bill Parliamentary Paper on the Exposure Draft* (June, 2021).
- 175 Greg Severinsen and Raewyn Peart *Reform of the Resource Management System: The Next Generation – Synthesis Report and Next Steps* (Summary for Policy Makers) (Environmental Defence Society, Auckland, February 2019) at 106. Compare Inga Carlman “The Resource Management Act through external eyes” (2007) 11 NZJEL 181.
- 176 Although the courts have hinted that the merits of applications close in time may sometimes be compared, on the understanding that one of them might better promote sustainable management; see *Central Plains Water Trust v Synlait Ltd* [2009] NZCA 609, [2010] 2 NZLR 363 (CA) at [89]; *Synlait Ltd v Central Plains Water Trust* [2010] NZSC 32, [2010] NZRMA 257; *Ngai Tahu Property Ltd v Central Plains Water Trust* [2009] NZSC 24.
- 177 Mining proposals on the Chatham Rise were not consented in the end, partly due to the presence of a benthic protection area, but there is no legislative requirement or even clear link for that to have to be considered under the EEZ Act.
- 178 This term is deliberate. It is not just the Crown that could do such things. Depending on institutional arrangements, this could involve partnership with mana whenua.
- 179 That is still true of aquaculture, despite a Treaty settlement guaranteeing Maori 20 percent of new aquaculture space (see spotlight below).
- 180 Greg Severinsen and Raewyn Peart *Reform of the Resource Management System: The Next Generation – Synthesis Report and Next Steps (Summary for Policy Makers)* (Environmental Defence Society, Auckland, February 2019).
- 181 New Zealand Productivity Commission *Better urban planning: Final Report* (February 2017) at 107.
- 182 Resource Management Act 1991, pt 7A; *Central Plains Water Trust v Synlait Ltd* [2009] NZCA 609, [2010] 2 NZLR 363.
- 183 See Ministry for Primary Industries “Fisheries New Zealand Review of Sustainability Measures: Overview of legislative and other requirements” <www.mpi.govt.nz>; and *New Zealand Recreational Fishing Council Inc v Sanford Ltd* [2009] NZSC 54, [2009] 3 NZLR 438.
- 184 That might exclude many fishers who are unable to upgrade their gear, but support could be forthcoming.
- 185 See Greenhouse Gas Geological Sequestration Act 2008 (Vic), ss 42, 96(1)(b), 98(2)(b); Meredith Gibbs “The Regulation of Geological Storage of Greenhouse Gases in Australia” in Ian Havercroft, Richard Macrory and Richard Stewart *Carbon Capture and Storage: Emerging Legal and Regulatory Issues* (Hart Publishing, Oxford, 2011) at 173.
- 186 *Ngāi Tai ki Tāmaki v Minister of Conservation* [2018] NZSC 122.
- 187 *LegaSea Rescue Fish Ika Rauora: A pathway to fish abundance and marine ecosystem recovery* (May 2020) at 19.
- 188 In Canada, some have suggested that a key reason for the success of MPA network rollout has been the set of management measures established around them, including school groups engaging in citizen science projects to survey beach and intertidal habitats. This could form part of a broader environmental research strategy suggested by the Parliamentary Commissioner for the Environment (see Chapter 10).
- 189 Greg Severinsen and Raewyn Peart *Reform of the Resource Management System: The Next Generation – Synthesis Report and Next Steps (Summary for Policy Makers)* (Environmental Defence Society, Auckland, February 2019) at 104.
- 190 New Zealand Productivity Commission *Better urban planning: Final Report* (February 2017) at 39.
- 191 New Zealand Productivity Commission *Better urban planning: Final Report* (February 2017) at 44.
- 192 Well-being of Future Generations (Wales) Act 2015, s 4.
- 193 Well-being of Future Generations (Wales) Act 2015, s 3(1).
- 194 For example, net gain offsetting, biobanking, development contributions, green taxes etc (see Chapter 8). Currently, the tools the RMA provides are overwhelmingly linked to preventing or mitigating adverse effects rather than making people provide benefits. See Resource Management Act 1991, s 108AA; Greg Severinsen “Glass half empty or glass half full? Adverse effects, positive effects and conditions under the Resource Management Act 1991 and Resource Legislation Amendment Bill 2015” (2016) 11(9) BRMB 110.
- 195 See the review by Bethan C O’Leary and others “Effective Coverage Targets for Ocean Protection” (2016) 9 Conservation Letters 398.
- 196 World Parks Congress “A strategy of innovative approaches and recommendations to enhance implementation of marine conservation in the next decade” (25 November 2014) IUCN <www.iucn.org>.
- 197 World Parks Congress “A strategy of innovative approaches and recommendations to enhance implementation of marine conservation in the next decade” (25 November 2014) IUCN <www.iucn.org>.
- 198 *Increasing marine protected area coverage for effective marine biodiversity conservation* WCC-2016-Res-050-EN (2016).
- 199 The High Ambition Coalition for Nature and People (HAC) was launched by France and Costa Rica at the One Planet Summit on Biodiversity in 2021. One of the key aims of the HAC is to advocate for the adoption of a new target for the protection of 30 percent of terrestrial and marine spaces by 2030 at the Conference of the Parties to the CBD to be held in 2021. Refer High Ambition Coalition “Why 30x30?” (12 January 2021) <www.hacfornatureandpeople.org/>

- 200 Open-ended Working Group on the Post-2020 Global Biodiversity Framework *Zero Draft of the Post-2020 Global Biodiversity Framework* (IUCN, Kunming, China, CBD/WG2020/2/3 February 2020).
- 201 Office of the Prime Minister's Chief Science Advisor *The Future of Commercial Fishing in Aotearoa New Zealand* (February 2021) at 116.
- 202 Although some exceptions are still provided for.
- 203 Department of Conservation *Te Mana o Te Taiao – Aotearoa New Zealand biodiversity Strategy 2020* (Department of Conservation, Wellington, August 2020) at 53, see objectives 10.3.3-10.5.3; objectives 10.6.1-10.6.3; and objectives 12.1.1-12.7.3.
- 204 Department of Conservation *Te Mana o Te Taiao – Aotearoa New Zealand biodiversity Strategy 2020* (Department of Conservation, Wellington, August 2020)
- 205 Raewyn Peart *Voices from the Sea: Managing New Zealand's Fisheries* (Environmental Defence Society, Auckland, 2018) at 101.
- 206 *Resolution on the Regulation of Ocean Fertilization* Res LC-LP.1 (2008) at [2].
- 207 *Resolution on the Regulation of Ocean Fertilization* Res LC-LP.1 (2008) at [8].
- 208 For an interesting account of the rise of agriculture, see Jared Diamond *Guns, Germs and Steel: The Fates of Human Societies* (WW Norton, New York, 1997).
- 209 See Raewyn Peart *Farming the sea* (Environmental Defence Society, Auckland, 2019), ch 7. For example, an industry itself (or parts of it) could collapse due to biosecurity incursions.
- 210 Raewyn Peart *Voices from the Sea: Managing New Zealand's Fisheries* (Environmental Defence Society, Auckland, 2018).
- 211 Jared Diamond *Collapse: How Societies Choose to Fail or Survive* (2nd Edition, Penguin Books, London, 2011).
- 212 See Ellie Hooper "What is seabed mining and why does it threaten the oceans?" (6 August 2020) Greenpeace NZ <www.greenpeace.org/aotearoa>; and Kiwis Against Seabed Mining "Protect Our Seabed" (19 January 2022) <www.kasm.org.nz>
- 213 Although the difference, of course, is that the Crown *owns* the petroleum resource so it is not a revolutionary idea to say that it can determine whether or not it is extracted.
- 214 For example, the objectives of the Climate Change Response Act are very specific (beyond just a general reduction of greenhouse gas emissions), and restrictions imposed to prevent the transmission of Covid-19 have been swift and severe (eg national lockdowns).
- 215 See Chapter 3.
- 216 See Robert Joseph and others *The Treaty, Tikanga Māori, Ecosystem-based Management, Mainstream Law and Power Sharing for Environmental Integrity in Aotearoa New Zealand – Possible Ways Forward* (Ko Ngā Moana Whakauka and Te Mata Hautū Taketake – the Māori and Indigenous Governance Centre, Waikato, 2019).
- 217 See Ministry for the Environment *Natural and Built Environments Bill* (Exposure Draft, 2021), cl 6; compare *Ngāi Tai Ki Tāmaki Tribal Trust v Minister of Conservation* [2018] NZSC 122.
- 218 *Ngāi Tai ki Tāmaki Tribal Trust v Minister of Conservation* [2018] NZSC 122.
- 219 *Ngāi Tai ki Tāmaki Tribal Trust v Minister of Conservation* [2018] NZSC 122.
- 220 By virtue of the direction in the Conservation Act applying to all legislation administered by the Department of Conservation.
- 221 Developments in this area remain to be fully tested in the courts. For example, an obligation to give effect to the principles of te Tiriti applies across all legislation administered by the Department of Conservation, including the Marine Reserves Act. To what extent that would prevent marine reserves from being deployed on a large scale is an interesting one.
- 222 Resource Management Act 1991, s 33; and Tūwharetoa Māori Trust Board "Section 33 Transfer with Waikato Regional Council" (media statement, 31 July 2020). The Council transferred water monitoring functions around Lake Taupō to Ngāti Tūwharetoa.
- 223 Waitangi Tribunal *Ko Aotearoa Tenei: A Report into Claims Concerning New Zealand Law and Policy Affecting Māori Culture and Identity* (Wai 262, 2011) at 286.
- 224 Resource Management Review Panel *New Directions for Resource Management in New Zealand* (June 2020), ch 3.
- 225 See New Zealand Productivity Commission *Low-emissions economy: Final Report* (August 2018) at 221, citing Richard Macrory *Regulation, Enforcement and Governance in Environmental Law* (2nd Edition, Bloomsbury Publishing, London 2014) at 264.
- 226 Robert Joseph and others *The Treaty, Tikanga Māori, Ecosystem-based Management, Mainstream Law and Power Sharing for Environmental Integrity in Aotearoa New Zealand – Possible Ways Forward* (Ko Ngā Moana Whakauka and Te Mata Hautū Taketake – the Māori and Indigenous Governance Centre, Waikato, 2019).

8 Reconsidering the toolkit



Mussel harvesting, Coromandel

8.1 Introduction

A “tool” is not a term that has a firm definition when it comes to the oceans management system. We are using it very broadly to mean any form of public intervention that influences people’s behaviour when interacting with the oceans. Tools can be as diverse as a biodiversity strategy, a regional coastal plan, a mining permit, a resource rental, a primary school syllabus, a fisheries sustainability measure, a rāhui, a green tax, a marine reserve or a disclosure requirement for companies. Even something like a property right can be considered to be a tool.

Sometimes tools exist in close-knit ecosystems that rely on connections between each other to function well (eg a regional coastal plan contains policies under which a resource consent application is considered). Other tools only stray loosely and occasionally into each other’s orbits (especially when they are housed within different statutes, such as fisheries plans and regional coastal plans). They can be deployed by different institutions including government ministries, councils, semi-autonomous agencies such as the EPA or Maritime New Zealand, or iwi/hapū.

Tools in the current system are too diverse to be able to provide a comprehensive list, but the table below provides some sense of what they include.¹ Many are provided for within statutory frameworks, although some (eg various strategies, subsidies or advertising campaigns) are non-statutory.

The existing system contains many things that can be regarded as “tools”. They can have quite different purposes, operate in fundamentally different ways, and be looked at in different degrees of granularity. All of them are ways in which the system intervenes to shape people’s behaviour.

Because we are concerned with the system as a whole, rather than the siloed reform of particular statutory frameworks,² we do not intend to provide an exhaustive account of every possible tool that could be reformed. We are primarily interested in the *types* of tools that could be used, and the *ways* in which they could be designed. Some specific options for reform emerge during discussion, but it is important not to lose the

RMA	EEZ Act	Conservation legislation	Maritime Transport Act	Biosecurity Act
NPS and the NZCPS	Regulations	Conservation management strategies	Marine protection rules	National and regional pest management plans
NESs	EEZ policy statements	Management plans	Maritime rules	National and regional pathway management plans
Regulations	Marine permits	Regulations	Navigation bylaws	Craft risk management standards and plans
Regional policy statements	Decommissioning plans	Wildlife sanctuaries	Maritime registry	Quarantine areas
Regional coastal plans	Enforcement orders and abatement notices	Concessions	Maritime levies	Agreements for readiness and response
Coastal permits and other resource consents		Marine reserves	Emergency dumping permits	Levy orders
Abatement notices and enforcement orders		Marine mammal sanctuaries	Marine oil spill response strategy and contingency plans	Restricted place and controlled area declarations
			NZ oil pollution fund	Biosecurity database

Figure 8.1: A selection of statutory tools provided for in the current system (this is a small snapshot of tools available)

forest for the trees. With this in mind, over the course of the next three chapters, we explore the toolkit through a series of categories (regulatory, non-regulatory, spatial and strategic/integrative tools) rather than traversing specific statutory frameworks (eg tools under the Fisheries Act, RMA or Marine Reserves Act).

That said, it is also useful to get a sense of what it would be possible to achieve without fundamental overhaul. If adding a new tool here or there – such as a resource rental, a new tax or an integrated oceans policy statement – might fix some problems, it might be preferable to do that rather than immediately jumping to more dramatic and expensive measures that may take many years to implement. Indeed, some have observed with regret that Aotearoa New Zealand tends to jump to complex legislative reform agendas when simpler solutions might suffice.³ We may not need to overhaul the entire toolkit.

This project is concerned with the toolkit at a systemic level, rather than exploring every single potential tool. We look at different categories, including regulatory, non-regulatory, spatial and strategic/integrative tools.

8.2 The relationship between tools and other themes

Before exploring different categories of tools, it is worth thinking about their relationship with other themes. First, tools can be designed to perform specific *roles* (see Chapter 7). Sometimes multiple tools can be used to perform a single kind of role. For example, environmental limits can be set via prohibited activity standards under the RMA, catch limits under the Fisheries Act, or the establishment of a marine reserve under the Marine Reserves Act. But a single tool can also multitask. This is particularly noticeable under the RMA where a regional coastal plan and associated consents can be concerned with limit setting, dispute resolution, making value-based trade-offs, protecting Māori interests and allocating some marine resources.

Secondly, every tool will be designed to achieve one or more *objectives* (see Chapter 7). This means that it is difficult to evaluate which tools would be “best” without first determining exactly what we are wanting them to achieve. For example, whether a market-based or permitting system for allocating fishing rights would be preferable might depend on whether efficiency and certainty, or equity and flexibility, were seen as more important.

Thirdly, principles can themselves be a kind of tool when they are “operational” in nature (see Chapter 7). For example, the purpose and principles of the RMA and Fisheries Act, including the Tiriti provisions, are themselves tools that influence the behaviour of decision-makers, and they need to be carefully defined.

Fourthly, different tools can reflect different worldviews (see Chapter 7). For example, some might consider it wrong to use tools that put a price on nature, even if charging people would be effective in reducing overall damage. Māori might consider that no-take MPAs are inappropriate if they sever ancestral connections, even if they are one way to improve biodiversity and fish stocks.

Finally, tools are intimately linked to legislative arrangements (see Chapter 11), because many require statutory backing. The statutory framework within which they are located, including the purpose for which they are deployed, is therefore significant for how they operate.

The toolkit of a future system is closely related to other themes, including the roles the system is expected to perform, the worldviews, principles and objectives that underpin it, and legislative design choices.



Oyster farm, Coromandel harbour

Raewyn Peart

8.3 Regulatory tools

Tools are about influencing people's behaviour, but they can do so in quite different ways. The defining feature of regulatory tools is that they have teeth – they can result in sanctions on people who do not comply with them. Usually, they tell people what they cannot do. Sometimes they tell people what they *must* do, although positive duties in the environmental context are usually triggered by people choosing to do something else first, which invokes a quid pro quo for doing so (eg an application for a coastal permit triggering a requirement to provide environmental enhancement).

Many might associate regulatory tools with command and control regulation such as that found in the provisions of the RMA (eg you cannot discharge a contaminant without consent) or Fisheries Act sustainability measures (eg you cannot use a particular method of fishing). Planning and consenting frameworks under the RMA and EEZ Act are, at root, comprised of regulatory tools and we look at those further below.⁴ However, regulatory tools in the oceans management system include a much wider diversity of interventions. For example, product stewardship schemes can be used to manage the lifecycle of plastic products that may end up in the marine environment. Some regulatory tools may not actually prevent or even control people's behaviour, but instead require reporting. For example, some have suggested that recreational fishers be licensed (a regulatory intervention), not to prevent their access to fish, but rather to (among other things) ensure that accurate information about their catch (and thus overall impacts on fish stocks) is obtained.

Regulatory tools will be vital in a future system. In particular, they are necessary to perform the role of setting environmental limits, although they can also be used to discharge other roles. We can use regulation to make trade-offs above limits (eg considering through consent processes whether the harm of a proposal is worth the benefit), fulfil te Tiriti obligations (eg including mana whenua in regulatory decision-making processes, protecting places of significance to Māori or imposing measures like rāhui), and make allocative choices (eg through a first in time consenting process or tendering process).

Below, we consider types of regulatory tool that could be used, or changed, in a future system. The term “regulation” covers many things, and there are myriad options for reform on this front. A lot of them involve targeted change to specific regulatory tools,⁵ and while we touch on some options below, this project is about higher-level questions and cannot cover all possibilities of detailed design. We begin by looking at how “framework” type regulatory tools – loosely described as planning

and consenting – could be deployed differently. We then consider the more specific tool of “environmental limits” and what that could look like in different legislative contexts, before considering how the concept of “rights” (including property rights, other rights to use resources, and human/environmental rights) could be rethought. Finally, we look at emergency orders, rāhui and some tools under “non-marine” legislation that warrant attention.

Regulatory tools tell people what they must do, must not do, or what they can do subject to conditions. They can be enforced. But regulatory tools are much more diverse than just command and control regulation. There are many options for reforming the regulatory toolkit on a whole range of fronts.

8.4 Planning and consenting

It is worth considering the place of wider planning frameworks in a future system. Overall, they are regulatory in nature, but they provide more flexibility for decision-making than just a list of prohibited activities and standards.⁶ A good place to start is with the RMA, which is generally regarded as the home of “planning”. Under the framework of the Act, a complex array of instruments has developed comprised of national direction, regional policy statements, various plans and resource consents. The RMA is set to be replaced by a new statute, the NBA. We explored some of its key proposed features in Chapter 4.

National direction

The NBA's intent to provide a more comprehensive package of national direction (a National Planning Framework) could be used as an opportunity to promulgate both national policy and regulations to fill notable gaps in marine management. Some are underway or have been previously suggested, such as an NES outlining common minimum standards for wastewater (and possibly stormwater) discharges, national direction on offshore aquaculture, and an NPS outlining how the te Tiriti relationship is intended to work under the NBA (including in the marine space). Other NPSs could be developed for marine biosecurity more generally, and/or for plastics (including risks to the marine environment).

A package of national direction could also be structured differently so that it gave proper attention to the interconnectedness of marine issues. The marine environment could be given greater focus by ensuring that, for example, an integrated set of domain-based policies formed a first layer of

provisions of the National Planning Framework (including the NZCPS and NPS for Freshwater Management). Other elements (eg sectoral policies and rules for forestry, urban development and wastewater disposal) would then be required to be consistent with or implement that first layer. That would help prevent potential misalignments arising between narrower sectoral regulations (eg NES for Plantation Forestry) and the policy intent of broader tools like the NZCPS, and ensure that the former were designed in a way that actively *gave effect* to the latter.

Existing national direction could be strengthened using a marine lens, for example to prohibit or phase out clear-felling of plantation forestry (or at least to require integrated catchment approaches to stagger planting and harvesting),⁷ to extend the NPS for Freshwater Management to include estuaries as management units (and strengthen provisions for sedimentation),⁸ and to link the concept of good urban design under a revised NPS on Urban Development to the benefits that such design can have for marine outcomes.⁹

While new national direction relevant to the marine environment could lead an independent existence (eg a new NES for wastewater), the NZCPS itself could also get a makeover. Provisions on sediment could be strengthened to complement the NPS for Freshwater Management.¹⁰ It could be made clear that all policies requiring the “avoidance” of adverse effects are deemed to be “limits” under the NBA, and it could deal more explicitly with the adverse effects of fishing activity on marine biodiversity, thereby signalling to councils the range of things that regional plans must deal with (see *Motiti* discussed in Chapter 6).¹¹



A National Planning Framework envisaged under the NBA provides an opportunity for marine matters to be more thoroughly integrated into other parts of national direction. New marine-related national direction could be created and existing documents reviewed through a marine lens. The NZCPS itself could be strengthened.

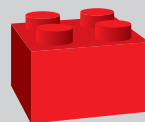
To improve implementation, the NZCPS could be linked with new national level regulations (equivalent to NESs) that are explicitly designed to give effect to its objectives and policies. We have had an NZCPS since 1994, but that has never been translated into complementary national regulations, instead being left to regional councils to implement.¹² This has led to uneven outcomes, with some councils yet to give effect to the 2010 NZCPS more than a decade after its promulgation.¹³

Individual pieces of national direction have also become adept at providing their own framing for the development of other implementation instruments (eg future development strategies under the NPS on Urban Development), and similar instruments could be created under the auspices of a revised NZCPS. This could, for example, see the development of a statutory “marine restoration strategy” to complement the conceptually similar “future development strategies”.

A spotlight on the proposed NPS for Indigenous Biodiversity

The NPS for Indigenous Biodiversity (under the RMA) is currently at draft stage. It does not directly apply to the coastal marine area, but regional biodiversity strategies made under it would include indigenous biodiversity in the coastal marine area, presumably as a way to link it to the policies in the NZCPS.¹⁴ Additionally, it would require councils to take an integrated approach, recognising the impact that terrestrial activities can have on the indigenous biodiversity of the coastal marine area.¹⁵

This high-level policy document sets standards and goals but does not describe the discrete mechanisms by which they will be achieved. One option would be to require the NPS for Indigenous Biodiversity or the NZCPS to include timebound implementation directions – specific actions that have to be done by a certain date – as has been done through the NPS on Freshwater Management.

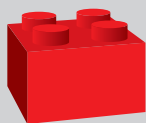


The NZCPS could be paired with new national level regulations (an NES) to give effect to its objectives and policies, and/or it could be strengthened to provide for more extensive “implementation” provisions.

National direction is not only important under the RMA/NBA. Since 2017, there has been the ability to promulgate an EEZ policy statement under the EEZ Act. Yet no such instrument has been made, despite broad criteria being included as to what the Minister must consider when deciding whether to do so. This means that consenting is largely undertaken in a policy vacuum other than the Act’s general purpose and principles. Ongoing difficulties and uncertainty about deep sea mining can be seen equally as the product of a lack of policy and strategy (whether, where and why we want mining to occur or not) as they are about scientific

uncertainty about what the impacts of mining operations would be.¹⁶ One wonders if applicants like Chatham Rock Phosphate and Trans-Tasman Resources would have gone to so much trouble and expense pursuing consent for mining in the places they did if there had been clearer policies outlining the places or contexts in which the impacts of mining are deemed unacceptable (eg on the Chatham Rise or in benthic protection areas) and where they are to be entertained or even encouraged.

An EEZ policy statement is clearly a tool that could be made much better use of. It could build on the relevant general provisions of the NZCPS, or target policy towards the activities likely to occur in the EEZ (eg mining, offshore aquaculture). It could also – even at a broad level – engage in a spatial sense by identifying valuable features of different areas (eg the Chatham Rise or around Rangitāhua/the Kermadec Islands).¹⁷



An EEZ policy statement could provide a much stronger framework for consenting in the EEZ, increasing certainty for applicants and the environment.

Planning

Proposed changes to plan-making under the NBA means that the resultant plans should become more effective tools for marine management. Notably, the production of an integrated combined plan for each region enables a more holistic approach for the coastal marine environment (including where particular land uses like plantation forestry are enabled and where urban growth is directed), and empowers iwi/hapū to have a stronger voice through partnership alongside councils in plan-making.

Planning also underpins the conservation system, through the development of conservation management strategies and plans, which in turn guide the Department of Conservation's operational work and the grant of concessions (including for tourism operations within marine reserves or interacting with protected marine species). Many of these documents are out of date. They lack clarity and there is a history of poor implementation.¹⁸ We are exploring reform of this system in our separate conservation law reform project. Of particular relevance here, though, is how such plans could potentially be reconfigured to strengthen marine management.

One approach could be to develop broader bio-regional marine conservation plans, that provide a strategic lens to the management of stresses on marine protected species and biodiversity more generally, and indicate spatially where protection is needed. Such plans would need to have a stronger implementation mechanism, such as clear milestones and timeframes, and strong links with funding and the grant of concessions. They would also need to interface with RMA plans, and this could be through the Minister of Conservation's role of signing off regional coastal plans. Part of such sign-off could be ensuring they comply with the relevant marine conservation plan.

There would likely be duplication between the two planning systems (due to the overlapping biodiversity functions of councils and the Department of Conservation), and this indicates that a better way forward might be to focus on a single higher-level marine spatial planning process, which we discuss in Chapter 10.

It is interesting to contrast the planning approach under the RMA/NBA and Conservation Act with that for fisheries. Fisheries plans can be created under section 11A of the Fisheries Act, and while the Minister must take them into account in making decisions, they are not mandatory and their purpose remains murky. They are nothing like the structured, focused and legally influential plans that are made under the RMA.¹⁹ As we have said previously:²⁰



Stand up paddle boarding, Wellington harbour

There is currently a very weak policy and planning framework to guide fisheries decision-making ... The legislative framework only provides for fisheries plans (not policy or standards), and the provisions are sketchy as to the purpose, content and preparation process for the plans. [However,] the development of such documents helps to engage a broader constituency in fisheries management, and to provide greater certainty as to how fisheries management will be effected in the public interest. Once the policy or plans are settled, it helps to reduce the politicisation of decision-making and the frequent u-turns which have characterised fisheries decision-making to date. A key matter to resolve is what the role of fisheries planning should be, the scope and content of such plans, and how they should be developed.

Fisheries plans could conceivably be made mandatory, their place/hierarchy in the system made clearer, and their content or at least their purpose prescribed. They could also be regional or local in their application (bioregional fisheries plans) as opposed to the general ones that have been prepared in the past,²¹ involve greater public input,²² and reflect the full range of values reflected in the purpose of the Fisheries Act.



Combined plans under the NBA should provide more effective tools for marine management. Conservation planning could be strengthened to have a focus on marine bio-regional areas. A future system could also see the creation of a more developed planning framework for fishing.

Consenting

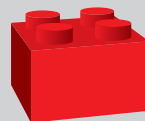
Consenting or permitting frameworks can be better at linking rights to obligations than property rights. Attempts to add responsibilities to property rights later on can be resisted because the market has evolved (prices have been set) in their absence. This raises the question of whether a more RMA-style consenting framework should be applied to fisheries alongside the QMS. The Fisheries Act already requires permits to fish commercially.²³ However, such permits are not like resource consents, and although they can be subject to wide ranging conditions, these are simply what are considered “appropriate” and are not linked to policies or objectives in a fishing plan or any other instrument.

Permits could, however, operate more in line with the RMA, where fishing could, depending on location and method (and therefore environmental impact), be a permitted, controlled or discretionary activity (and where consents would be assessed in light of objectives and policies of a place-

based fisheries plan). This would take a more structured, policy-driven approach to sustainability measures to deal with the impacts of fishing, such as more nuanced control over methods like bottom trawling (tailored conditions to determine when and where a particular operation could do so) and the use of mitigation devices (eg conditions for the design requirements of nets and longlines to reduce seabird and marine mammal bycatch). It could also allow a more sensitive approach to interactions between fishing and existing activities like aquaculture, where conditions could manage what acceptable impacts might look like on both, rather than just having an adverse effects test for new aquaculture operations.

However, if imitation is the sincerest form of flattery, this begs the question as to why an RMA-like planning and consenting framework would be developed in parallel under the Fisheries Act when the RMA itself might be more suited to the task, or (if there is an irreconcilably difficult legislative boundary between them) whether the two might not be combined in a single Oceans Act (see Chapter 11). There may be valid reasons (eg the spatial scale of QMAs differ greatly from regional boundaries, and a proactive tool like a TAC may not be a comfortable fit within a largely reactive planning framework). But if regional coastal plans were to encompass fisheries management tools, it then begs the question as to why we would not simply have a single, overarching plan for the whole marine area. Indeed, that would be possible, and we look at marine spatial planning in Chapter 10.

It might also be worth considering what other things could benefit from a consenting framework. What about waste, where the Waste Minimisation Act could require consent to produce particular types of product rather than just prohibit them or provide product stewardship schemes? This tool could even be incorporated into the RMA, linking to objectives relating to marine pollution such as minimising plastics, and reaching into the Fisheries Act by controlling the use of gear like nets (which make up a large proportion of marine plastic pollution).



Fishing permits could be brought under a more environmentally policy-driven framework, which could operate alongside the QMS.



Consenting could be applied more broadly to waste minimisation frameworks.

8.5 Environmental limits

Environmental limits are one kind of regulatory tool that warrants particular attention. In Chapter 3 we pointed out that the current system does not really set limits in a systemic way, including under the RMA, Fisheries Act, or conservation legislation. In Chapter 7 we identified limit setting as a distinct role a future system could play and explored what its objectives might be.

But an environmental limit can be regarded not just as a general name for a role the system could play (that the system must clearly articulate lines in the sand beyond which harm is not allowed), but also as a specific label for a kind of regulatory tool that could be embedded in a new system.²⁴ This would be a form of regulation with a special legal status. Any provision having this label could not be eroded or balanced against other considerations and would have its own dedicated and uncompromising purpose.

The concept of an environmental limit could be deployed across a wide range of statutes. There could, for example, be an umbrella statute (eg an Oceans Act) that defines what the consequences of designation as a “limit” means,²⁵ with various regulations under other statutes then being deemed to *be* limits. Mandatory limits could relate to many different things, including mortality to threatened species specified in conservation legislation, bycatch of marine mammals and seabirds, and proxy measures of ecological integrity.

Environmental limits and the NBA

A spotlight on limits under the proposed NBA

The concept of an “environmental limit” under the NBA is a new tool that could be used to achieve better marine outcomes. It has arisen in response to criticisms that the RMA – even after the *King Salmon* clarification – may not be fit for purpose in setting a comprehensive range of strict limits.²⁶ Despite its rhetoric in section 5, the RMA remains riddled with trade-offs. Some have pointed to the absence of national bottom lines for estuaries in the NZCPS.²⁷ True national level limits in the marine space are largely limited to prohibitions on dumping, driven by international obligations under the London Dumping Protocol (see Chapter 3).

Limits under the exposure draft of the proposed NBA are, in contrast, envisaged to be mandatory and must be set for a wide range of things. Three of those are “coastal waters”, “estuaries” and the overlapping concepts of “biodiversity, habitats and ecosystems”.²⁸ However, that arguably lacks the specificity to require the things that really matter to be addressed, and could be strengthened by providing a schedule outlining the elements of the marine environment that require biophysically-focused limits (eg sediment, nutrients, wastewater, chemicals, habitat protection etc) and the limits on human activities required to defend them (eg forestry, agriculture, urban development). The risk of leaving this tool vague is that it gets narrowed down to only some indicators and pressures (eg coastal water quality) and not others that may be harder to measure and achieve (eg the decline of biodiversity in a particular ecosystem).

It is not clear what a “limit” for marine biodiversity or ecosystems would look like. But it may require spatial expression – specific areas being mapped and protected, not just general prohibited activity status for an activity. This might create a duty to ensure that the NBA is used as a mechanism to create a network of MPAs, rather than relying on things like marine reserves or bespoke marine protection legislation. Indeed, while it is not in the exposure draft of the Bill, the Randerson Panel recommended that an active duty be placed on the Minister to identify and prescribe significant habitats, which might translate to an obligation to *map* them and include them in the more integrated setting of a regional plan. This has, to some extent, been achieved already in bespoke legislation created for Fiordland which draws on the RMA zoning tools.²⁹ We discuss MPAs, and the potential role of the NBA in achieving them, further in Chapter 9.

Raewyn Peart



Salmon farm, Marlborough Sounds

As indicated in the spotlight, environmental limits are a core tool under the proposed NBA, and this could be mirrored in the EEZ Act. A number of questions arise when thinking about how regulatory limits might be designed under the NBA. For instance:

1. Should hard limits be set for specific sectors and activities, or should limits focus only on describing unacceptable effects? For example, a limit could prohibit oil and gas exploration, carbon geo-sequestration, ocean fertilisation or bottom trawling. This may depend on how specific the system's objectives should be (see Chapter 7), and whether we depart from an effects-based system to a prescriptive, activity-based one.
2. At what spatial scale should limits be set? Would it, for example, be appropriate to enhance shellfish beds or create a marine reserve in one estuary in exchange for allowing greater sedimentation and fishing in another? That may influence how (or whether) a marine biobanking system and offsetting framework is designed.³⁰
3. What should the purpose of regulatory limits be? As described in Chapter 7, this could range from low ambition (eg preventing extinction, ecosystem collapse and hospitalisations) to the more ambitious (eg a higher conservation status in the threat classification system and "swimmable" and "edible" estuaries) or linked to te ao Māori (eg safeguarding the mauri of the moana).
4. Should limits be imposed only through actual regulation (eg prohibited activity rules, catch limits, marine reserve regulations) or should they also be contained in policies (eg "rule like" policies in the NZCPS)? The latter might provide some flexibility, but would not be a replacement for actual regulatory controls.
5. What should be the legal consequences of something being a limit? For example, they could include:
 - An inability to progress a private plan change to alter prohibited activity standards.
 - An additional level of scrutiny by an independent regulator when limits are being created and amended.
 - Reversing the burden of proof, or requiring a higher standard of proof, to weaken a limit (reflecting the principle of non-regression).

- More robust monitoring requirements.
- Mandatory public notification of any relevant consent applications.

As mentioned in Chapter 7, there are some challenges in operationalising the concept of a "limit" in the marine environment. For one, it is difficult to measure and observe environmental change or understand interconnected and unpredictable ecological processes and responses. It is hard to know what a minimum state or tipping point actually is.³¹ However, a precautionary approach to limit setting under the NBA – choosing to establish a point beyond which there is an unacceptable risk of significant or irreversible harm³² – does not require full or perfect information. We frequently set limits based on imperfect information for human health, biosecurity and in the context of fisheries stock management. And as the Cawthron Institute has pointed out, "where an environmental limit [in the sense of a biophysical indicator] cannot be clearly specified and directly managed, complementary measures will be needed to protect environmental integrity".³³ This does not make a limit based on environmental integrity less valuable; it may simply mean that instead of numerical standards specifying the minimum state of every aspect of an environment (eg degree of sedimentation), limits are expressed as strongly worded policies or precautionary prohibitions, moratoria, or standardised conditions on activities known to pose an unacceptable risk (eg clearfell harvesting of an entire catchment). In some cases, there are existing tools that might be able to assist in translating minimum outcomes to measurable limits at place (such as the New Zealand Estuary Trophic Index).³⁴

The concept of an "environmental limit" might be treated as a specific tool, not just a role the system needs to play. This is contemplated in the proposed NBA.



Environmental limits contemplated by the NBA could be more targeted to the marine context, including by being more specific about what things limits must be created for. To be useful, a provision classed as a limit would need to have clear consequences different to other provisions.



Trawler, Hauraki Gulf

Environmental limits and fisheries

A range of regulatory tools under the Fisheries Act could be re-characterised as environmental limits. These would be mandatory – a legal obligation rather than a political choice – and could have their own separate, more protective purpose. Legislative provisions would make it clear they could not be eroded or undermined by shorter-term considerations around economic or social benefits.³⁵ In particular, it is worth thinking about whether some existing sustainability measures might be rethought as “limits”.

A spotlight on sustainability measures

Sustainability measures enabled under the Fisheries Act can encompass a wide range of things, including:

- Setting the TAC and TACC;
- Restricting the size, sex or biological state of the species harvested;
- Restricting the areas from which any species may be harvested;
- Restricting the fishing methods that can be used to harvest any stock or which are deployed in any area;

- Restricting the fishing seasons that apply to any stock, any area, any fishing method or any fishing vessel;
- Other methods not specifically described which are aimed at managing the effects of fishing on any stock or on the marine environment.

Despite this very broad range of management tools, relatively few have been deployed since the QMS was introduced. In many places, management of commercial fishing has largely focused on the setting and (in some cases) adjusting of TACCs. Management of recreational fishing has largely focused on imposing bag limits and minimum harvest sizes, followed by closures when stocks collapse.³⁶ Measures such as closing areas for habitat protection, and requiring the adoption of less damaging fishing gear, have been much less noticeable.

The tools to address a wide range of environmental impacts caused by fishing are clearly there. For example, additional benthic and other protected areas could be created, and minimum requirements mandated for fishing gear (coupled with incentives for innovation and uptake).³⁷ It would also be possible for regulations to simply prohibit or phase out bottom trawling and dredging in coastal waters and on seamounts, although for some fishers that may cause hardship and may need to be accompanied by supporting measures. Some have also suggested banning other methods like purse seining (on the grounds that this would enable fish closer to the surface to flourish too, providing more food for seabirds)³⁸ and set netting (to protect dolphins and vulnerable reef fish). And in addition to setting a TAC for a stock, one option would be to impose controls on fishing effort in certain areas, as well as a potential shift to more selective fishing methods more broadly (eg cages and long lining).

A less ambitious option would be to freeze the current dredge and trawl footprint, at least until additional protective measures for benthic habitats could be designed.³⁹ Impacts on seabirds could be reduced by requiring the adoption of innovative mitigation technologies (such as underwater line setting devices) as well as further mandating other devices that scare or deter birds from risk areas.⁴⁰ All such things can be done under current tools. One paper has identified four key types of action that could be focused on: technical measures, spatial controls, impact quotas, and effort control (see Figure 8.2).

Class	Option	Objective
Technical measures	Modify or adapt existing bottom trawl gear	Reduce seabed impacts and maintain or increase catchability of target species
Spatial controls	Prohibition by gear type	Eliminate high impact gears in a defined region
	Freeze trawl footprint	Confine impacts to previously impacted areas
	Nearshore restrictions and zoning	Reduce bottom trawling in shallow, sensitive habitats and minimise gear conflicts
	Prohibition by habitat type	Protect sensitive areas
Impact quotas	Multipurpose habitat management	Protect essential, representative and vulnerable habitats
	Invertebrate bycatch quotas	Reduce bycatch of benthic invertebrates
Effort control	Habitat impact quotas	Habitat conservation to protect benthic organisms
	Removal of fishing effort	Reduce impact by reducing fishing activity

Figure 8.2: Options to reduce habitat impacts of fishing⁴¹

When it comes to mechanisms for reducing the impacts of fishing on the broader marine environment, the problem is not so much that the tools are lacking, but that there is a lack of will to use them.⁴² That is compounded by a (at least perceived) lack of certainty around the nature of property rights in fish (see Chapter 6). There are constant debates about whether a particular sustainability measure taken to protect the environment is “taking” or “eroding” a property right or not, whether some form of compensation should be payable, and whether te Tiriti rights are being undermined by stealth. The Act is torn between using a reasonably robust toolbox to achieve one part of its purpose, and the defence of property rights underpinning the other. While the

sustainability principles of the Act sound firm on paper, they fall short of the directive approach to protection increasingly being taken under the RMA (eg for fresh water) and proposed under the NBA. This is somewhat curious, as the property rights on land which are affected by RMA provisions are legally much stronger than those attached to fisheries quota.

It is therefore worth considering whether a similar approach to that proposed in the NBA should also inform the reform of the Fisheries Act – the idea of having a mandatory, comprehensive set of national-level regulatory limits rather than just a toolbox of sustainability measures to be deployed in a selective or discretionary manner. This could provide clarity as to what kinds of measures *are* legitimate and necessary to achieve the purpose of the Act, even if they reduce the value of property rights, and when they trigger compensation or support. It could involve a simple direction that the responsible Minister *must* deploy sustainability measures and standards necessary to achieve the purpose of the Act.

The idea of regulatory “limits” in the Fisheries Act is not a radical prospect. The Fisheries Task Force established to inform the 1996 Act promoted the establishment of environmental bottom lines which would set out “the limits beyond which the activity of fishing should not push the environment” to achieve the public good.⁴³ That would include strategies for maintaining stocks at permissible levels as well as controlling the methods and location of fishing to avoid adverse effects. The ability



Snapper in Danish seine net, Hauraki Gulf

Raewyn Peart

for the Minister to promulgate binding environmental standards was even included in the initial Bill, but this was removed before it passed into law, and replaced with general principles and mostly discretionary sustainability measures. The reason given was that the TAC was “the main environmental standard for most wild fisheries”, and the QMS was “the principal fisheries management mechanism”.⁴⁴ This goes back to the question of whether property rights or other regulatory mechanisms are more appropriate for setting environmental limits. However, our experience in fisheries has demonstrated the problems that can occur with strongly relying on property rights to address environmental impacts.⁴⁵

A spotlight on environmental limits for fisheries in the United States

United States federal fisheries legislation has,⁴⁶ since 1996, required the identification and, since 2002, the mapping of essential fish habitat and the minimisation, to the extent practical, of the adverse effects on such habitat caused by fishing. Other actions to encourage the conservation and enhancement of these areas must also be identified. Essential fish habitat is defined as “those waters and substrate necessary to fish [which includes all marine life other than marine mammals and seabirds] for spawning, breeding, feeding or growth to maturity”.

The mapped areas are included in regional fisheries management plans and have resulted in extensive bans on trawling and other bottom-disturbing fishing methods in some places. For example, in 2005, bottom trawling was prohibited from 95 per cent of the fisheries management area around the Aleutian Islands in Alaska, with six Habitat Conservation Zones being closed to all bottom contact fishing gear due to the high density of coral and sponge habitat. Bottom trawling for all groundfish species was also prohibited in 10 designated areas along the continental shelf of the Gulf of Alaska, among a raft of other protections.⁴⁷



There are many regulatory tools available under the Fisheries Act that have been underutilised. A future system could provide more structure and direction around how (and why) they are to be deployed, and could characterise some of them as environmental limits.

Limits with respect to fish stocks

It is also worth considering the toolkit for setting catch limits for fish stocks themselves. This is arguably already a form of limit managed squarely under the Fisheries Act through the setting of a TAC and TACC.⁴⁸ However, various changes could be made in a future system.

One option would be for the Harvest Strategy Standard, which is currently used as a non-statutory guide when setting catch limits, to be formalised in legislation as a core part of the system. A recent High Court decision has confirmed that this instrument cannot be ignored (see the spotlight below). However, the Standard does not have legislative status; in future it could be recognised under the Fisheries Act to ensure there is rigour around setting commercial catch limits. This might go some way to remedying the lack of formal policy instruments under the Act (as discussed above), and could even evolve to add more value-based principles for setting catch limits (ie when to aim for something other than MSY).

A spotlight on the Tarakihi decision: The place of the Harvest Strategy Standard

In June 2021, the High Court issued its judgement on a challenge by the Royal Forest and Bird Society of decisions made by the Minister of Fisheries which set the TAC and TACC for East Coast tarakihi fish stocks.⁴⁹ For the 2015-6 fishing year, the size of the stock was estimated to be just 17 per cent of virgin biomass, further reducing to 15.9 per cent by the time of an April 2019 stock assessment. In response, the Minister made decisions to reduce the TAC and TACC for the stock for the 2018 and 2019 fishing years resulting in a combined reduction of 22.3 percent for the East Coast tarakihi stock.⁵⁰

The Court found that the Minister had made an error of law (in setting the rebuild period for the stock), failed to take into account a mandatory consideration (the Harvest Strategy Standard), and had regard to an irrelevant consideration (an Industry Rebuild Plan) when making the decisions. Although the decisions were not set aside, the findings of the Court will guide the Minister's future decisions for the stock, and will no doubt have flow on effects for decisions on other stocks which require a rebuild.

The decision provides useful clarity on the application of several provisions of the Fisheries Act to stock management, and the way in which environmental limits are to be applied in that context.

The Court made it clear that stocks are not to be managed below MSY, thereby confirming it as a firm limit. It dismissed the argument that social, cultural or economic considerations could be taken into account when determining the period of rebuild, on the basis that this was “not a tenable interpretation” of the relevant section (s13(2)(b)(ii)) because it would enable stocks to be “perpetually” maintained below MSY.⁵¹ This was in the context of the Minister being influenced by an Industry Rebuild Plan when setting a longer period for stock rebuild.⁵²

The Court also made it clear that the Act required more than the Minister simply “moving in the right direction” when stocks were below MSY.⁵³ The Minister is required to identify the rebuild target (ie target stock size), the period of rebuild (which must be appropriate for the particular stock), and the probability of achieving the target which are all “essential elements of the rebuild plan”.⁵⁴

One of the matters successfully argued by Forest and Bird was that the Harvest Strategy Standard and accompanying Operational Guidelines were a mandatory consideration for the Minister when making decisions on setting the TAC/TACC. This was despite there being no mention of the Standard (or the setting of any policy or standards) in the Act itself.

The Harvest Strategy Standard was developed by the Ministry for Primary Industries in 2008. It establishes default limits and standards for fish stocks including management targets, soft limits (which when breached generate a rebuild plan) and hard limits (which when breached may generate closure of the fishery). When applied to the tarakihi stock, it indicates a management target of 40 per cent of virgin biomass (with a 70 per cent probability), a soft limit of 20 per cent and a hard limit of 10 per cent. It also indicates a rebuild time of 10 years. Any departures from these default settings “must be justified in terms of the particular circumstances that warrant such departure”.⁵⁵

The Court found that although the Standard did not have legislative force, it constituted “best practice”, was an “established and recognised body of opinion” and therefore was the “best

available information” under section 10 of the Fisheries Act. This section sets out a number of information principles that “must be taken into account” by decision-makers, with subsection (a) stating that “decisions should be based on the best available information”. This meant that the Minister must take the Standard into account, although he is not required to comply with its provisions.

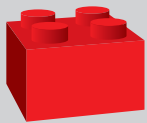
However, the Court did highlight that, in this case, the weight to be given to the Standard “is not solely at the Minister’s discretion” as “while the HSS [Harvest Strategy Standard] does not have legislative force, there is no counter argument from the respondents to the HSS statement that one cannot be satisfied that rebuild is complete until there is at least a 70 per cent probability that the target has been achieved”.⁵⁶ This indicates that the Minister cannot decide to depart from the Standard unless there is a solid scientific basis for doing so.

An indication that the Harvest Strategy Standard has not been factored into fisheries decisions to date is indicated by the advice given to the Minister by Fisheries New Zealand on the tarakihi stock. In 2018, it provided the Minister with three options to rebuild the stock. Only one (Option 1) was estimated to rebuild the stock within the 10 year target indicated by the Harvest Strategy Standard (requiring a 55 percent reduction in TAC). Option 2 was projected to require a 20 year rebuild (through a 35 percent reduction in TAC), and Option 3 which included a lower TAC reduction (20 percent) had no estimate of rebuild time. The Minister adopted Option 3.

The 2019 advice paper contained four options provided by Fisheries New Zealand, none which met the requirements in the Standard. Option 1 (31 percent reduction in TACC unevenly spread) had a rebuild period of 12 years and Option 2 (35 percent reduction evenly spread) 11 years, both with only with a 50 (rather than 70) percent probability; Option 3 (no TACC reduction but voluntary industry measures) had a reduced rebuild target (of 35 rather than 40 percent of virgin biomass), a rebuild target of 20 years and no associated probability; Option 4 (10 percent reduction and voluntary industry measures) had a rebuild period of 25 years (with 50 percent probability) and of more than 30 years with a 70 percent probability. The Minister adopted Option 4.

At the time of writing, the High Court decision was under appeal. But if the decision stands, the Harvest Strategy Standard may need to be considered as a policy document that establishes

default limits for the setting of the TAC and TACC for stocks, and these can only be departed from on the basis of sound scientific grounds. However, that begs the question whether the legislation itself should have such a document at its heart. Something like the RMA has a mandatory core document – the NZCPS – to assist in decision-making about limits. Fisheries is not conceptually that different. There is also a deeper normative question here (see Chapter 7): should the limits at the heart of the TAC be aiming to achieve MSY or something different?



The Harvest Strategy Standard, which provides a more nuanced approach to setting TACs, could be formally incorporated into legislation.

The scale or granularity with which limits are set also needs attention. That applies to all frameworks (eg zones under the RMA and the coverage of MPAs) but is particularly relevant to limits on taking fish. This is both spatial (where the boundaries of QMAs should be) and species-related (whether limits apply to individual or connected stocks).



Commercial crayfishing, Mercury Bay

A spotlight on an ecosystem-based approach to stock assessment

Some commentators have argued that fisheries management tends to focus on single species rather than taking account of interactions between species or with the ecosystems within which fish species live.⁵⁷ Within the food web, fish function both as predator and prey, and in those roles, they affect the structure and function of their habitats. When fish are removed by harvest, this reduces the size of their own population, but also in turn affects the populations of other biota and the relationships amongst them within complex food webs. Focusing management primarily on the size of the harvested stock (and “counting fish”) has the potential to lead to the wider effects of fishing being ignored, including changes in ecosystem components that may affect the managed stock itself in the longer run.

The Fisheries Act does not limit management to a single species approach. In fact, section 9 requires that decisions take account of the need to sustain associated or dependent species, the maintenance of biological diversity of the aquatic environment and protection of habitat of particular significance for fisheries management. This is evident in the plenary assessments of fisheries stocks⁵⁸ including its use of the Aquatic Environment and Biodiversity Annual Reviews,⁵⁹ which provide summaries of scientific data and analysis of wider ecosystem effects and relationships including bycatch, benthic effects of fishing and ecosystem status.

However, in practice and despite the Annual Reviews, the wider effects are less well understood because the research focus and investment has largely been on valuable commercial species rather than non-commercial species or the quality of the marine environment in a wider sense. The Ministry for the Environment with Statistics New Zealand note that about half of Aotearoa New Zealand’s fish stocks (mainly minor fished species) have too little information to reliably assess their stock status.⁶⁰ Yet these are often vital components of the broader marine food web.

One commentator suggests some of the problem is attributable to the “user-pays” funding model for research because it relies on a levy on quota owners who can influence what the research is focused on.⁶¹ In addition, the costs are levied back to quota in individual stocks, and some of the smaller less valuable stocks

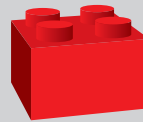
do not produce enough revenue to make investment in their research financially viable (which raises the interesting question of whether we should be fishing stocks if we cannot afford the research to understand the effects of doing so). Quota owners are understandably more focussed on the stock they own quota in, and on those that are of the most commercial value.

Alternative funding models are likely to provide a better basis for science aimed at achieving sustainability in its widest sense. For example, funding might still come partially from a levy on quota owners but with research guided by an independent panel with a wider societal remit, and more thoroughly integrated into a coherent strategy and plan for environmental research.⁶² And other sources of funding might be looked at, too, recognising that there are considerable recreational fishing and broader public interest in having a strong information base. For example, one option could be to provide a licensing regime for recreational fishers and ringfence revenue for ecological rather than stock-based research. Our system for environmental research, information and science has been looked at recently by the Parliamentary Commissioner for the Environment,⁶³ and oceans are an important part of that given how little we know and how difficult things can be to observe. Arguably there needs to be a more fundamental rethink of our funding model.

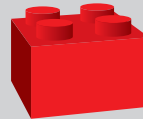
Other limits might be considered with respect to fisheries management as well. For instance, while the TAC is intended to be a hard limit, it does not currently really operate as one. This is because controls on recreational catch (eg through bag limits per person) are set based on an estimate of how many fish are likely to be caught. Unlike commercial fishing, there is no hard cap at which point fishing must stop, although as the fish become harder to catch the amount of recreational fishing tends to decline.

One option would be to create such a cap, although that would require a more intrusive system by which recreational fishers must be licensed and report their catch. It would also require a more complex allocative mechanism (if there is a hard limit, who gets to fish? Is it first in, first served? A system of tradeable rights? Some preference for iwi?). This may risk fraught conversations about fairness, freedom, te Tiriti obligations, and cultural values.

An alternative could be a spatial distinction between commercial and recreational catch limits. This could see the establishment of dedicated recreational fishing areas close to the shore that would have their own catch limits (which could be set at a more ecologically sensitive and “experience” focused objective⁶⁴ than MSY), with separate areas open to commercial fishing.⁶⁵ Such “recreational fishing parks” were proposed for the Hauraki Gulf and Marlborough Sounds in the government’s 2015 discussion document on new MPA legislation but have not been proceeded with.



A hard “cap” could be placed on recreational take (a “total allowable recreational take”) as well as a commercial TACC. However, that could have challenges.



Greater spatial separation could be created between recreational and commercial fishing activities by creating dedicated recreational fishing areas.

The scale at which catch limits are set is also important. At present, QMA’s are very large and contain a wide variety of ecosystems within them. Local depletion can not only cause difficulties for recreational and customary harvest, but also impacts on localised ecosystems that rely on a delicate balance within a food chain. For example, efforts to control fishing under the RMA around Motiti Island have been focused on a much more granular marine environment, not the QMA of which it is a part.



Commercial longlining, Hauraki Gulf

Raewyn Peart

A spotlight on QMAs

The Fisheries Act spatially divides the sea up into differently configured administrative units (called “fisheries management” areas). Most of these extend from the shore out to the edge of the EEZ. Seven areas are adjacent to the mainland coast and another four cover offshore areas surrounding island groups. QMAs for fish stocks have largely been defined on the basis of these fisheries management areas. These areas were largely developed for administrative efficiency (albeit based on broad understandings of the differing characteristics of the marine environment) and do not coincide with many biological fish stocks, even though such stocks are the main focus of fisheries management effort.⁶⁶

It is not clear the extent to which local units *within* a QMA can be managed separately through apportioning part of the TAC and TACC to them, or whether this needs to be achieved by establishing no take protected areas or tools like taiāpure or mātaimai. More spatially granular catch limits might help reduce pressure in sensitive places, reducing the need for a domino effect of rāhui and closures where localised depletion is deemed too much by mana whenua and local communities. The recent introduction of real-time electronic catch reporting would make more spatially granular management much more feasible than in the past.

While some stocks may require more *localised* management, others may require broader spatial management than under individual QMAs. In addition, the space in which related species live and move is constantly changing. This calls for a more agile *process* to be embedded in legislation by which QMAs, and more nuanced spatial areas within them, could (and must) be changed over time. That may be valuable, not just to transition to a more ecosystem-based system but also to reflect new information and ongoing changes in habitats and population dynamics driven by climate change and other stressors (eg pollution from land).⁶⁷ However, the impacts on individual transferable quota (ITQ) would need to be carefully managed, as these are currently spatially linked to a specific QMA. While there is power in the Fisheries Act to change QMA boundaries, it has hardly ever been used.⁶⁸

Instead of just changing management boundaries, another option might be to create layers of management in which existing units of space can be managed together in a more coordinated and adaptive way to recognise interdependencies between them (see rock lobster spotlight).

A spotlight on interdependent stocks: the case of the rock lobster

Rock lobsters are widely distributed in coastal waters around Aotearoa New Zealand. The red or spiny rock lobster (*Jasus edwardsii*) supports important coastal fisheries, which have been managed within the QMS since 1990.⁶⁹ The national fishery is currently divided into nine management areas labelled “CRA”. Rock lobsters within each CRA are assumed to constitute separate stocks for assessment and management purposes.⁷⁰

This oversimplifies the biological reality. Rock lobsters have one of the longest larval stages of any marine species.⁷¹ After hatching in shallow coastal waters, larvae drift offshore, where they spend 12 to 24 months transitioning through different development stages.⁷² During the extended offshore phase, larvae can be transported considerable distances by ocean currents and subsequently settle and recruit to other geographic areas.⁷³ Consequently, stocks that are linked by prevailing ocean currents are interdependent and the depletion of one geographically defined stock can lead to reduced recruitment in other stocks.⁷⁴ Put simply, they move around QMAs.

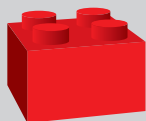
One study identified important relationships between rock lobster stocks on the east coast of the North Island where recruitment is generally lower than in other parts of the country.⁷⁵ For example, CRAs 2 to 4 receive nearly all of their settlement from ‘up-current stocks’ including CRA 1 (Northland), while CRA 1 is thought to receive the majority of its own settlement from CRA 9 (Westland/Taranaki).⁷⁶

The findings of recent stock assessments suggest the failure to recognise the interdependence of stocks has contributed to poor outcomes in north-eastern rock lobster fisheries. Significant declines in stock levels have been observed in CRA 1 (Northland) and CRA 2 (Hauraki Gulf / Bay of Plenty) since the 1940s.⁷⁷ In 2017 it was found that the spawning biomass in CRA 2 was critically low, at only 18 percent of the unfished reference level, and a formal rebuilding plan was implemented for the stock.⁷⁸ The latest stock assessment for CRA 1 shows the stock is in a state of persistent depletion, at a level that is only marginally higher than the historic low point.⁷⁹ An increase in the overall biomass of CRA 1 is necessary to support the rebuild of CRA 2 and other down-

current stocks. In short, they are all connected. However the recent sustainability proposal for the stock fails to recognise this.

A more suitable management approach would recognise the relative importance of relationships between stocks and implement stronger measures to protect stocks that are important sources of larvae for other geographic areas. Such an approach is already available under the Fisheries Act. Indeed, the Minister is required to take into account the interdependence of stocks when setting a TAC for a fishery. To date, a narrow interpretation of s 13(2) has been adopted by considering relationships between different species (ie trophic webs). However, a more scientific approach would include consideration of different stocks of a single species (ie across QMAs).

Management boundaries for fishing in the current system do not necessarily reflect the biological reality of fish stocks or their ecosystems.



A future system could provide more framing around how to set localised catch limits within QMAs, requiring boundaries to be redrawn based on ecological factors, or providing a more agile process (and trigger points) by which QMAs are (or must be) revised.

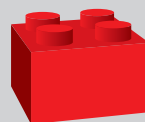
Environmental limits and conservation

Because their purposes are already more uncompromisingly protective, regulatory tools under marine conservation legislation could also be regarded as “limits”. However, statutes like the Marine Reserves Act, Marine Mammals Protection Act and Wildlife Act have a number of problems, and would benefit from redesign from the ground up.⁸⁰ A particularly egregious aspect is that, although they sound highly protective, they often lack the “bite” to impose true environmental limits. That could change in a future system so that the species and area-based protections imposed under such frameworks are both mandatory and powerful.

For example, population management planning is a tool that is currently available to address the bycatch of protected species in fisheries but it has never been successfully used and is arguably not fit for purpose. Since 1996, the Marine Mammals Protection Act and Wildlife Act have both provided for the creation of these plans. They are designed to ensure the recovery of *threatened* species to non-threatened status, or to prevent populations declining, but they can be created for non-threatened species as well. Plans can specify a maximum amount of fishing-related mortality for a species. When a population management plan is approved, the Minister of Fisheries is required to take all reasonable steps to ensure that the maximum allowable fishing-related mortality set is not exceeded.⁸¹

The plans require the joint sign-off of the Ministers of Conservation and Fisheries, and the Minister of Fisheries is required to consider the impact of the plan on commercial fishing.⁸² Thus, while such plans can impose limits once created, they are not really systemic limits because they can involve trading off the economic benefits of fishing with the desire to protect marine mammals.

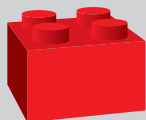
There have been several attempts to create population management plans for the New Zealand sea lion and Hector’s dolphins, but none have been completed. The reasons for this include an overly complex statutory process for their development, the need for cross-agency support which was difficult to obtain at the time, and the targets set in the legislation being unworkable.⁸³ Instead, agencies have focused on developing non-statutory plans such as national plans of action for seabirds and sharks and threat management plans for Hector’s and Māui dolphins. These often leave the implementation of fishing bycatch controls to the provisions of the Fisheries Act, which is potentially problematic, given that the purpose of that Act is the sustainable *utilisation* of fisheries, and it is not focused on the absolute protection of marine mammals or other marine wildlife.



Tools under conservation legislation could be strengthened so that they provide for more powerful species-based environmental limits. In particular, the process for creating population management plans could be made simpler and/or focused only on the biological needs of protected species (rather than the impact on other users of the sea).

Enforcement of limits

It is one thing to set limits. But consideration also needs to be given to what happens if a limit *is* infringed. This partly relates to the need for better compliance, monitoring and enforcement, and the ability to revoke or extinguish rights for repeated or reckless non-compliance with regulatory requirements (notably under the RMA). But where breaches of minimum standards of environmental health cannot be directly attributed to one or more particular people (ie it is not an compliance or enforcement issue), legislation could still provide for other forms of action to be taken. Breaching a sedimentation limit in estuaries might, for example, result in a moratorium on clearfell harvesting in a catchment (or the use of a staggered catchment-wide harvest management plan).



The system could provide that a breach of environmental limits has clear and immediate consequences as a matter of law, including (to the extent necessary) overriding existing land use rights.

8.6 Legal rights

“Rights” are ultimately a regulatory tool, even though they are not usually called by this name. If a person is granted a legal right, then any infringement of it by another person can lead to compulsion and coercion (eg through court action). But rather than public authorities forcing or preventing people from doing things, rights are about allowing (and, indeed, relying on) the holders of rights to defend them.

Rights come in many shapes and forms, and are not just lofty and inalienable things to be found in something like the Bill of Rights Act. The system can also consciously create rights, including *property* rights, as a means to achieve broader public policy objectives.⁸⁴ As tools that can be used to change behaviour, they become of particular interest for reform.

As described in previous chapters, there are many rights in the current system. In Chapter 6 we looked at whether these – or at least some of them, such as private marine title or fishing quota – should be treated as sacrosanct and “off limits” for reform from the outset. Assuming that a conversation about them is at least on the table, our concern in this chapter is now quite different: *how* rights (including but not limited to existing ones) might be used or changed as part of the toolkit to better achieve our objectives.

Property rights

One reason the system might create property rights in marine resources (by privatising them and allowing them to be traded) is:⁸⁵

to incentivise people to protect and use resources wisely or sustainably. This assumes that if people have a property or other stake in a resource (or aspect of the environment), they will manage it in a way that reflects both their own interests and that of society as a whole. It is a way to avoid a “tragedy of the commons”.

This “enclosure” approach to the commons was arguably a strong theoretical foundation for establishing quota in fisheries, and reflects an anthropocentric and instrumentalist view of the moana.⁸⁶ As we have said elsewhere, relying on property rights and markets as a tool to achieve broad and environmental goals:

relies on people’s rational self-interest to manage resources sustainably... [and] using them as an allocative mechanism assumes that markets will distribute the value of resources efficiently and equitably (or that equity doesn’t really matter). Both of these are highly questionable propositions and we must be wary of going down the route of further privatisation... The neoclassical ideal of the rational economic actor is often not the reality. If people cannot always be relied on to look after their own property, then they cannot be relied on to secure the interests of the broader public, future generations or nature itself.

So while tradeable property rights can be a tool to achieve efficiency in allocation (it ends up at its highest value use, at least in monetary terms),⁸⁷ it may be naïve to expect them (on their own) to achieve public interest outcomes in a future system. This can be seen in the case of commercial fishing quota.



Commercial fishing vessels, Whitianga

Raewyn Peart

A spotlight on fishing quota

There was some expectation that privatising fishing rights through the QMS would lead to better sustainability of commercial stocks. Owners would have an incentive to better manage them, and new entrants could buy others' rights rather than putting pressure on regulators to simply allow more fishing. But its core function was arguably an allocative one. In creating a market, quota enabled fishers to exit the industry (with capital) and promoted a rationalisation of the fishing fleet. It also provided the currency to settle the Tiriti claims.

In theory, the QMS resulted in quota being owned by those who valued the right to fish most highly, thereby promoting economic efficiency. The system also had an element of equity, in that new entrants could buy their way into the market (although this assumed a well-functioning market which has not necessarily been the case). The market determines, quite simply, who gets what. Once quota have been issued, other than some limits on aggregation, the only real ongoing allocative decision left to public intervention is the relative rights of the commercial, customary and recreational sectors when the Minister sets a TACC.

Although allocation is largely left to the market, setting the overall harvest limit (TAC) and other public interest sustainability measures remain with the Minister. At times, the country flirted with idea of devolving fisheries management to quota owners, on the basis that they were best placed to ensure the sustainability of stocks. But this has generally not happened. The market has not been left to determine what is sustainable, only what is efficient and fair. The QMS is therefore a long way off the Coase Theorem described in Chapter 7 (where *environmental* wellbeing is assumed to flow from clear and unambiguous ownership).⁸⁸

However, the tension between the different potential functions of the QMS has remained to some extent, creating uncertainty as to when property rights should be used as a tool, not only to allocate rights, but also to set environmental limits and manage stocks. This can be seen in the lack of clarity around when fisheries planning (and other management tools like setting the TACC) should be devolved to industry groups to self-regulate and when it should be wielded by central government.

This is not to suggest that devolution is always a bad thing. Indeed, while there have been notable failures, industry self-management in some fisheries has had benefits.⁸⁹ It is simply to say that a future system could be clearer about the role that different tools – including private property rights like quota – are intended to play in a future system. Are they purely allocative, or do we rely on them as incentives to protect the environment?

That is particularly relevant when it comes to sustainability measures necessary to prevent harm to the broader marine environment from fishing activities (ie not just stock management), and to protect habitats of importance to fisheries. Both of these things are core to the purpose and principles of the Act. There are already ministerial powers to impose far reaching controls, but they have for the most part not been used.⁹⁰ For some, this can partly be put down to a lack of political will and lobbying pressure (due to complex incentives in the market for quota),⁹¹ but it speaks also to persisting differences in philosophy. Interviewees who were deeply embedded in the fisheries management system at the time environmental regulatory standards were being considered reported that action on environmental matters was stymied by an unresolved debate about whether such impacts needed to be regulated under a QMS system at all.

There are risks in relying on property rights to achieve public interest environmental outcomes, especially when it comes to setting environmental limits. There is still an underlying philosophical debate about the extent to which private property rights under the QMS should be relied on to protect the marine environment, and to what extent that should be the role of clear and proactive regulation like sustainability measures. This will need to be resolved within a future toolkit.

Tradeable rights exist beyond just fisheries quota. For example, cap and trade systems for diffuse pollution (eg nutrients in catchments) that can impact the marine environment (particularly estuaries) have been set up in some places, and such things are supported by some economists.⁹² Tradeable property rights have also been the main tool of choice for restricting the emission of greenhouse gases under the emissions trading

scheme.⁹³ Other property rights in the marine environment have not been created to achieve broader policy goals, but are either historical artifacts or due to (arguably outdated) assumptions that resources must be owned by “someone”. For example, private title over marine space still exists in some places, including for properties subject to coastal erosion. And while subsurface minerals (including oil and gas) have not been privatised, they *have* been nationalised. Extraction is still (in theory)⁹⁴ managed for the public interest, but this relies on the transfer of property rights in minerals from the Crown to private entities. In other words, the Crown’s custodial role is linked to its ownership of the resource, not as an environmental steward, and management under the Crown Minerals Act largely reflects that.

This is not necessarily the way of the future. If the public trust doctrine were to apply in Aotearoa New Zealand, there would be an ethical obligation to manage all forms of property, including minerals, in ways that serve environmental and social outcomes (especially when it comes to climate change). Crown property rights in particular could come with broader obligations as to their management, not just efficiency and financial return.

Some other private property rights exist in the marine environment due to historical reasons, such as private title over the seabed and Crown ownership of some minerals like oil and gas. Especially where Crown owned, property rights could be more firmly linked to broader environmental and social outcomes.

In many other cases, however, enclosure and property rights in the moana have been strongly resisted. The MACA Act is very clear that the common marine and coastal area (the majority of the foreshore and seabed itself) cannot be owned at all,⁹⁵ even though customary marine title (where recognised) confers some rights similar to property interests.⁹⁶ Even fisheries themselves have not been “privatised” as such. Most fish remain a shared resource. Property rights only exist for commercial fishing and are not relied upon at all to manage the sustainability of recreational and customary take. And commercial fishing quota only provide a right to *take a proportion* of a maximum allowable catch set by the Minister, not a property interest in the fish themselves.

Under the RMA, the law also takes pains to emphasise that rights to do things, like occupy the seabed or discharge contaminants, are not *property* rights to be owned or traded.⁹⁷ They cannot be frustrated or derogated from by granting competing rights to others, but they are time limited, rights of transfer are controlled, and conditions can be reviewed and changed. Regulatory obligations (conditions) are firmly linked to these rights in a way that does not occur with property rights in fisheries quota and minerals. More broadly, the law tends to presume no ownership of wildlife unless within the control of a person (eg domesticated).

The grant of private property rights in the marine environment has not been as widespread as on land. Rights under the RMA and EEZ Act are deemed not to be property, there are no property rights in wild fish themselves (only in commercial rights to take a proportion of a TACC) and wild species are not “owned”. The marine space is a mix of private and public interests.

When it comes to the use of property rights in a future system, we have many options. While there is unlikely to be appetite for a conversation about reinstating or extending Crown or private ownership of the foreshore and seabed itself (although the nature of mana whenua interests is by no means resolved legally or politically),⁹⁸ property rights could be created or expanded in other ways. For example:

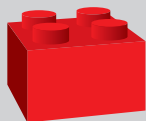


The QMS could be expanded to include commercial operators of recreational fishing activities (eg charter boats), by requiring such operators to cover their catch by purchasing ACE.

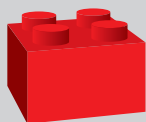


Seawall, Milford, Auckland

That might address allegations of unfairness that some operators are indirectly earning money from “free” fish (and where large scale and concentrated recreational fishing can have impacts akin to small-scale commercial harvest) while others (especially struggling commercial fishers reliant on ACE) have to pay for them. Going further:

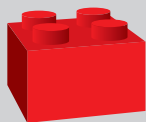


A parallel system of quota could be established for all recreational fishers (replacing tools like bag limits).⁹⁹

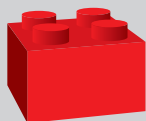


Recreational fishing could be included in the same market as commercial quota, so (at least in theory) fisheries would go to their highest value use.

However, these measures would run up against a strongly held belief by some that the freedom to access fish for non-commercial purposes (including by many Māori to feed the marae or whānau) should be available to all, not just those who can afford to pay for quota. In this sense, fishing may be seen as not too different from access to the transport network or drinking water infrastructure, and as much about the social and spiritual value as the instrumental value of fish. On another front:

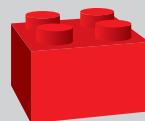


Coastal permits under the RMA and EEZ Act could be made more akin to property rights by allowing greater tradability and longer duration, especially when it comes to aquaculture and other activities requiring a long-term presence (eg wind turbines affixed to the seabed or desalination facilities).



Property rights in aquaculture could be established that are not linked to particular places or the need for coastal occupation (eg for mobile aquaculture operations based on a particular biomass rather than the area of operation).¹⁰⁰

At present, aquaculture proponents are facing challenges where spatially fixed consents are not able to move easily when conditions change (eg such as seawater warming in the Marlborough Sounds). Tradeable



Aquaculture rights could be made more fungible with quota rights, meaning that trading of rights could occur across sectors.

A quota holder might choose to trade perpetual quota for long-term aquaculture rights, while large scale aquaculture operations could have a mechanism to “buy out” ITQ rather than be refused consent on the basis they will have adverse impacts on fishing. This would involve immensely challenging design questions¹⁰¹ and potentially unpredictable consequences, and would ultimately depend on what we were trying to achieve. But it may be worth exploring. Finally:



Cap and trade markets for some forms of diffuse pollution (eg nutrients) could be rolled out more proactively across relevant catchments, and include estuaries.¹⁰² Depending on the ability to measure or estimate runoff from individual properties, that could include sediment.



Recreational fishing, Firth of Thames

Raewyn Peart

The use of property rights and markets could be expanded in a future system. That might direct resources to their highest value use. However, it might run up against issues of equity of access (particularly to fish) and some public expectation about the nature of the marine environment as a “commons” or shared space.

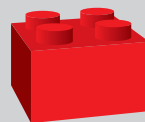
A future system could, alternatively, head in the opposite direction by questioning whether property rights in the marine space are a useful tool at all, or rather a relic of a past neoliberal age which should be undone across the board. For example, there could be a mechanism to remove privately held title in the coastal marine area (eg through gradual buyback and extinguishment of such titles), reflecting the idea that the ocean should be space owned by no one.

Seabed minerals could, like most resources in the ocean, also be owned by no one. They could be managed (and not owned) by the Crown, potentially in partnership with mana whenua, and according to more robust environmental (and climate) principles.¹⁰³ That could be achieved via amendment to the Crown Minerals Act.

The limitations of a market-based system under the emissions trading scheme are also more obvious at sea than on land. It can be harder to calculate or estimate emissions from activities like bottom trawling and sinks from planting and restoration (eg kelp and seaweed) in an environment that is extremely difficult to observe.¹⁰⁴ It is likely that a future system will need to use other regulatory and non-regulatory tools

to prevent climate change impacts (eg narrowing trawl corridors) and to incentivise mitigation measures (eg subsidies for seaweed farming).

Moreover, “property” rights need not be created in aquaculture at all, especially if coastal permits were to become more flexible (eg by being able to move them to “backup” sites in the event of environmental change, with such sites already identified in the context of a broader marine spatial plan).¹⁰⁵ In short, property rights may be useful if carefully linked to regulatory obligations, but they are not the only options in a future toolkit.



Property rights could be eschewed in a future system by removing “ownership” over some things (eg buyback of private title and a different status for Crown owned minerals), and by declining to use market based tools for others (eg greenhouse gas emissions and occupation rights for aquaculture).

While there may be ethical hurdles to *increasing* the use of property rights (we may not want to further privatise the ocean), there are arguably more formidable practical hurdles in unravelling existing rights. This is because many property rights have significant value and eroding them would understandably meet resistance. It is also because some of them have been used as a tool to implement Te Tiriti settlements. And there are questions as to whether doing so would be just or equitable. Changing the status of Crown property rights, like in some minerals, might not prove too challenging. But private property rights, including for fishing quota, are quite different.



Seafood processing plant, Motueka

A spotlight on the Legasea fisheries proposal

Some have suggested that a property rights (market) approach to the allocation of fishing rights should be undone, not just because of the practical expectations it can create (resistance to environmental regulations that can reduce property value) but also because of alleged injustices in allocative decisions themselves (ie that rights and the value share that flow from them are distributed in an unfair way). Here, we need to keep in mind that if or how commercial fishing rights are reformed depends on what the objectives of the system are. After all, markets are not necessarily expected to create what some may see as perfectly equitable results, and it is not clear whether a complete alternative to the QMS would create better environmental outcomes.

On the one hand, the QMS has proved successful in reducing fishing effort and in enabling the rebuilding of some fish stocks. It has underpinned the settlement of Māori fishing claims and associated economic revival of iwi. It has also supported the development of considerable financial capital in the fishing industry. To dismantle it would be extremely difficult, due not just to the extensive property rights involved, but also the fact quota has been used as currency for full and final Treaty settlement purposes. Furthermore, arguably it is not the QMS itself that has caused environmental impacts; it is the absence of supporting sustainability measures under the Fisheries Act and the RMA which could still be strengthened while maintaining a system based on quota.

That said, the social outcomes of the QMS have been far from uniformly positive.¹⁰⁶ Four main corporate entities control large holdings, and those wishing to harvest the fish have to pay to lease the right to fish from them on an annual basis. The operation of the market, including the relationship between the TACC, deemed values, ACE and port prices as well as the vertical integration of the industry (large corporate quota owners generally lease ACE out on the basis that landed fish will be processed at their own plant) is extremely complex and can produce both hardship and efficiencies. The overall result has been a divergence over time between the share of value going to corporate quota holders/retail stores and fishers relying on ACE.¹⁰⁷ That is on top of the exclusion of a number of fishers (eg deckhands and part timers) from quota holdings when the QMS was first established.

It is also arguable that the existence of strong and perpetual property rights without closely associated responsibilities¹⁰⁸ is a factor in *why* some sustainability measures have not been taken under the Act. The separation of quota ownership from those doing the fishing (through the creation of ACE) has arguably diluted some of the stewardship benefits of creating property rights in a resource, due to many fishers no longer having a long-term stake in the industry.¹⁰⁹ The quite different management mechanisms for commercial, recreational and customary fishing have also arguably exacerbated tensions between them, which remain unresolved.¹¹⁰ And market dynamics have in part contributed to a reduced ability for ACE reliant fishers to reinvest in more environmentally sustainable boats, gear and technology.

This is a field for rich debate, and many different perspectives are possible. We do not resolve that here. Yet it is worth noting that some, such as Legasea, have proposed unravelling the QMS through the mass buyback of quota and instead authorising commercial fishing through a permitting process, including associated environmental conditions and a financial return to the public and Māori.¹¹¹ This “Rescue Fish” policy is described as requiring:¹¹²

the Government to buy back existing quota rights in the inshore fisheries at fair value. The estimated buyback cost is between \$0.76 and and \$3.1 billion... Commercial fishing will be subsequently managed by a permitting regime. Permits will be leased, time limited and have a resource rental attached. Rental income will be collected by the Crown and shared with Māori. A new Fisheries Act will both prioritise the maintenance of healthy fish stocks and exclude bottom trawling and dredging from inshore waters. Priority will be given to Māori customary and public fishing.

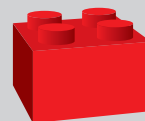
This measure would go well beyond using the “existing” toolkit better; it would be about fundamentally changing or replacing one of the tools in the toolbox. To some, it might be beyond the proper rationale for intervention, and more akin to extinguishing the Torrens system for real estate on land (see Chapter 6). Achieving such a thing may also prove very challenging, not least because of the implications for the Tiriti obligations. Has the QMS become so embedded in the Tiriti settlement framework that it is practically impossible to unwind? Would changing the terms of the Māori Fisheries Settlement open the floodgates to relitigating a whole raft of broader Tiriti settlements? Some have called for that to happen, but it has significant risks and would be fraught.¹¹³

It is not necessarily the case that an alternative permit-based system for commercial fishing would be without its own issues, or preferable to more targeted changes to Fisheries Act tools. A more modest option might be the buyback and retirement¹¹⁴ or redistribution (eg by tender) of a smaller portion of quota by the Crown on a willing seller basis, to address issues of equity. The Crown (or an arm's length agency) could become a significant perpetual quota holder, itself leasing out quota or ACE in order to achieve positive social, environmental or other public outcomes. It could be leased out at less than market value for fishers using environmentally friendly techniques or gathering scientific information; to small scale or artisanal fishers to support local fishing communities and reduce localised impacts of large vessels; and at reasonable prices for choke species that might otherwise prevent independent fishers landing their target catch.

ACE could even be leased for longer, more secure periods of time, giving more certainty of tenure for smaller fishers to invest in boats/gear and provide fishing families with more economic security. Rights could be leased at below market prices to soften the impact of any reductions in TACC on those most vulnerable. Although the Crown already holds quota, it does so for only a limited number of reasons and is not an active market participant to shape or “distort” outcomes.¹¹⁵ If there were to be public quota holder, property rights in the QMS would remain as a tool, but a public interest intervenor would operate *within* that market to soften its impacts.

Alternatively, stronger aggregation limits could be imposed in some fisheries, reducing maximum holdings (for example) to 10 or 20 percent across inshore stocks.¹¹⁶ The power for the Minister to allow holdings theoretically up to 100 percent of a fishery could be removed, taking away the potential to authorise uncompetitive markets and exacerbate access issues.¹¹⁷ Some aggregation controls might even be accompanied by regulations that reserve a portion of quota for less efficient but (arguably) more sustainable or socially equitable forms of fishing (eg artisanal fishers). Greater market intervention to protect smaller operators and sustain fishing communities can be seen in Iceland, where a portion of ITQ has been earmarked for small boats and those using hook and line systems.¹¹⁸ In Aotearoa New Zealand, does efficiency of harvest and faith in the market outweigh broader social and environmental outcomes that might be achieved by regulatory intervention? That depends on one's values.

Economic regulation might even break up the industry, as has been done in the electricity sector, by controlling the extent to which large quota owners can engage in downstream retail activities or by limiting contractual clauses requiring the use of particular processing facilities by ACE holders. That could be a complex thing to do, and whether it would be a good idea may, again, depend on whether social objectives are legitimate ones for the system to pursue (see Chapters 6 and 7).



Some have suggested altering or even replacing the property rights based QMS system. It could be undone through buyback of quota and implementing a permitting system. Alternatively, more targeted changes (eg more aggregation controls, creation of a public quota holder, and earmarking some quota for particular types of commercial fishers) could be made to soften the social impacts of market forces and incentivise environmental improvements.

Ultimately, whether property rights should be used more or less in a future system is an ethical matter as much as it is a discussion about what would be most effective in achieving social, economic and environmental outcomes. Property rights in the marine space are by no means inconsistent with improving outcomes (including environmental ones), and one way forward would be to more closely link existing (and new) property rights with other regulatory tools designed to safeguard the public interest. Care will need to be taken, however, as once established such rights can be hard to undo or alter later on.

Non-property rights

While not all private rights are “property”, some form of rights are necessary in a future system. This is because people require a degree of certainty that they may use resources – whether catching fish, occupying space, or something else – for a viable period of time. Secure rights may, indeed, be important to ensure the delivery of positive outcomes both by the public and private sector (eg food security from aquaculture, energy security from offshore renewables development, economic value through minerals development, and infrastructure like ports). In particular, the private sector will not invest in large scale activities without adequate security of tenure.

This does not mean that such rights need to be absolute. They can be specifically constrained (eg time limited), and associated with obligations (eg offsetting requirements).¹¹⁹ If a future system has a role in setting environmental limits, it will be important that such rights are not permitted to threaten them, especially where activities are ongoing and cumulative effects only become apparent over time. As such, it may be prudent (in some contexts) to shift from the language of “rights” to those of “privileges”, to highlight that rights conferred are only vis a vis other people and not the environment.

In particular, it should not be assumed that an existing right causing harm will continue simply because it has existed in the past. The Randerson Panel has pointed out that defending such rights could imperil the ability to prevent environmental degradation over long periods of time.¹²⁰ It suggested that existing use rights might be overridden where they threaten environmental limits, so it will be important for those limits to include specific elements relating to the marine environment.¹²¹

Rights to use resources will be necessary in a future system, to provide the level of certainty needed for the private and public sector to invest, and to enable important social, cultural and economic outcomes. However, they cannot be absolute if environmental limits are to be defended.



Existing use rights for land could be overridden where environmental limits were threatened (eg to avoid significant impacts on protected areas in or near estuaries).

There is also the issue of the duration of rights. Many permits granted under the RMA and EEZ Act have limited duration. Some activities require a longer (or potentially indefinite) period of time to provide adequate commercial certainty and viability, and arguably should not face the risk of full reconsenting when consent expires. One example is offshore wind energy, where operations may last many decades. Another might be the potential use of the marine space for carbon farming in the future, where that “use” may need to endure in perpetuity or the very long term. On land, such security can be provided by ownership or long-term lease of land, but at sea such rights are currently limited to a maximum of 35 years.¹²² As uses of the marine space diversify, more nuance may be needed when it comes to how long rights endure for different activities, especially where they provide significant public benefit and do not prevent compatible uses (such as recreation, fishing and tourism).



National guidance could state what durations for resource rights are appropriate for different activities, in order to provide adequate commercial certainty, while also avoiding locking in sub-optimal uses.

Allocation of resource rights

While the features of resource rights are important (eg their flexibility and duration), so too is the mechanism by which they are allocated (or reallocated) between people or uses. Many options would be possible on this front, and different resources could be treated differently. The “best” mechanisms for doing so will depend on what the system is trying to achieve when performing its allocative role (see Chapter 7), such as equity, efficiency and the Tiriti obligations.

Some rights can be conferred to all. Here, rights become “freedoms” (eg rights of access to the sea and rights to go fishing recreationally). However, where activities require exclusive use, or where rights are scarce, the system needs a mechanism by which one use is shared with another (eg commercial and recreational fishing) or where one person is preferred for the same use (eg multiple aquaculture proponents seeking limited space). The history of allocating a scarce resource in the context of aquaculture is particularly interesting.



Jetty, Pakatoa Island

Raewyn Peart

A spotlight on aquaculture

Historically, regional councils had little ability to control allocation for aquaculture activities under the dual system of the Fisheries Act and the RMA's first in, first served model.¹²³ Alternative methods were considered, but largely rejected by councils, as it was concluded that the RMA's allocation model did not sufficiently enable councils to devise alternatives (such as balloting).¹²⁴

During the 1990s, there was a sharp acceleration in the growth of the aquaculture industry. With a first in, first served allocation system, and no coastal occupation charges, there was a strong incentive to apply for “free” valuable space ahead of others. This led to a “gold rush” of applications, which gave rise to the Tiriti claims about rights to aquaculture space,¹²⁵ and prompted the government to place a moratorium on processing applications in 2002.¹²⁶

In January 2005, a new regime came into force which required all new farms to be located within AMAs.¹²⁷ It was envisaged that councils would spatially define such areas in their regional coastal plans as a single sector form of marine spatial planning. Space within AMAs could then be tendered. Under the aquaculture Treaty settlement (see Chapter 3), 20 per cent of space was to be provided to iwi, and the AMA approach provided a clear way of achieving that.¹²⁸ There was also provision for private plan changes enabling aspiring marine farmers to propose AMAs, but none came forward. Several councils attempted to create AMAs, but the processes became bogged down, and none were ultimately created. The process proved to be costly, uncertain and politically controversial. Some councils, such as Northland Regional Council, identified areas off-limits to aquaculture in the event a private plan change application was to be made.

The lack of new space under these provisions prompted further law reform, and in 2011, the requirement for locating a marine farm within an AMA was removed.¹²⁹ Aquaculture reverted to being treated in a similar way to other activities in the coastal marine area (dealt with on a reactive basis by assessing proposals as they are received), albeit with a requirement to undergo an undue adverse effects test in regards to wild fisheries. The aquaculture to Tiriti settlement is now to be achieved through the use of regional agreements, which estimate prospective future space and

provide the option for a monetary settlement in lieu of (uncertain) future space. Despite some progress, this has proved a fraught process and conceptually more challenging than the proactive establishment of defined aquaculture space through AMAs.¹³⁰

Councils have, however, been equipped with a broader range of tools (tendering and other competitive processes) to allocate space, in an attempt to prevent a recurrence of the gold-rush scenario.¹³¹ This provides a lot of flexibility for councils *if they choose to use it*, and can be regarded as a positive thing. It is a relatively low-cost way to find out the value potential users attach to resources and where it can be used most efficiently. It can also provide a financial return for the use of public space.¹³² However, while there has been some use of such methods (eg for fish farms off Coromandel harbour),¹³³ overall uptake has been patchy, with industry choosing to focus its efforts on national action (eg an NES dealing with consenting of existing marine farms) and specific regulations for relocation.¹³⁴

Similarly, tendering for rights to take, remove, reclaim and drain in the coastal marine area has existed since the inception of the RMA, and the Act provides a process whereby the Crown can sell exclusive rights to apply for coastal permits where there is likely to be competition. However, this has also seldom been used.

The debate about allocating rights to the fixed use of space has in the past largely centred on aquaculture, but has implications for many other activities as uses of the marine environment diversify (eg offshore energy generation or an expansion of MPAs).



Oyster farm, Mahurangi harbour

Raewyn Peart

One option for the future would be to continue to grant rights on a first in time basis. Here, a permitting process designed to assess impacts on the environment effectively doubles as an allocative process; the first person to make a complete application has priority rights to the resource. That has been described as a somewhat unintentional, suboptimal, and “bureaucratic” mechanism for allocation.¹³⁵ It might be adequate where there is little competition for resources, but not when there are competing users.



One option for allocating marine resources would be to use a first in time permitting system, whereby the first user to apply receives rights as long as the environmental impacts of an activity are acceptable. However, that can have a number of issues.

More direct tools for allocation could be provided. Legislation could set out a competitive process by which two or more real-world applications could be compared on a structured basis. This is what already happens under the Crown Minerals Act for minerals like petroleum (through block offers), when more than one firm is interested in obtaining rights.¹³⁶ One benefit of a competitive approach is that there would be a degree of certainty that an activity would actually go ahead. A downside is that the most financially viable activities in the present day (and those for which applications would be received) may not be the ones most in the long-term public interest (eg if suitable space is required for offshore wind farms a decade later).

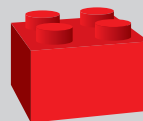
As described in the spotlight above, tools like tendering are already *available* to allocate coastal space and other resources under the RMA,¹³⁷ but the dominant mechanism is still the consenting process (first in, first served). Tendering or an invitation for multiple applications could be made mandatory, or national policy could be created for when that approach was required or encouraged.¹³⁸ That would necessitate clearer principles about the outcomes sought and the weight to be given to different attributes of an activity (eg the relative importance of financial return, employment, environmental enhancement, social expectations and equity).

As mentioned in Chapter 7, the RMA has very little normative guidance as to *why* one tender should be accepted over another. Nor does the Fisheries Act provide guidance when the Minister is setting the TACC for a stock to determine the relative rights of commercial and recreational fishing. Conservation legislation also lacks allocative principles when it

comes to deciding who should be conferred rights to undertake activities in protected areas or interact with protected species.

If more structured mechanisms for allocation were to be used in a systemic way, the impact of te Tiriti would need to be made clearer. Although there is a settlement granting 20 percent of new aquaculture space to mana whenua, it remains unclear the extent to which broader Treaty principles would give priority to Māori for *other* uses of the marine space, including tourism operations and offshore energy generation. As described in Chapter 7, the courts have said that a direction to *give effect* to the principles of te Tiriti has strong allocative implications even where no settlement exists (with consideration given to whether mana whenua should be given preferential rights to concessions).¹³⁹ Yet uncertainty remains as to how that might apply in practice in different places (including a marine setting),¹⁴⁰ or in contexts beyond the Conservation Act. The Department of Conservation is continuing to grapple with what it means and legislative guidance may be helpful.

Mandatory attribute weighted tendering could be applied beyond the context of occupation rights. For example, it could be a tool to allocate commercial fishing rights if the QMS were to be replaced (or if a portion of quota or ACE owned by a public quota holder were to be tendered). Alternatively, rights (or a portion of them) could be auctioned. Purely financial approaches like this – basing allocation on who can pay the most for a right – do not consider equity of access or wider values, such as cultural considerations. There is also the risk of corporatising rights in a few large users, forcing out others, constraining new entrants, and undermining overall community wellbeing.¹⁴¹

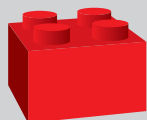


More proactive, structured and competitive allocative mechanisms could be used (or made mandatory) in a future system, such as auctioning or attribute weighted tendering.

Another option could be to use activity-based zoning to allocate spatial rights to different sectors or uses in advance.¹⁴² As we explained in *Farming the Sea*, for example, it is by no means clear that the concept of AMAs (see the spotlight above) was a bad one, and many other reasons can explain its lack of uptake.¹⁴³ The idea of proactively identifying suitable areas for particular activities has been notable in the Sea Change Tai Timu Tai Pari initiative in the Hauraki Gulf.¹⁴⁴ Such an approach may allow easier fulfilment of the aquaculture settlement (and resolve other potential te

Tiriti claims), because a percentage of actual space could be identified, mapped and valued in advance.

Spatial allocations (to identify which *uses* are appropriate, rather than who gets to undertake that use) over larger areas could be progressed through a prior process of marine spatial planning rolled out across the country (see Chapter 10). That would provide an opportunity, not just for sharp spatial boundaries between incompatible uses to be established (eg shipping avoiding future aquaculture areas for biosecurity reasons, or excluding fishing or some fishing methods from protected areas), but also for compatible activities to occur in spatially overlapping ways (eg tourism and recreational fishing; or offshore wind farms and aquaculture). It could also allow for areas to be reserved for foreseeable uses that are deemed to be in the public interest but have not yet become viable, such as desalination or offshore wind energy.¹⁴⁵ However, care would need to be taken to ensure that any collaborative spatial planning process designed to allocate marine space did not become just a negotiation between sectors; *mana whenua* and the public also have strong interest in how resources are used and would need to have a strong voice.¹⁴⁶



A more proactive allocation of rights in particular spaces or zones could be achieved through marine spatial planning. This could distribute rights between different *uses* based on public interest principles, and potentially stakeholder consensus, although not necessarily different *users*.

Reallocation of rights

While an initial allocation of rights is important, equally so is how the system allows rights to be reallocated over time. This is reasonably straightforward where there are perpetual property rights capable of being traded on an open market (eg fishing quota or cap and trade systems for nutrient pollution in catchments).¹⁴⁷ As long as the market is operating well, rights can be purchased and sold freely.¹⁴⁸ As mentioned earlier, public authorities could even operate in such markets by purchasing quota or pollution rights and retiring them or leasing them to others.

The system also needs to contemplate reallocation when such markets don't operate. The current position is that a later consent cannot interfere with the conditions of an earlier one,¹⁴⁹ and that consented rights form part of the “existing environment” against which new proposals must be assessed.¹⁵⁰ In other words, there is a robust principle of non-derogation within the RMA (and, by implication, the EEZ Act).¹⁵¹

The concept of non-derogation also spans statutory boundaries. It can be seen, for example, in the provision that aquaculture operations must not have significant adverse impacts on fishing. Here, such impacts are essentially treated as a derogation of the property right in fishing quota, and the law contemplates agreements being reached and compensation being payable by aquaculture proponents.¹⁵² Notably, there are no provisions for impacts going the other way (ie fishing activities impacting on aquaculture).

It is fairly unusual to have a statutory mechanism for private agreements to be reached between two different uses in a “public” or “shared” space, but it is an approach which could be considered for wider use. However, such a loose “dispute resolution” process may become unwieldy if there were multiple parties involved in an increasingly congested space. It may also become inappropriate without a broader exercise guided by the *public* interest in how the marine space should be used.

Similar limitations can be seen in tools for allocating the use of subsurface space and minerals, which is premised on the assumption that mining is the only potential commercial use. Carbon capture and storage might be an alternative in the future, necessitating some form of agreement or compromise – a mechanism for reallocating subsurface space.¹⁵³

At the more directive end of the spectrum, thought could be given to the role of something like the Public Works Act, or access arrangements familiar to the mining context, to facilitate the deployment of publicly important activities even if they have some impacts on, or overlap



Dolphin watching vessel, Kaikōura